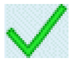
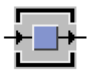
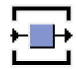
If the transparency color is not part of the color palette proper, double-click any field in the *Color Palette* that contains a color you do not need, and enter `0, 128, 128` into the dialog **Colors > Custom** and click **OK**.

- Make sure that **Icon** > **Activate Transparency** is active.
- Click **Apply Changes**  to make the area transparent.

| Not transparent | Transparent |
|---|---|
|  Station |  Station |

- To use the changed icon as the current icon of the object, make sure that Current is selected.

**Back to Work with Object Icons**

# Viewing and Visualizing Statistics

## View and Visualize Statistics

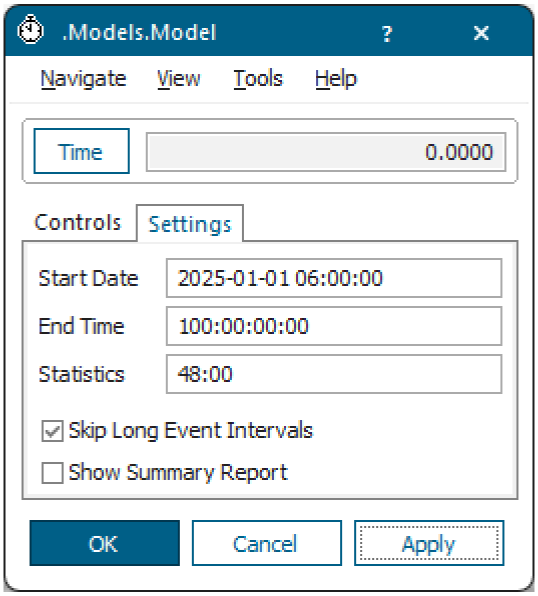
By default, all material flow objects collect statistics values from the beginning to the end of the simulation runs.

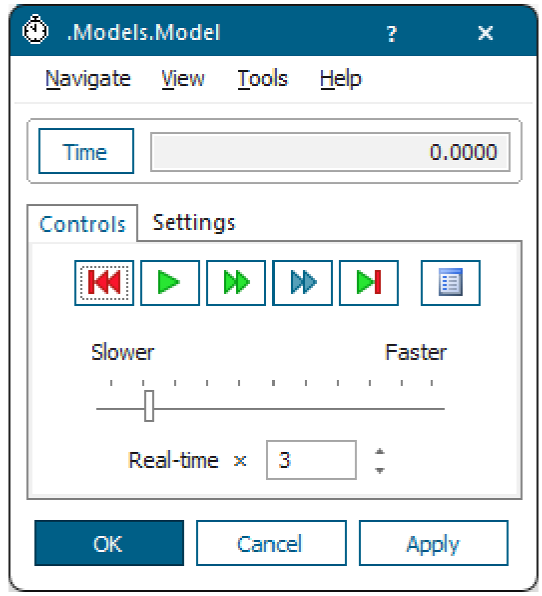
If you don't want them to do so, deactivate statistics collection by clearing the check box **Resource Statistics** Resource statistics on the tab **Statistics** or **Product Statistics** Product statistics on the tab **Product Statistics**.

In addition, the material flow objects provide **Energy Statistics** values.

The *EventController* controls when statistics collection starts and when it will be reset, i.e., started anew.

- Click the tab **Settings**. Click in the text box **Statistics** and enter the time at which the *EventController* resets statistics and deletes all statistics values it collected up so that time, compare **Statistics Collection Period**. *Plant Simulation* starts collecting statistical data for all material flow objects anew from this time on. This way you can discard data collected during warm-up phases, which might distort the validity of the simulation results.




- When you reset your simulation model, by clicking **Reset Simulation** , *Plant Simulation* deletes statistics data of all objects and resets the statistics values to 0. During a simulation run you can achieve the same effect with the method **initStat**. This way you can restart the collection of statistical information after a warm-up period of a machine, for example.

Covering *Plant Simulation* statistics in detail would go beyond the scope of the *Plant Simulation Step-by-Step Help*. It is described under the topic **Statistics of the Material Flow Objects** in the *Plant Simulation Reference Help*.

**Compare Statistics of the Mobile Objects**

**Back to Animate the Simulation Model**

## Viewing Statistics in the Dialogs of the Objects

### View Statistics in the Dialogs of the Objects

In its most basic form you can view the statistics, which the objects collected during a simulation run, in their dialogs, either on the tab **Statistics** or by opening the statistics report of the objects

Proceed as follows:

- The **material flow objects** proper show the most important statistical data they collect on the tab **Statistics**.

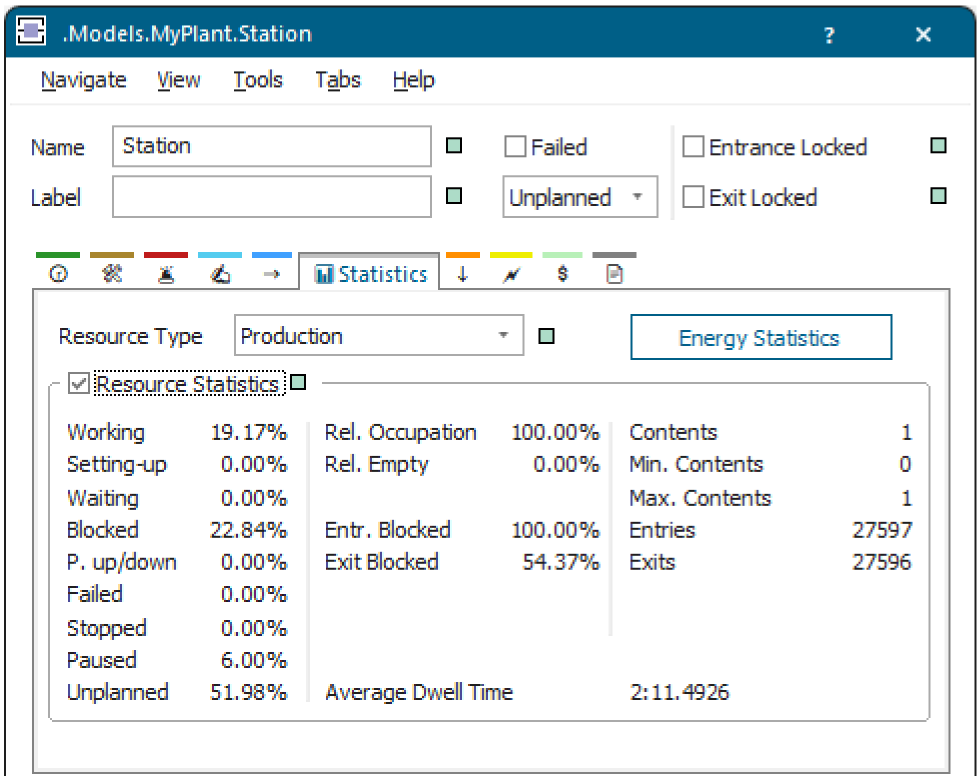
  By default, the resource type used most often is pre-selected:

  - **Production**: For the *Station*, *ParallelStation*, the *AssemblyStation*, and for the *DismantleStation*.

- **Transport**: For the *PlaceBuffer*, *Buffer*, *Sorter*, *Track*, *TwoLaneTrack*, *Conveyor*, *Transporter*, and for the *Container*.
- **Storage**: For the *Store*.

The resource type you select here affects statistics of mobile objects.

> **Note**
>
> The percentages for **Working**, **Setting-Up**, **Waiting**, **Blocked**, **Powering up/down**, **Failed**, **Stopped**, **Paused**, and **Unplanned** should add up to 100 percent.




> **Note**
>
> *Plant Simulation* does not dynamically update the values during a simulation run when the dialog box is open. To accomplish this, select **Refresh** in the **View** menu of the object or press **F5**.

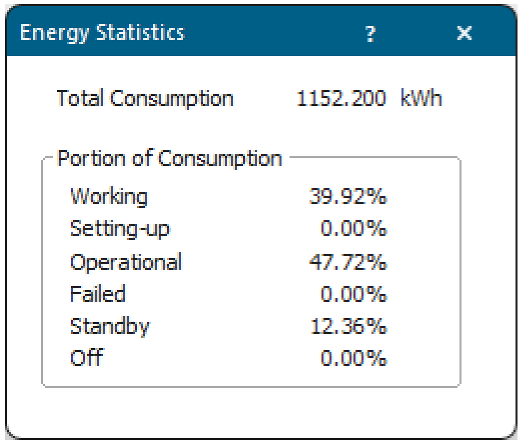
| Item | Description |
|---|---|
| **Working** | Shows the portion of the statistics collection period during which the object was **Working**. |
| **Setting-up** | Shows the portion of the statistics collection period during which the object was **Setting-Up**. |
| **Waiting** | Shows the portion of the statistics collection period during which the object was **Waiting**. |

| Item | Description |
|---|---|
| **Blocked** | Shows the portion of the statistics collection period during which the object was **Blocked**. |
| **Powering up/down** | Shows the portion of the statistics collection period during which the object was **Powering up/down**. |
| **Failed** | Shows the portion of the statistics collection period during which the object was **Failed**. |
| **Stopped** | Shows the portion of the statistics collection period during which the object was **Stopped** by a **LockoutZone**. |
| **Paused** | Shows the portion of the statistics collection period during which the object was **Paused**. |
| **Unplanned** | Shows the portion of the statistics collection period during which the object was **Unplanned**, i.e., is not scheduled to work. |
| **Relative Occupation** | Shows the capacity-based portion of the time during which the object was occupied, not paused and not failed to the time the object was not paused and not failed as a *real* number. |
| **Relatively Empty** | Shows the portion of the statistics collection period during which the object was **Empty** in relation to the time during which the object was available. |
| **Contents** | Shows the number of *MUs*, which are located on the object, as an *integer* number. The rules described under **Entries** below apply. |
| **Minimum Contents** | Shows the minimum number of *MUs* that was located on the object, as an *integer* number. The rules described under **Entries** below apply. |
| **Maximum Contents** | Shows the maximum number of *MUs* that was located on the object, as an *integer* number. The rules described under **Entries** below apply. |
| **Entries** | Shows the number of *MUs* that entered the object as an *integer* number. When a *MU* enters the object, it only counts the *MU* itself, not its contents: When a *Container* holding several *MUs* enters the object, the number of entries increases by 1, not by the number of *MUs* the *Container* transports! |
| **Exits** | Shows the number of *MUs* that exited the object as an *integer* number. When a *MU* exits the object, it only counts the *MU* itself, not its contents: When a *Container* holding several *MUs* leaves the object, the number of exits increases by 1, not by the number of *MUs* the *Container* transports! |
| **Average Dwell Time** | Shows the average time of all parts which stayed on the station. *Plant Simulation* only counts times during which the object was not paused and not unplanned. |

To view the statistics values, which the object collected, click **Show Statistics Report** on the **Home** ribbon tab. You can query most of these values with the *Methods* that are listed next to the name of the value in the description of the statistics table.

To view **Energy Statistics** values, click 

. *Plant Simulation* show these values in a dialog of its own:



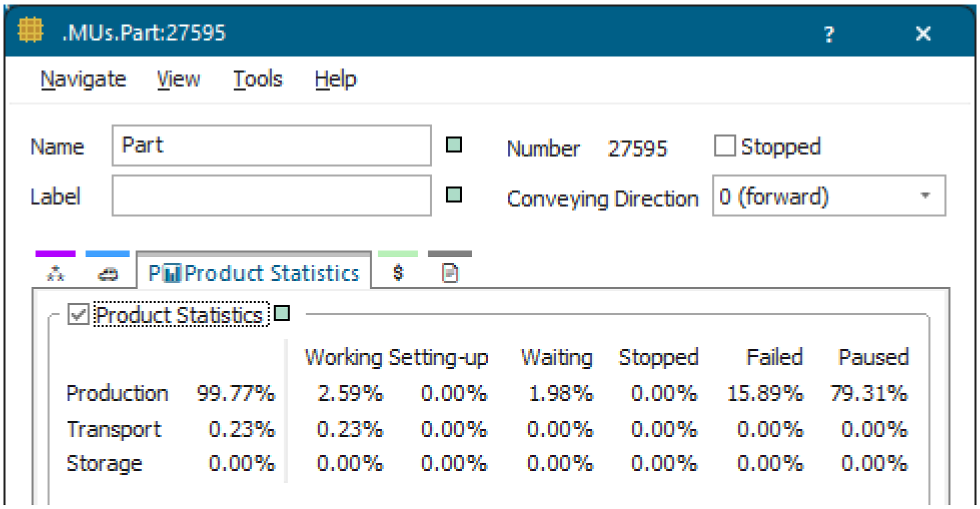

| Item | Description |
|---|---|
| **Total Consumption** | Shows the sum of the total energy consumption. |
| **Working** | Shows the portion of the total energy consumption during which the object was **working**. |
| **Setting-up** | Shows the portion of the total energy consumption during which the object was **setting-up**. |
| **Operational** | Shows the portion of the total energy consumption during which the object was **operational**. |
| **Failed** | Shows the portion of the total energy consumption during which the object was **failed**. |
| **Standby** | Shows the portion of the total energy consumption during which the object was on **standby**. |
| **Off** | Shows the portion of the total energy consumption during which the object was **off**. |

**Note**

The energy states of the material flow objects differ from the resource states with the same name. The values for the resource states refer to the statistics collection period, while the values for the energy states refer to the total energy consumption.

To view **Energy Statistics — Energy Statistics** in the *Statistics Report*, click the object in the *Frame*, and press **F6**.

- The **mobile objects** (*MUs*) show the most important statistical data they collect on the tab **Product Statistics**.




You can:

- **Check How Many Parts Were Introduced into the Plant**
- **Check How Many Parts Left the Plant**
- **Check Statistics of the Individual Stations**
- **Check the Contents List of the Stations**
- **Check Product Statistics of Parts**
- **Check Statistics of Exporter and Worker**
- **Viewing the Statistics Report**

**Compare Show Statistics in a Chart**

**Compare Show Statistics and Other Values in a Report**

**Compare Show Values During the Simulation with the Display**

**Back to Accessing Statistics with SimTalk**

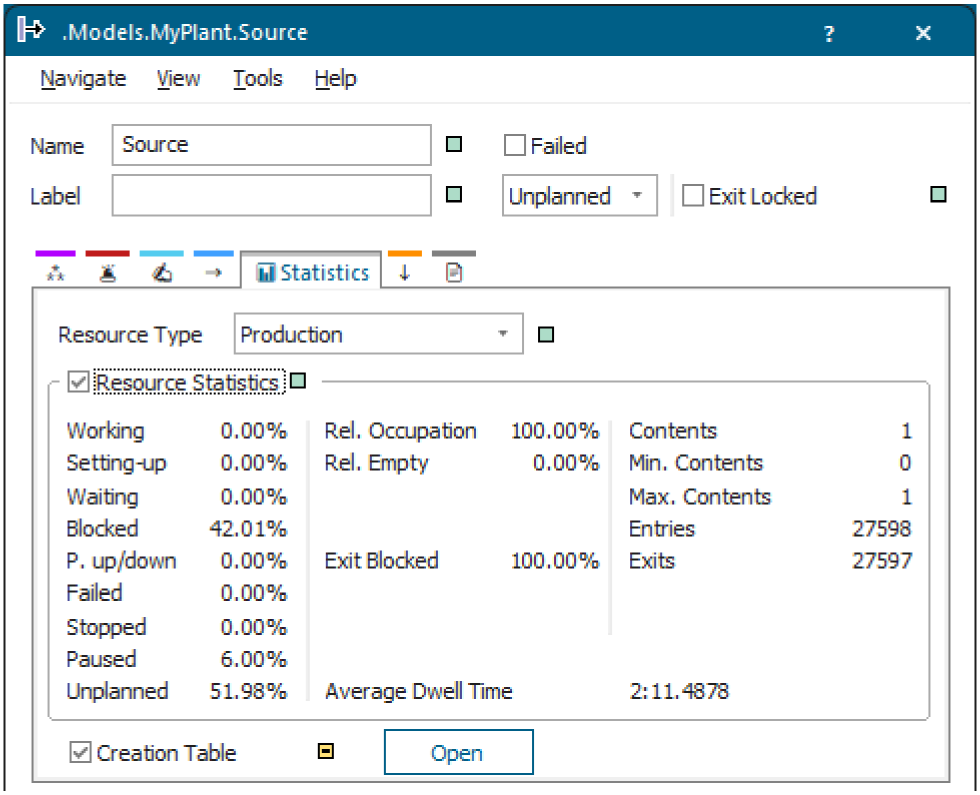
## Check How Many Parts Were Introduced into the Plant

You can check how many parts were introduced into the plant.

Next to the values described under **View Statistics in the Dialogs of the Objects**, you can make the *Source* record which *MUs* it actually created at which point in time.

- To write all events for which the *Source* produced *MUs* during a simulation run to a table, select **Creation table** on the tab **Statistics**.

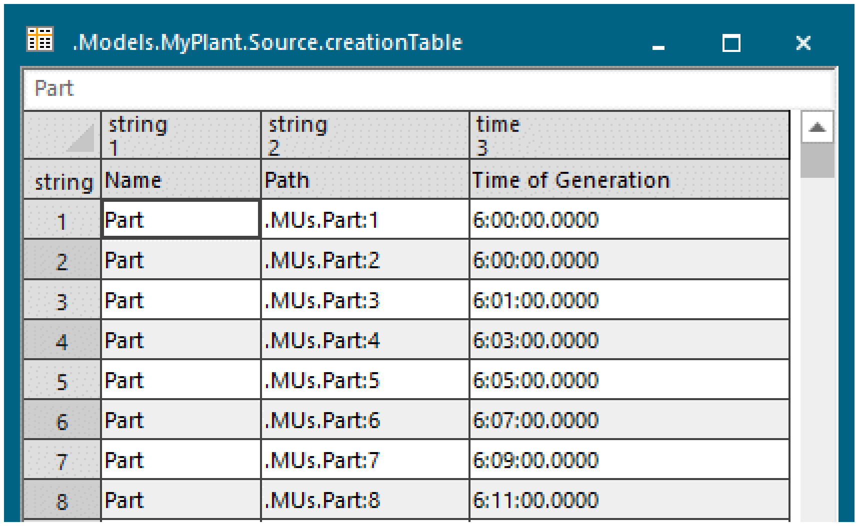
> **Note**
>
> The *Source* records these events in addition to the resource statistics it collects in any case.




- To open the creation table, click **Open**.

  Each row on the table shows an event for which the *Source* produced a *MU*:

  - The column **Name** shows the name of the produced *MU* class.
  - The column **Path** shows the path of the *MU* class, the name of the *MU* and its number.
  - The column **Time of Generation** shows the time at which the *Source* produced the individual *MU*.



.Models.MyPlant.Source.creationTable

Part

| | string 1 | string 2 | time 3 |
|---|---|---|---|
| string | Name | Path | Time of Generation |
| 1 | Part | .MUs.Part:1 | 6:00:00.0000 |
| 2 | Part | .MUs.Part:2 | 6:00:00.0000 |
| 3 | Part | .MUs.Part:3 | 6:01:00.0000 |
| 4 | Part | .MUs.Part:4 | 6:03:00.0000 |
| 5 | Part | .MUs.Part:5 | 6:05:00.0000 |
| 6 | Part | .MUs.Part:6 | 6:07:00.0000 |
| 7 | Part | .MUs.Part:7 | 6:09:00.0000 |
| 8 | Part | .MUs.Part:8 | 6:11:00.0000 |

**Compare Produce Parts with the Source**

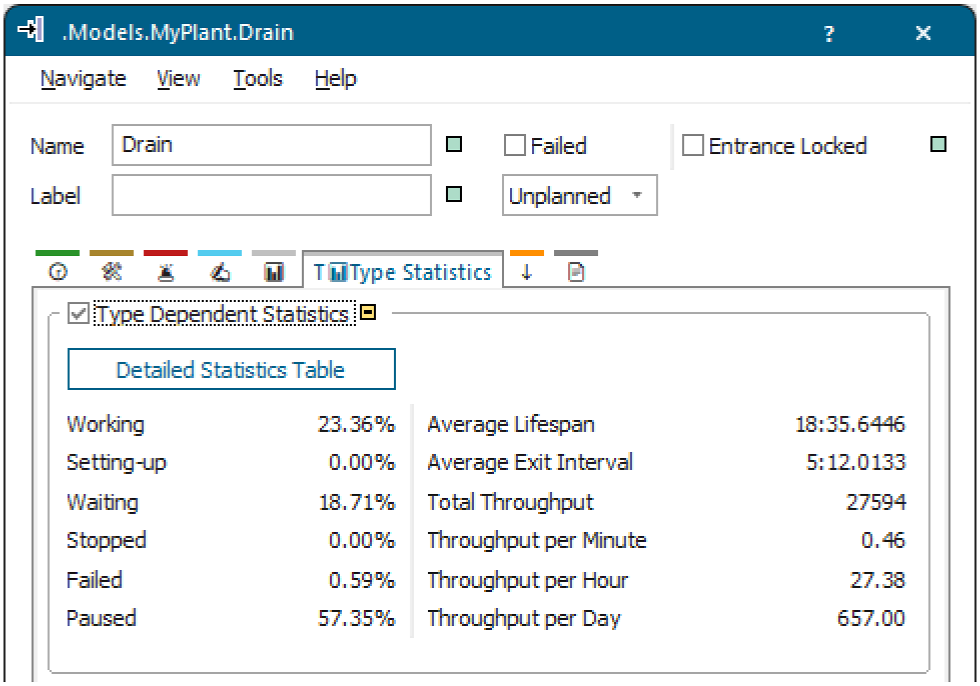
**Back to View Statistics in the Dialogs of the Objects**

## Check How Many Parts Left the Plant

You can check how many parts left the plant.

Next to the values described under **View Statistics in the Dialogs of the Objects**, you can make the *Drain* record statistics for *MUs* with the same name, i.e., product statistics, which move through the simulation model.

The numbers for the average exit interval and for the throughput are of special interest for production planning and production control specialists.




To collect statistics data depending on the type of *MU*, select **Type Dependent Statistics**. *Plant Simulation* will show this statistical data after the simulation run on the tab:

| Item | Description |
|---|---|
| **Working** | Shows the percentage of the sum of the times during which the *MUs* were located on a **working** object, in relation to the statistics collection periods of all *MUs*. In general, this is the life time of the *MUs*. |
| **Setting-up** | Shows the percentage of the times that the objects were **setting-Up** for the *MUs*, in relation to the statistics collection periods of all *MUs*. In general, this is the life time of the *MUs*. |
| **Waiting** | Shows the percentage of the times during which the objects were **waiting** for the *MUs*, in relation to the statistics collection periods of all *MUs*. In general, this is the life time of the *MUs*. |
| **Stopped** | Shows the percentage of the sum of the times during which the *MUs* were located on an **stopped** object, in relation to the statistics collection periods of all *MUs*. In general, this is the life time of the *MUs*. |
| **Failed** | Shows the percentage of the sum of the times that the *MUs* were located on a **failed** object, in relation to the statistics collection periods of all *MUs*. In general, this is the life time of the *MUs*. |
| **Paused** | Shows the percentage of the sum of the times that the *MUs* were located on a **paused** object, in relation to the statistics collection periods of all *MUs*. In general, this is the life time of the *MUs*. |

| Item | Description |
|---|---|
| **Average Lifespan** | Shows the average life-span of *MUs*, which were created and removed from the plant during the statistics collection period. Only those *MUs* are counted whose product statistics was active. |
| **Average Exit Interval** | Shows the average time interval between exits of the *MUs*, which the *Drain* removed from the plant. It results from adding up the times of all intervals, and then dividing them by the number of intervals. Statistics only starts counting from the time the first *MU* arrived! |
| **Total Throughput** | Shows the number of *MUs*, which the *Drain* removed from the plant, starting at the time, at which you activated **Type dependent statistics**. |
| **Throughput per Hour** | Shows the number of *MUs*, which the *Drain* removed from the plant in an hour on average, covering all observed times during which *MUs* arrived. |
| **Throughput per Day** | Shows the number of *MUs*, which the *Drain* removed from the plant in a day on average, covering all observed times during which *MUs* arrived. |

To view the statistics, which the *Drain* collected, click **Show Statistics Report** on the **Home** ribbon tab. You can query most of these values with the *Methods* that are listed next to the name of the value in the description of the statistics table.

**Back to View Statistics in the Dialogs of the Objects**

## Check Statistics of the Individual Stations

You can check statistics of the individual stations.

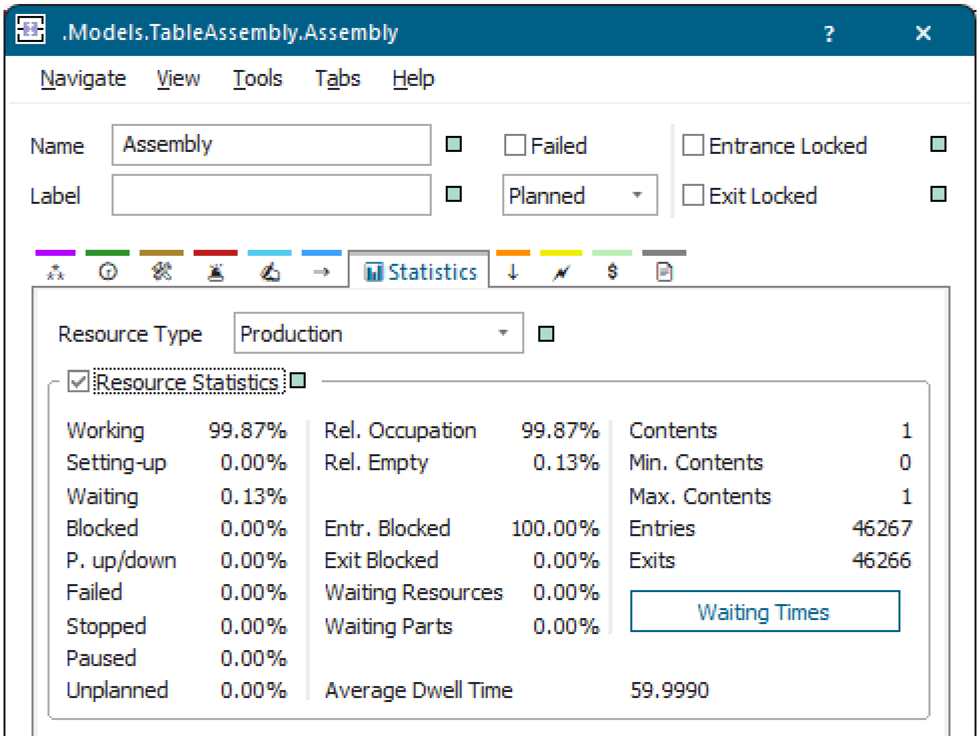
All material flow object show the values described under **View Statistics in the Dialogs of the Objects** on the tab **Statistics**.

The **AssemblyStation** and the **DismantleStation** show additional statistical values.

The *AssemblyStation* records the percentage of the time it was:

- **Waiting for parts**, i.e., for mounting parts in relation of the statistics collection period.
- **Waiting for resources**. This is the portion of the time during which the *AssemblyStation* was waiting before the *main MU* entered into it, or from the time set-up is finished, until assembly starts in relation of the statistics collection period.

Click the button to open the table **Waiting Times**, which shows the sum of the waiting times of the *MUs* for each predecessor.




**Back to View Statistics in the Dialogs of the Objects**

## Check the Contents List of the Stations

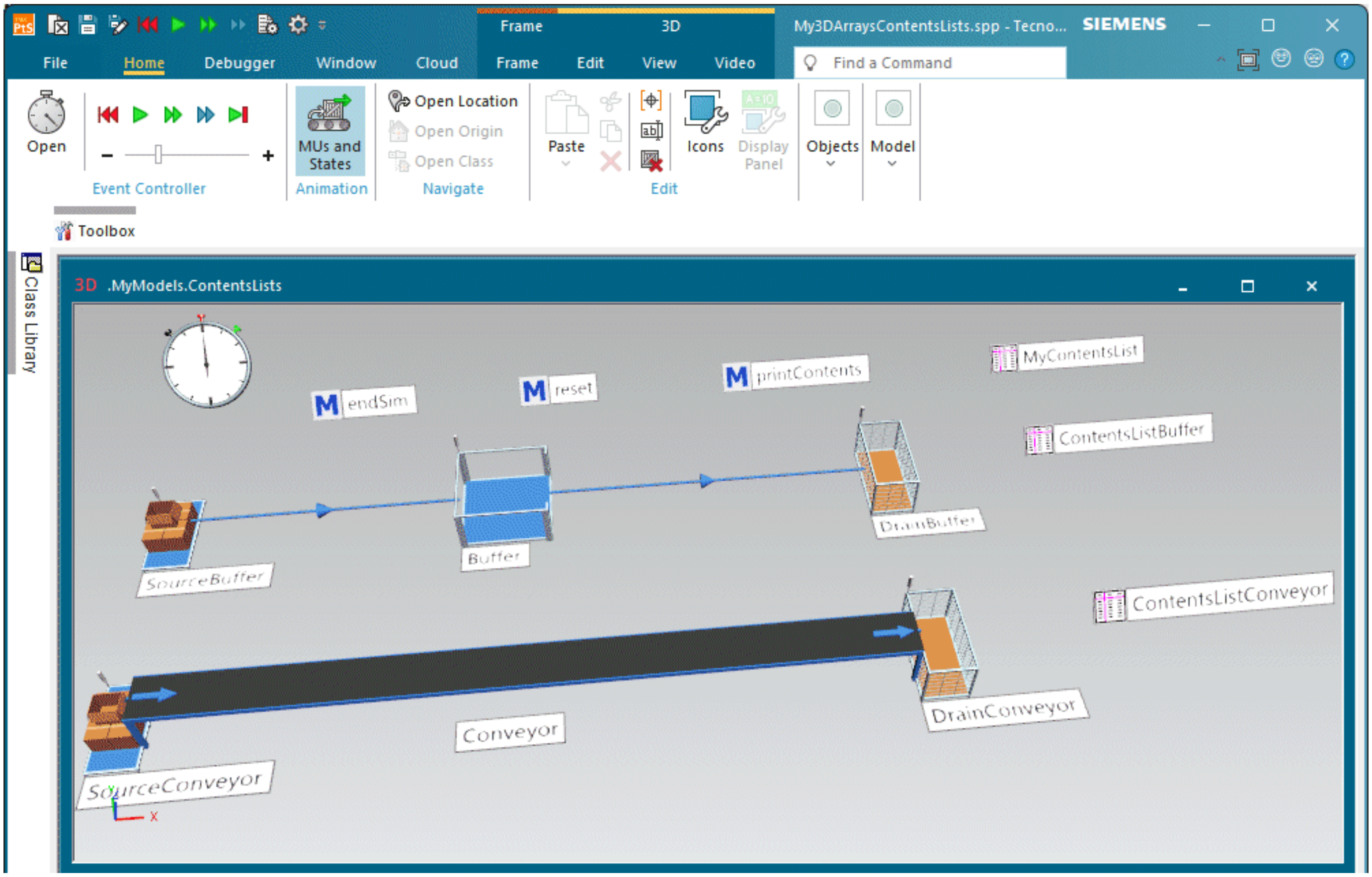
## Check the Contents List of the Stations

You can check and work with the **Contents List** of the individual material flow objects in a number of ways.

You can:

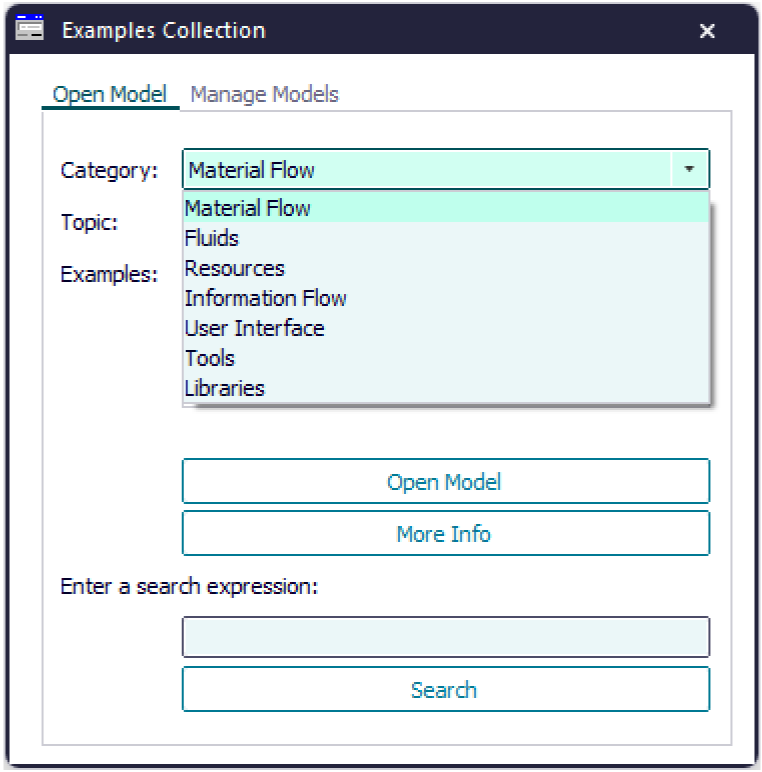
- **Open the Contents List in the Dialog of the Material Flow Objects**
- **Write the Content List into a Table for Further Processing**
- **Print the Contents List to the Console as an Array**

We demonstrate how to work with the contents list with the objects from the sample model below.




You can also use these techniques for working with the Forward Blocking List, the Exit Blocking List, and the Backward Blocking List of the individual material flow objects.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



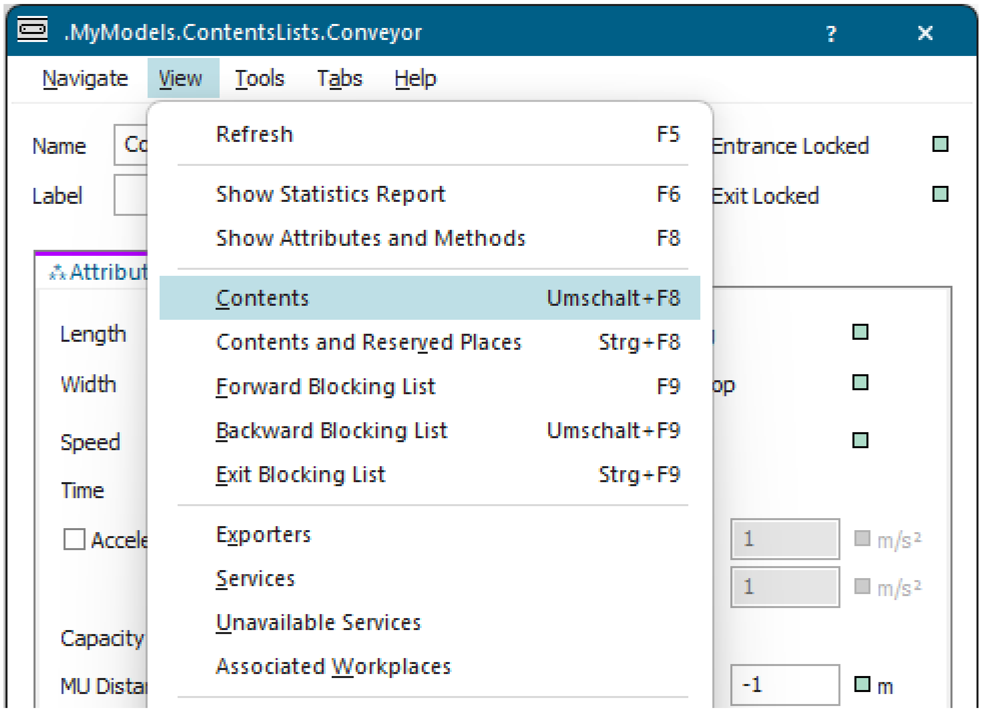

**See also** Contents [material flow objects]

**Back to** View Statistics in the Dialogs of the Objects

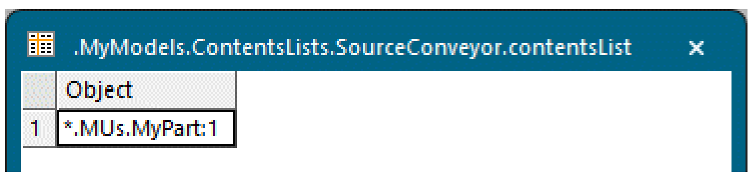
## Open the Contents List in the Dialog of the Material Flow Objects

You can open the contents list in the dialog of the material flow objects.

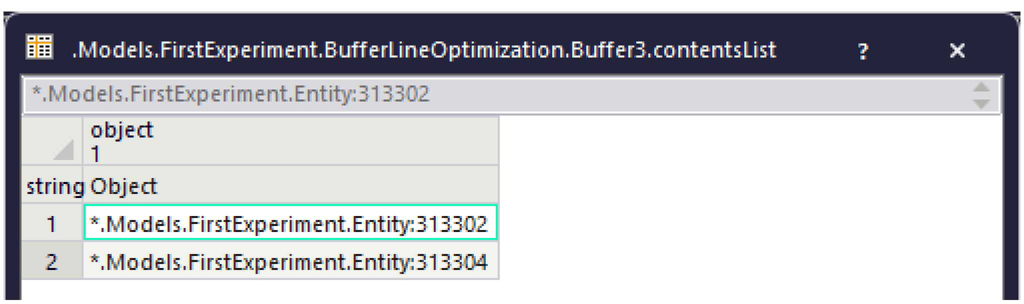
To open the Contents List of the individual material flow objects in your simulation model during or after the simulation run, open the dialog of the respective object and select **View > Contents**.



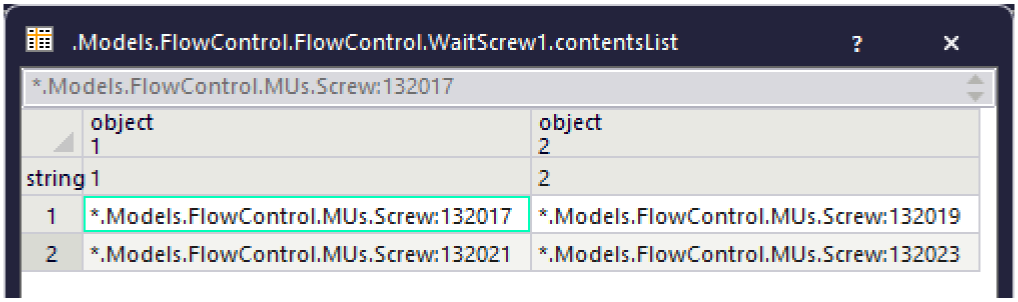

- The point-oriented objects with a capacity of 1 part show the path in the list, i.e., the name and the identifying number of the current part.




- The point-oriented objects with a capacity of more than 1 part show the path in the list, i.e., the names and the identifying numbers of the parts that are located on it. The capacity or the number of parts located on the object determines how many parts it shows.




- The *ParallelStation* shows the path, i.e., the name and the identifying number of the current parts in the list. The position of the part in the table matches the position of the part in its x-y coordinate system. The data type of the columns is *object*.



.Models.FlowControl.FlowControl.WaitScrew1.contentsList

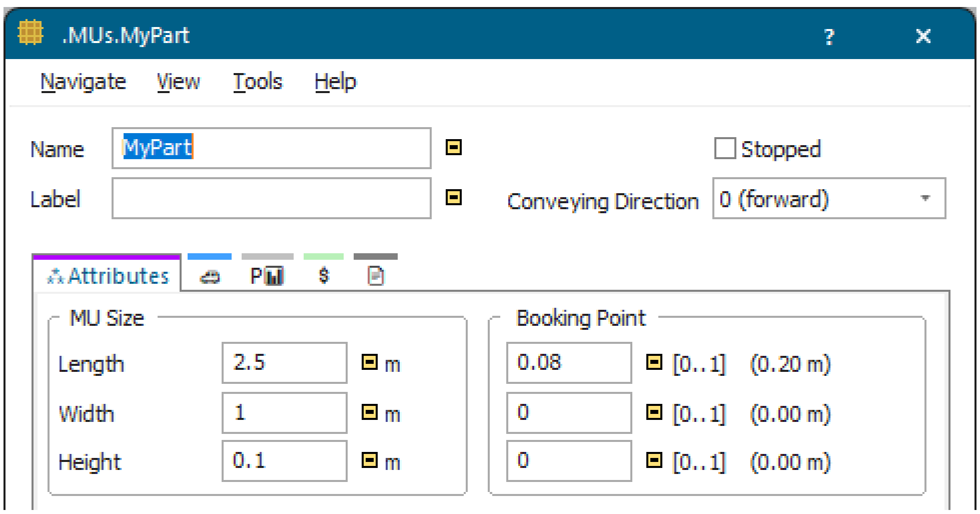
*.Models.FlowControl.MUs.Screw:132017

| | object 1 | object 2 |
|---|---|---|
| string | 1 | 2 |
| 1 | *.Models.FlowControl.MUs.Screw:132017 | *.Models.FlowControl.MUs.Screw:132019 |
| 2 | *.Models.FlowControl.MUs.Screw:132021 | *.Models.FlowControl.MUs.Screw:132023 |

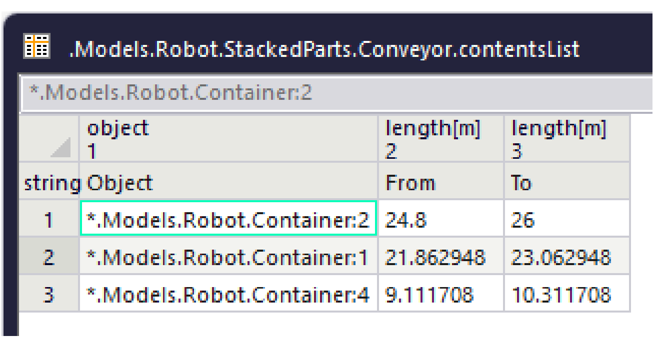
- The length-oriented objects (*Conveyor*, *Track*, *TwoLaneTrack*, and *Transporter*) show the path in the contents table, i.e., the name and the identifying number of the parts, that are located on the object. In addition it shows the position of the *MUs* in the object.

  The column **Object** shows the path to the *part*. The column **From** shows the start position of the *MU*. The column **To** shows the end position in the length unit you selected under **File > Model Settings/ Preferences > Units > Length**. The number of rows matches the number of *MU parts* located on the object.

  In our example, the *Conveyor* is 15 meters long, while each *part* is 2.5 meters long.




The *Conveyor* can thus accommodate 5 parts in their entirety, which the contents list shows.



.Models.Robot.StackedParts.Conveyor.contentsList

*.Models.Robot.Container:2

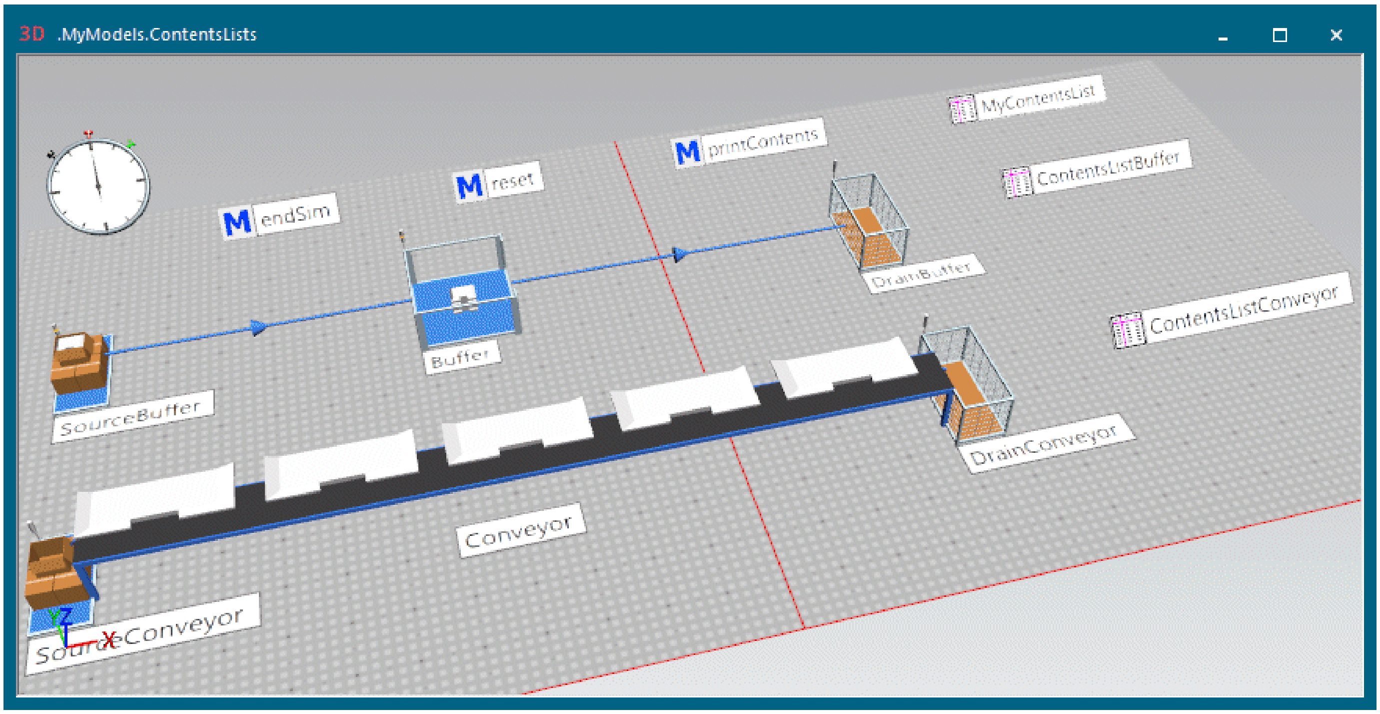
| | object 1 | length[m] 2 | length[m] 3 |
|---|---|---|---|
| string | Object | From | To |
| 1 | *.Models.Robot.Container:2 | 24.8 | 26 |
| 2 | *.Models.Robot.Container:1 | 21.862948 | 23.062948 |
| 3 | *.Models.Robot.Container:4 | 9.111708 | 10.311708 |


**Back to** Check the Contents List of the Stations

## Write the Content List into a Table for Further Processing

As the internal **Contents List** of the objects, which the menu command **View > Contents** opens, will be deleted when you reset your simulation model, you have to save it, if you need this data later on for further processing.

As the default return value of the **Contents List**, among others of the method **contentsList**, is an *array*, we are going write this *array* into a table with the method **copyToTable**.

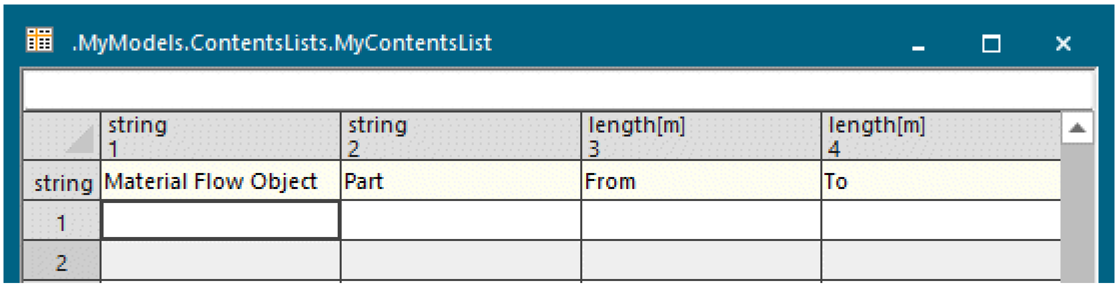



To accomplish this, we entered the following source code into an *endSim* method:

```
// Copies the contents lists into the table MyContentsList
// as an array with the method copyToTable
var x,y : real; var row : integer

MyContentsList.delete
var a : any := Buffer.contentsList
MyContentsList[1,1] := "Buffer"
a.copyToTable(MyContentsList,2,1)
row := MyContentsList.yDim + 2 // inserts an empty line

MyContentsList[1,row] := "Conveyor"
print a // prints the contents list as an array to the console
print "Number of items: ",a.yDim
var b := Conveyor.contentsList // local variable is of data type any
print b
print "Number of items: ",b.yDim
b.copyToTable(MyContentsList, 2,row)
MyContentsList.openDialog
```

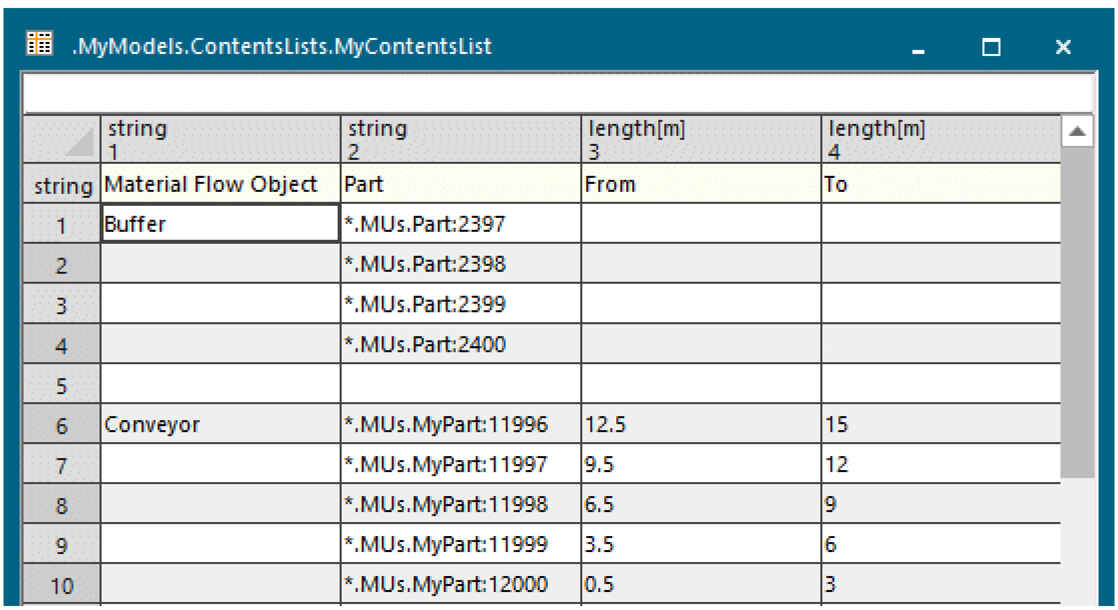
```
// Writes the contentsList into a table
// as previous versions of Plant Simulation did
Buffer.contentsList(ContentsListBuffer)
Conveyor.contentsList(ContentsListConveyor)
```

Then, we manually formatted the table *MyContentsList* by selecting the correct data types for the columns and by entering the column index items as shown below.


.MyModels.ContentsLists.MyContentsList

| | string 1 | string 2 | length[m] 3 | length[m] 4 |
|---|---|---|---|---|
| string | Material Flow Object | Part | From | To |
| 1 | | | | |
| 2 | | | | |

After running the simulation, the *endSim* method writes the contents lists to the table:

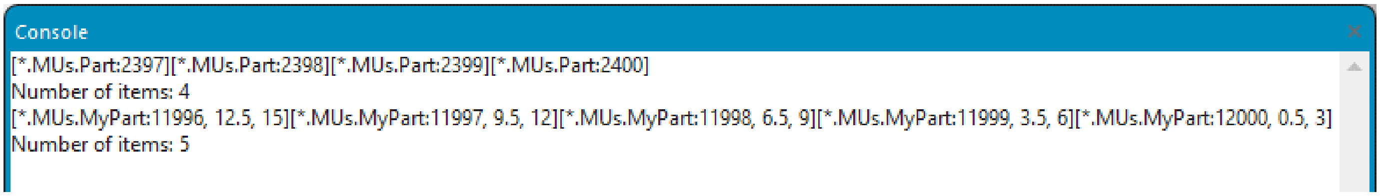

.MyModels.ContentsLists.MyContentsList

| | string 1 | string 2 | length[m] 3 | length[m] 4 |
|---|---|---|---|---|
| string | Material Flow Object | Part | From | To |
| 1 | Buffer | *.MUs.Part:2397 | | |
| 2 | | *.MUs.Part:2398 | | |
| 3 | | *.MUs.Part:2399 | | |
| 4 | | *.MUs.Part:2400 | | |
| 5 | | | | |
| 6 | Conveyor | *.MUs.MyPart:11996 | 12.5 | 15 |
| 7 | | *.MUs.MyPart:11997 | 9.5 | 12 |
| 8 | | *.MUs.MyPart:11998 | 6.5 | 9 |
| 9 | | *.MUs.MyPart:11999 | 3.5 | 6 |
| 10 | | *.MUs.MyPart:12000 | 0.5 | 3 |

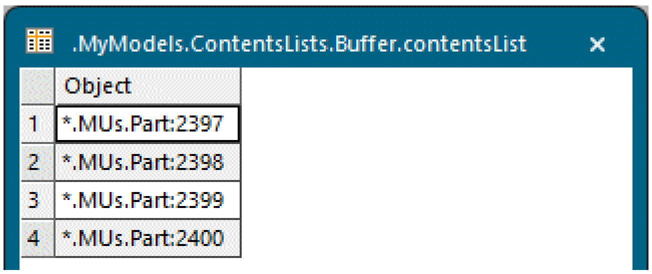
To delete the contents of the contents list, you can enter the following source code into a *reset method*.

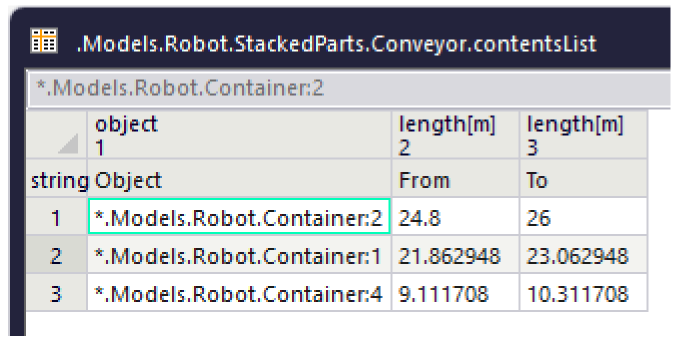
```
MyContentsList.delete
```

In addition, it writes the contents lists as an *array* each for the objects *Buffer* and *Conveyor* to the *Console*.

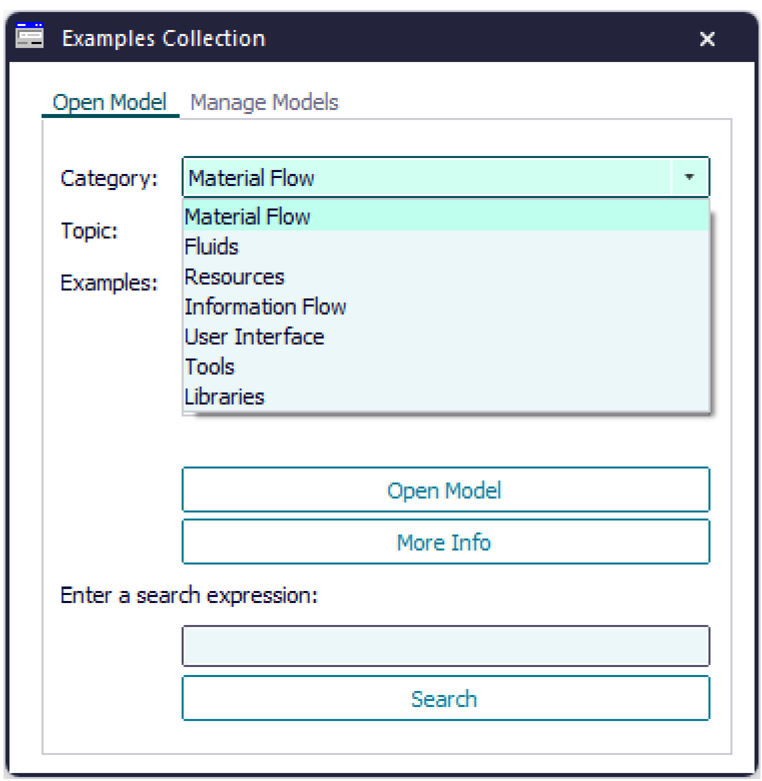



Finally, the *endSim* method writes the contents lists of the *Buffer* and the *Conveyor* into the tables *ContentsListBuffer* and *ContentsListConveyor*, which we inserted into the *Frame*.




Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



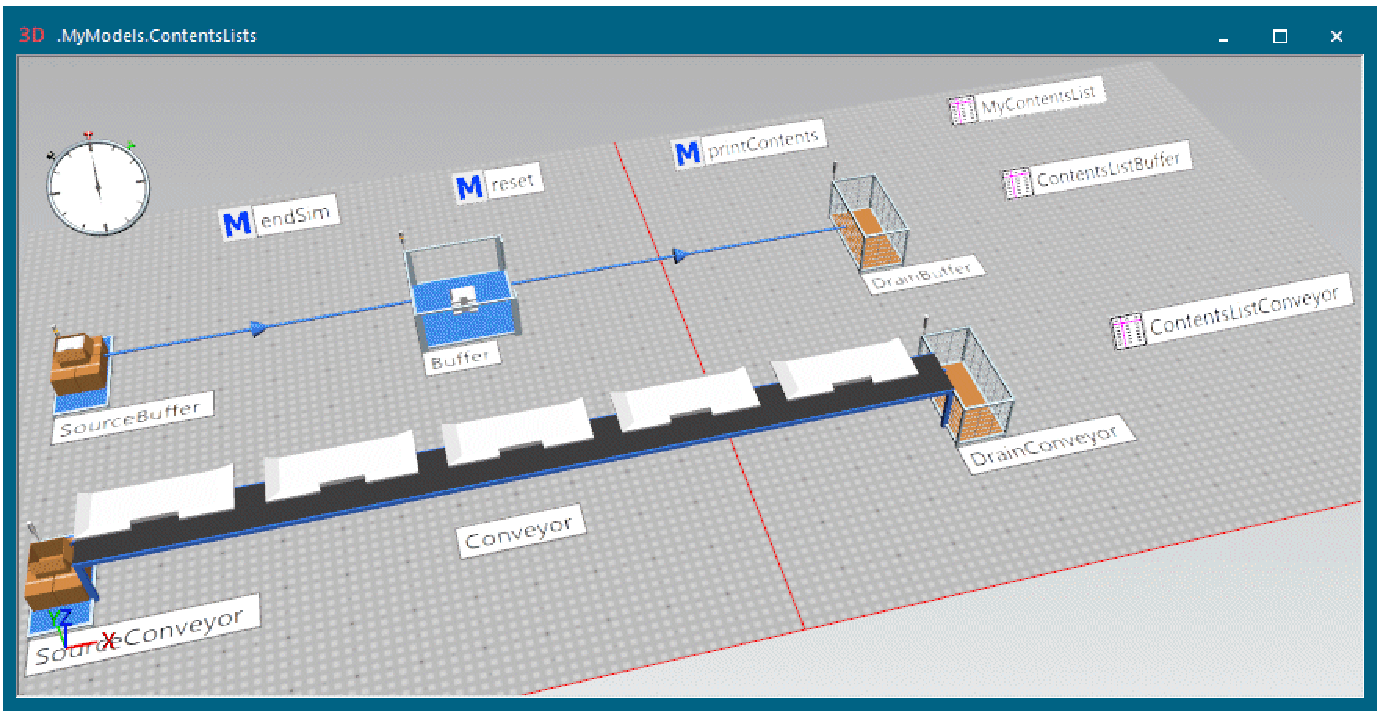
**Back to Check the Contents List of the Stations**

## Print the Contents List to the Console as an Array

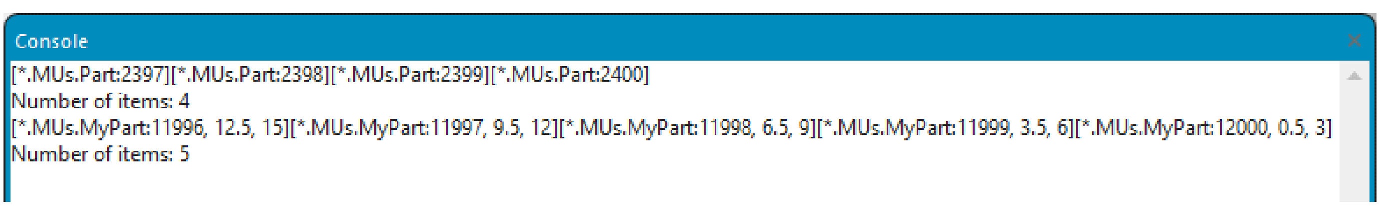
You can print the contents list to the *Console* as an array with a *Method*.

To print the **Contents List** of the objects *Buffer* and *Conveyor* from our example to the *Console* as an *array* each, type the following source code into a *Method* and run the method.

```
print Buffer.contentsList
print Conveyor.contentsList
```



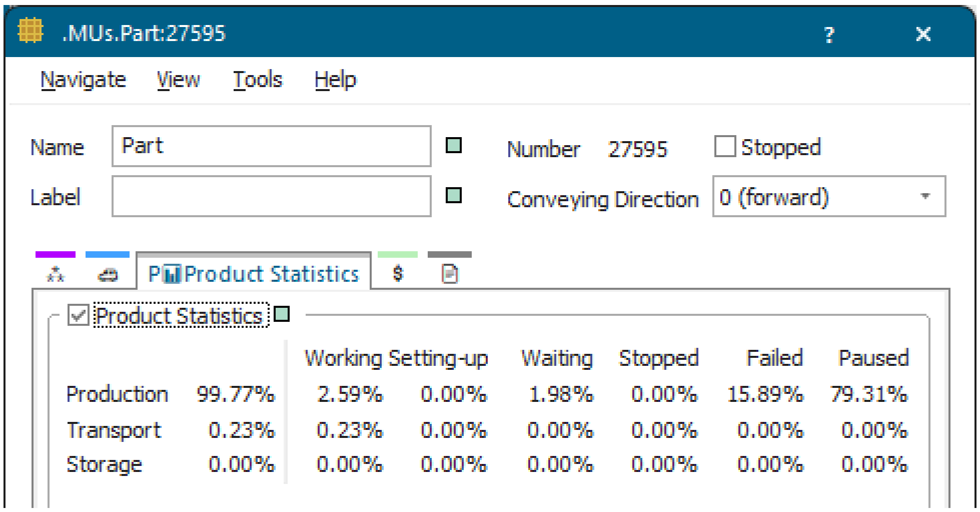

The result looks like this:



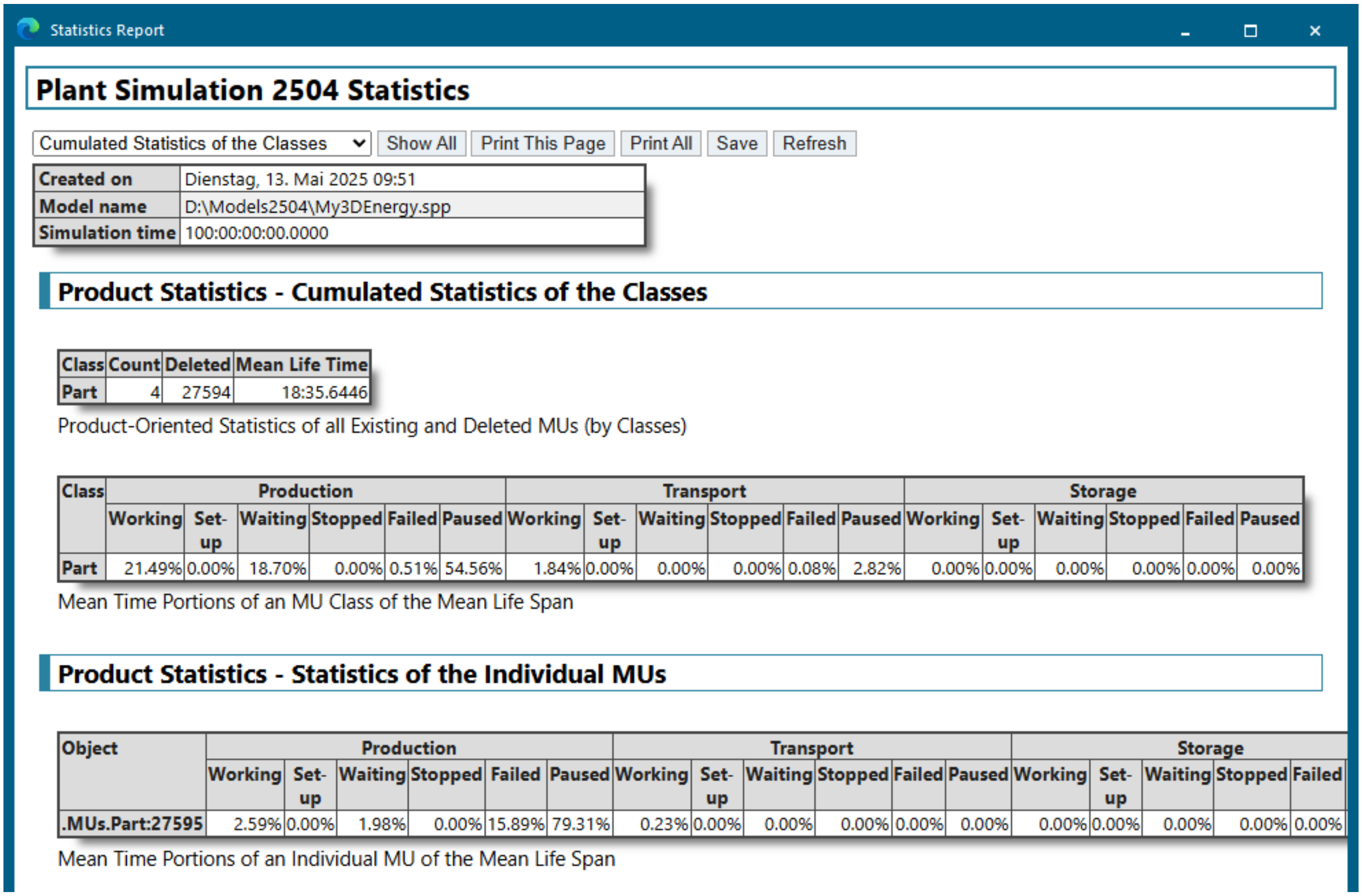

**Back to Check the Contents List of the Stations**

## Check Product Statistics of Parts

All *MUs*, represented by the *Part*, *Container*, and *Transporter*, show the percentages of the times the part spent **waiting** or **working** on a resource of type **Production**, **Transport** or **Storage** on the tab **Product Statistics**.




To view all product-related statistics data, which the *MU* collected, click **Home** > **Show Statistics Report**. You can query most of these values with the *read-only attributes* that are listed next to the name of the value in the description of the statistics report.


Statistics Report

**Plant Simulation 2504 Statistics**

Cumulated Statistics of the Classes | Show All | Print This Page | Print All | Save | Refresh

| | |
|---|---|
| **Created on** | Dienstag, 13. Mai 2025 09:51 |
| **Model name** | D:\Models2504\My3DEnergy.spp |
| **Simulation time** | 100:00:00:00.0000 |

**Product Statistics - Cumulated Statistics of the Classes**

| Class | Count | Deleted | Mean Life Time |
|---|---|---|---|
| Part | 4 | 27594 | 18:35.6446 |

Product-Oriented Statistics of all Existing and Deleted MUs (by Classes)

| Class | Production | | | | | | Transport | | | | | | Storage | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Working | Set-up | Waiting | Stopped | Failed | Paused | Working | Set-up | Waiting | Stopped | Failed | Paused | Working | Set-up | Waiting | Stopped | Failed | Paused |
| Part | 21.49% | 0.00% | 18.70% | 0.00% | 0.51% | 54.56% | 1.84% | 0.00% | 0.00% | 0.00% | 0.08% | 2.82% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |

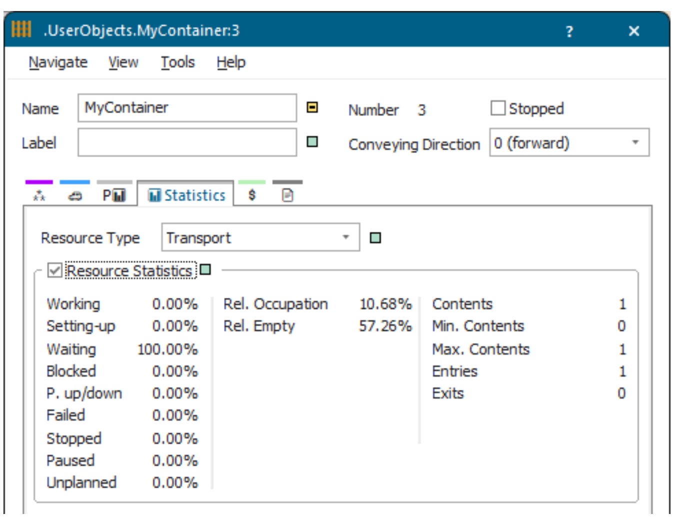
Mean Time Portions of an MU Class of the Mean Life Span

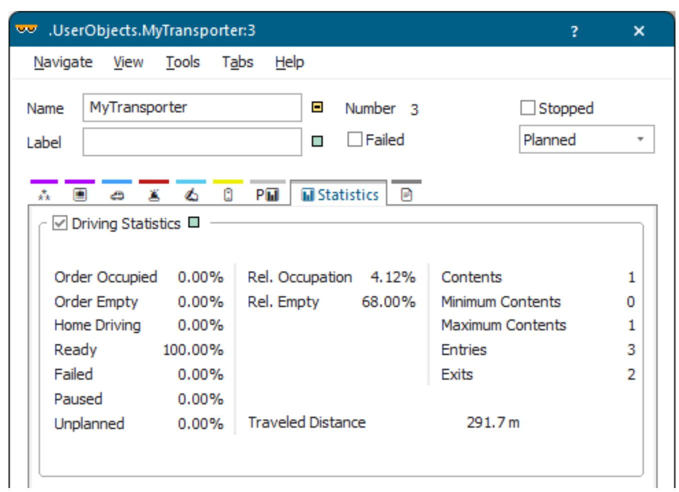
**Product Statistics - Statistics of the Individual MUs**

| Object | Production | | | | | | Transport | | | | | | Storage | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Working | Set-up | Waiting | Stopped | Failed | Paused | Working | Set-up | Waiting | Stopped | Failed | Paused | Working | Set-up | Waiting | Stopped | Failed |
| .MUs.Part:27595 | 2.59% | 0.00% | 1.98% | 0.00% | 15.89% | 79.31% | 0.23% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |

Mean Time Portions of an Individual MU of the Mean Life Span

As the *Container* and the *Transporter* can load and transport other parts, they also provide resource statistics, compare **View Statistics in the Dialogs of the Objects**.



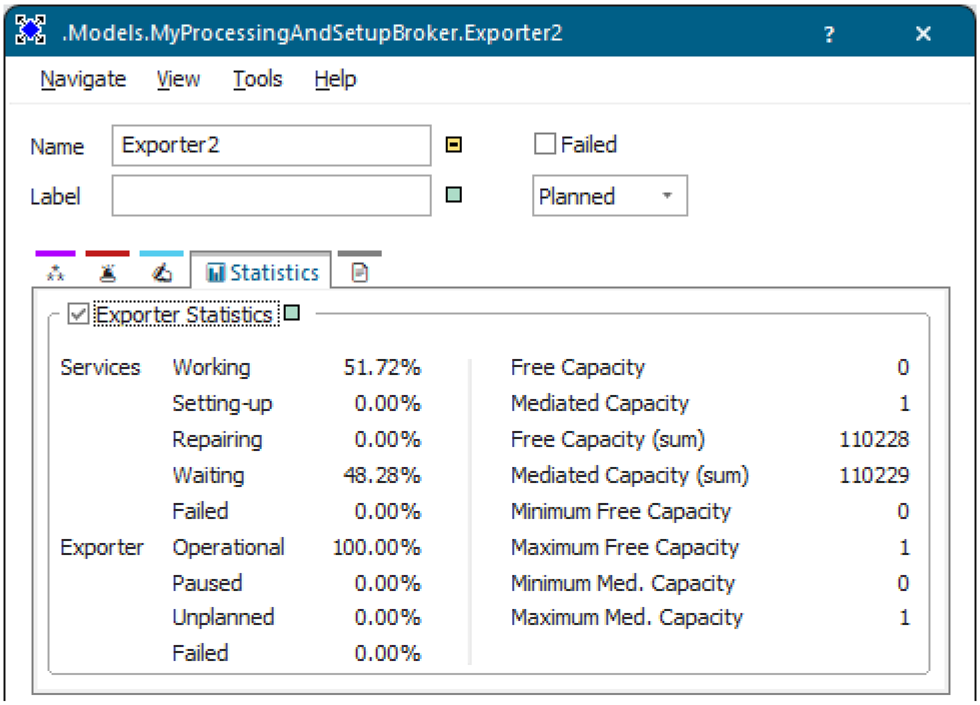




**Back to** **View Statistics in the Dialogs of the Objects**

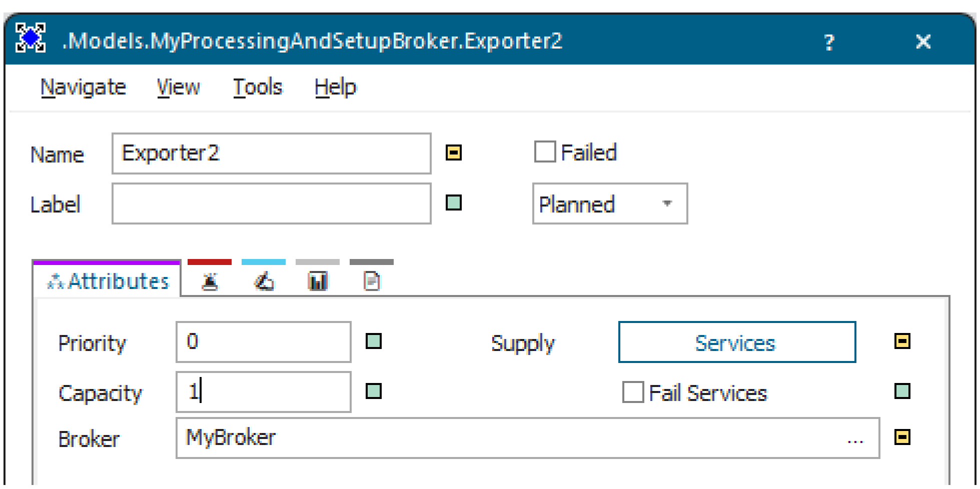
## Check Statistics of Exporter and Worker

For the *Exporter* and the *Worker* the tab **Statistics** shows the values for the *Services* and the *Exporter*.

For each of these blocks on the **Tab Statistics**, adding up the individual values yields 100 percent. You can query most of these values with the *read-only attributes* that are listed next to the name of the value in the description of the **Statistics Report**.




When you select the check box **Fail Services** on the tab **Attributes**, the *Exporter* collects the failure times of the *Services* and of the *Exporter*. When you deactivate it, the *Exporter* only collects the failure times of the *Exporter*.



The *Exporter* only collects *failed* times that take place during the processing and the set-up time, i.e., times that are located outside of the *paused* time and of the *unplanned* time.

*Waiting* times only accumulate during the times, when the *Exporter* is available, i.e., when these times are located outside of the *paused*, the *unplanned*, and of the *failed* times.

| Item | Description |
|---|---|
| **Services Setting-up** | Shows the portion of the set-up time for services of the statistics collection period of the *Exporter*, weighted with the capacity. |
| **Services Processing** | Shows the portion of the working time of the services of the statistics collection period of the *Exporter*, weighted with the capacity. |
| **Services Repairing** | Shows the portion of the statistics collection period, which the *services* spent for repairs, weighted with the capacity. |
| **Services Waiting** | Shows the portion of the statistics collection period, which the *services* spent waiting for an *importer* plus the time the *services* spent waiting for a *MU* at the *importer*, weighted with the capacity. |
| **Services Failed** | Shows the portion of the statistics collection period during which the *services* were failed. |
| **Exporter Operational** | Shows the portion of the statistics collection period during which the *Exporter* was working, weighted with the capacity. |
| **Exporter Paused** | Shows the portion of the statistics collection period during which the *Exporter* was paused. |
| **Exporter Unplanned** | Shows the portion of the statistics collection period during which the *Exporter* was unplanned. |
| **Exporter Failed** | Shows the portion of the statistics collection period during which the *Exporter* was failed. |
| **Free Capacity** | Shows the capacity that is available at the moment. |
| **Mediated Capacity** | Shows the capacity that is brokered at the moment. |
| **Free Capacity (sum)** | Shows the sum of the released capacity that the *Exporter* placed with any *importer*. |
| **Mediated Capacity (sum)** | Shows the sum of the brokered capacity. |
| **Minimum Free Capacity** | Shows the minimum capacity that is available. When the *Exporter* did not collect any valid value yet, the tab shows -1. |
| **Maximum free capacity** | Shows the maximum capacity that is available. When the *Exporter* did not collect any valid value yet, the tab shows -1. |

| Item | Description |
|---|---|
| **Minimum Mediated Capacity** | Shows the minimum brokered capacity. When the *Exporter* did not collect any valid value yet, the tab shows -1. |
| **Maximum Mediated Capacity** | Shows the maximum brokered capacity. When the *Exporter* did not collect any valid value yet, the tab shows -1. |

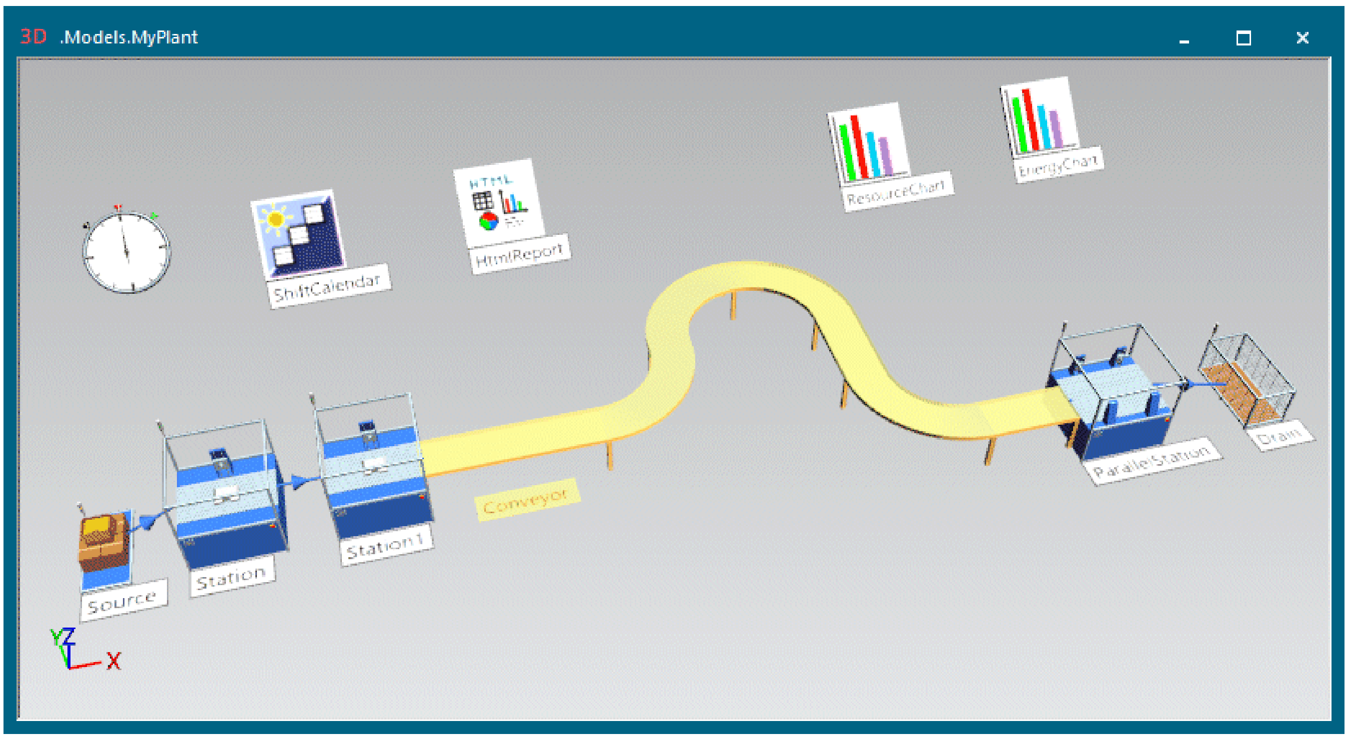
**Back to View Statistics in the Dialogs of the Objects**

## Viewing the Statistics Report

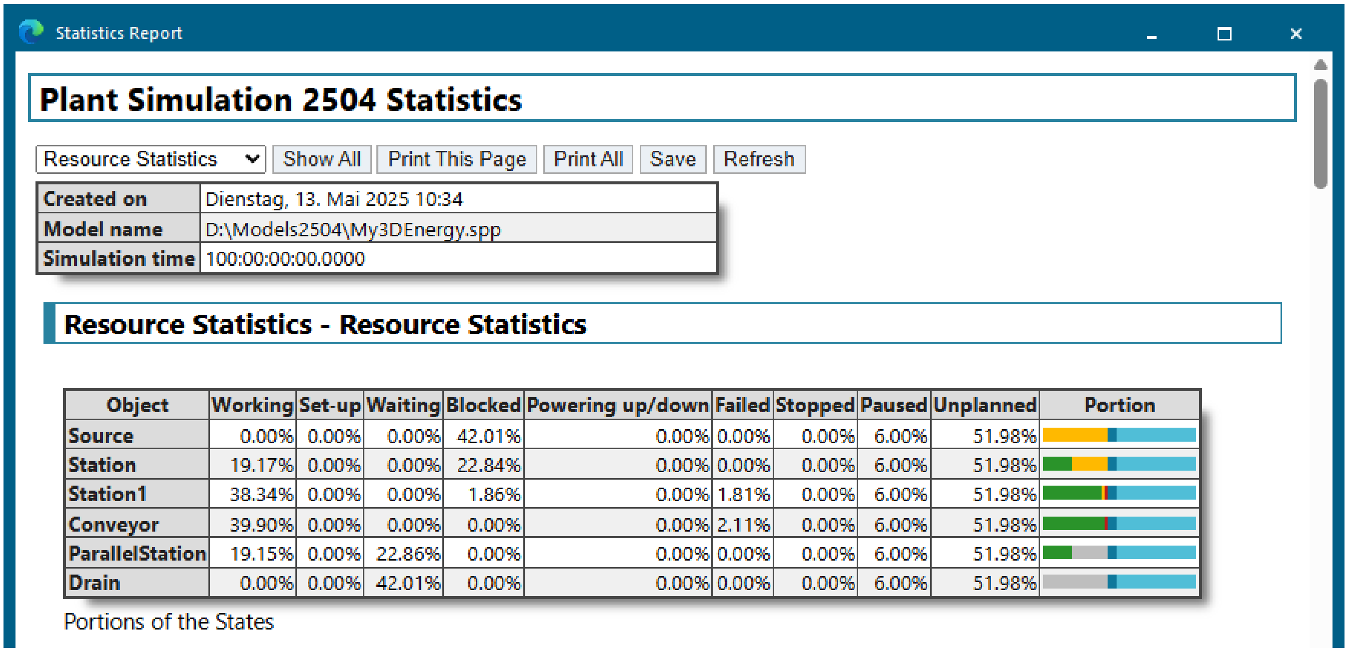
The most comfortable way of viewing statistics values of one or of several objects is to open the *Statistics Report*.

Below we demonstrate how to use it.

- Let us assume that we want to show the statistics of several objects in a single report. To do so, we select them (**Shift**+click, or drag a marquee over them) and press **F6**. Compare this example:




- This opens the **Statistics Report**, which looks like this:



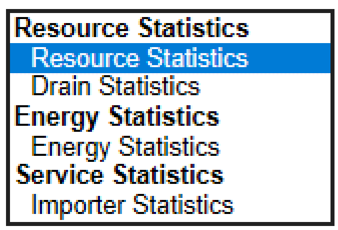

| | |
|---|---|
| Created on | Dienstag, 13. Mai 2025 10:34 |
| Model name | D:\Models2504\My3DEnergy.spp |
| Simulation time | 100:00:00:00.0000 |

Resource Statistics - Resource Statistics

| Object | Working | Set-up | Waiting | Blocked | Powering up/down | Failed | Stopped | Paused | Unplanned | Portion |
|---|---|---|---|---|---|---|---|---|---|---|
| Source | 0.00% | 0.00% | 0.00% | 42.01% | 0.00% | 0.00% | 0.00% | 6.00% | 51.98% | |
| Station | 19.17% | 0.00% | 0.00% | 22.84% | 0.00% | 0.00% | 0.00% | 6.00% | 51.98% | |
| Station1 | 38.34% | 0.00% | 0.00% | 1.86% | 0.00% | 1.81% | 0.00% | 6.00% | 51.98% | |
| Conveyor | 39.90% | 0.00% | 0.00% | 0.00% | 0.00% | 2.11% | 0.00% | 6.00% | 51.98% | |
| ParallelStation | 19.15% | 0.00% | 22.86% | 0.00% | 0.00% | 0.00% | 0.00% | 6.00% | 51.98% | |
| Drain | 0.00% | 0.00% | 42.01% | 0.00% | 0.00% | 0.00% | 0.00% | 6.00% | 51.98% | |

Portions of the States

- The report adds the statistics of the objects, which you selected, to the drop-down list in the top left corner of the display window. In our example these are the statistics values of the **Stationary resources Source**, **Station**, **Station1**, **Conveyor**, **ParallelStation**, and **Drain** as well as the **Importer Statistics**. To jump to any of the topics, which interest you, select that topic from the drop-down list:




  - One of the main topic shows the **resource statistics** of the material flow objects split up by the states these objects were in, i.e., the **working**, the **waiting**, the **set-up**, the **blocked**, the **powering up/down**, the **failed**, the **stopped**, the **paused**, and the **unplanned time**. These times add up to 100 percent. The set-up time and the empty time, which follow, may be part of the **working**, the **waiting**, or of the **blocked time**.

    The subtopics further down in the report show details about the **working**, the **waiting**, the **blocked**, the **failed**, the **stopped**, the **paused**, the **unplanned**, the **set-up**, and the **empty time** in the order in which they appear here.

    The bar on the right hand side shows the respective **portions** of the times.

| A bar with this color | Means that the object is | Looks like this English | Looks like this German |
|---|---|---|---|
| green | **Working** | Working | Arbeitend |
| brown | **Setting-Up** | Setting-up | Rüstend |

| A bar with this color | Means that the object is | Looks like this English | Looks like this German |
|---|---|---|---|
| gray | **Waiting** | Waiting | Wartend |
| yellow | **Blocked** | Blocked | Blockiert |
| purple | **Powering up/down** | PoweringUpDown | HochRunterfahrend |
| red | **Failed** | Failed | Gestört |
| pink | **Stopped** | Stopped | Angehalten |
| blue | **Paused** | Paused | Pausiert |
| light blue | **Unplanned** | Unplanned | Ungeplant |

- This is followed by material flow values of general interest.

| Object | Number of Entries | Number of Exits | Minimum Contents | Maximum Contents | Relative Empty | Relative Full | Relative Occupation without Interruptions | Relative Occupation with Interruptions |
|---|---|---|---|---|---|---|---|---|
| **Source** | 27598 | 27597 | 0 | 1 | 0.00% | - | 100.00% | 99.75% |
| **Station** | 27597 | 27596 | 0 | 1 | 0.00% | - | 100.00% | 99.75% |
| **Station1** | 27596 | 27595 | 0 | 1 | 0.00% | - | 100.00% | 99.75% |
| **Conveyor** | 27595 | 27595 | 0 | 2 | 83.54% | - | 0.40% | 0.37% |
| **ParallelStation** | 27595 | 27594 | 0 | 2 | 54.41% | - | 11.41% | 10.13% |

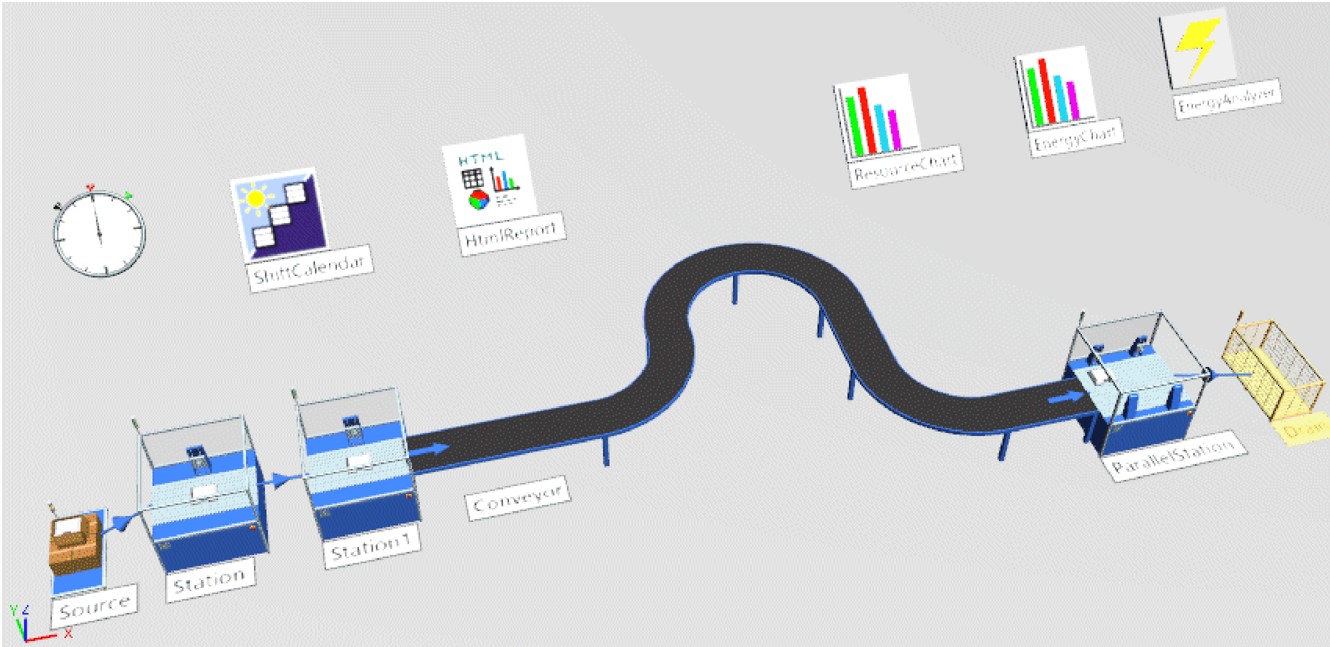
Material Flow Properties

- The subtopics show details of the **working**, the **waiting**, the **blocked**, the **failed**, the **paused**, the **unplanned**, the **set-up**, and the **empty time** in the order in which they appear in the first table. For details compare the **Statistics Report**.

| Object | Portion | Count | Sum | Mean Value | Standard Deviation |
|---|---|---|---|---|---|
| **Source** | 0.00% | 0 | 0.0000 | 0.0000 | 0.0000 |
| **Station** | 19.16% | 27597 | 19:03:57:00.0000 | 1:00.0000 | 0.0000 |
| **Station1** | 38.33% | 27596 | 38:07:51:15.8438 | 1:59.9984 | 0.2658 |
| **Conveyor** | 39.89% | 1 | 39:21:20:50.9568 | 39:21:20:50.9568 | 0.0000 |
| **ParallelStation** | 19.15% | 27567 | 19:03:31:56.3372 | 1:00.0107 | 0.3952 |

Working Time

- To add one or several objects in your model, which collect statistics to an open report, select it/them in the *Frame* and press **F6**. In our example we added the *Drain*.

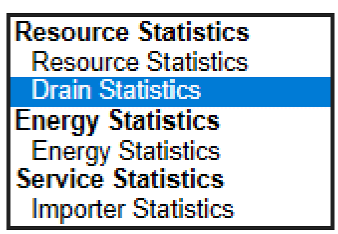


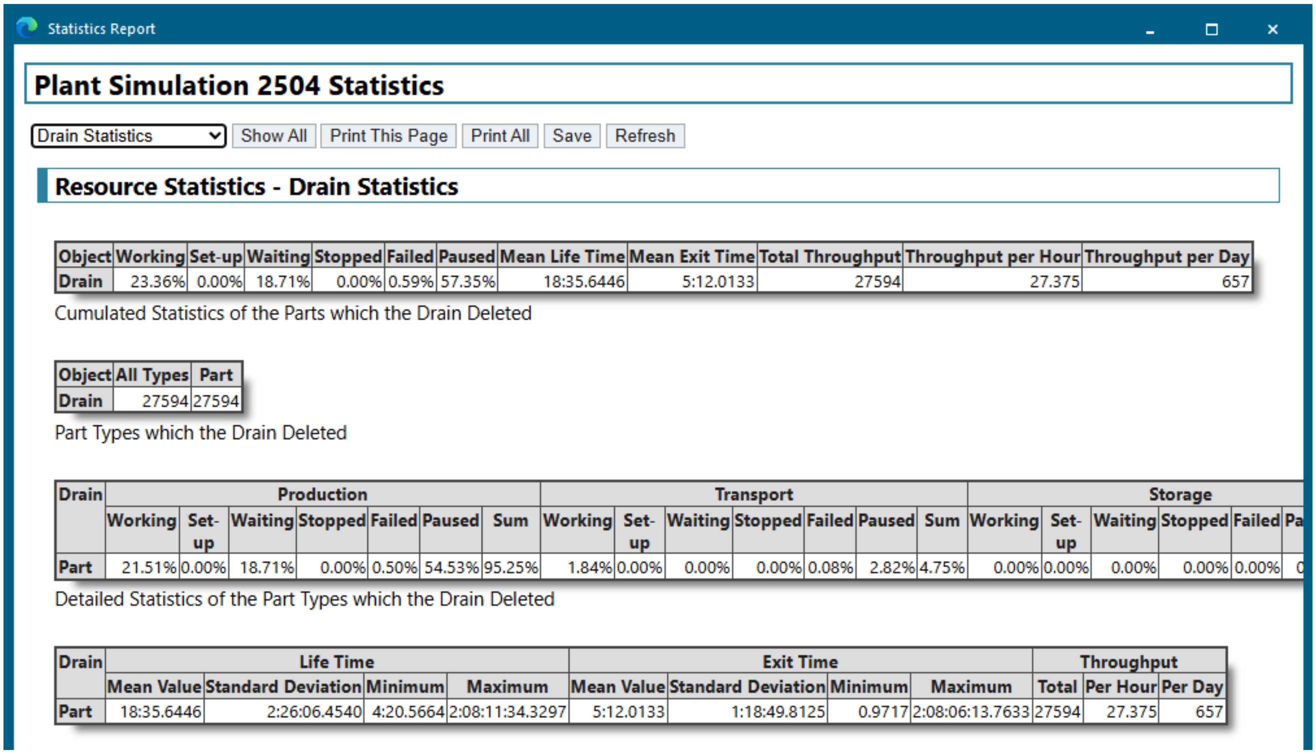

You will notice that the report added the object *Drain* to the list of stationary resources as well as to the drop-down list of available topics.

| Object | Working | Set-up | Waiting | Blocked | Powering up/down | Failed | Stopped | Paused | Unplanned | Portion |
|---|---|---|---|---|---|---|---|---|---|---|
| **Source** | 0.00% | 0.00% | 0.00% | 42.00% | 0.00% | 0.00% | 0.00% | 6.00% | 52.00% | |
| **Station** | 19.16% | 0.00% | 0.00% | 22.84% | 0.00% | 0.00% | 0.00% | 6.00% | 52.00% | |
| **Station1** | 38.33% | 0.00% | 0.00% | 1.86% | 0.00% | 1.81% | 0.00% | 6.00% | 52.00% | |
| **Conveyor** | 39.89% | 0.00% | 0.00% | 0.00% | 0.00% | 2.11% | 0.00% | 6.00% | 52.00% | |
| **ParallelStation** | 19.15% | 0.00% | 22.85% | 0.00% | 0.00% | 0.00% | 0.00% | 6.00% | 52.00% | |
| **Drain** | 0.00% | 0.00% | 42.00% | 0.00% | 0.00% | 0.00% | 0.00% | 6.00% | 52.00% | |

Portions of the States

- To jump to the statistics of this *Drain*, select **Drain Statistics** from the drop-down list. For details compare the **Statistics Report**.




Statistics Report

Plant Simulation 2504 Statistics

Drain Statistics | Show All | Print This Page | Print All | Save | Refresh

Resource Statistics - Drain Statistics

| Object | Working | Set-up | Waiting | Stopped | Failed | Paused | Mean Life Time | Mean Exit Time | Total Throughput | Throughput per Hour | Throughput per Day |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Drain | 23.36% | 0.00% | 18.71% | 0.00% | 0.59% | 57.35% | 18:35.6446 | 5:12.0133 | 27594 | 27.375 | 657 |

Cumulated Statistics of the Parts which the Drain Deleted

| Object | All Types | Part |
|---|---|---|
| Drain | 27594 | 27594 |

Part Types which the Drain Deleted

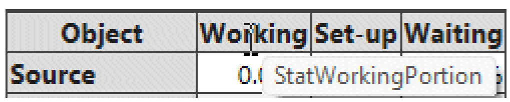
| Drain | Production | | | | | | | Transport | | | | | | | Storage | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Working | Set-up | Waiting | Stopped | Failed | Paused | Sum | Working | Set-up | Waiting | Stopped | Failed | Paused | Sum | Working | Set-up | Waiting | Stopped | Failed | Pa |
| Part | 21.51% | 0.00% | 18.71% | 0.00% | 0.50% | 54.53% | 95.25% | 1.84% | 0.00% | 0.00% | 0.00% | 0.08% | 2.82% | 4.75% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0 |

Detailed Statistics of the Part Types which the Drain Deleted

| Drain | Life Time | | | | Exit Time | | | | Throughput | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Mean Value | Standard Deviation | Minimum | Maximum | Mean Value | Standard Deviation | Minimum | Maximum | Total | Per Hour | Per Day |
| Part | 18:35.6446 | 2:26:06.4540 | 4:20.5664 | 2:08:11:34.3297 | 5:12.0133 | 1:18:49.8125 | 0.9717 | 2:08:06:13.7633 | 27594 | 27.375 | 657 |

- You can add any of the objects, which collect statistics, to the report. It then adds these objects to the drop-down list.

- To show a *Tooltip* with the name of the respective read-only attribute for this statistics value, drag the mouse over the column header.




- To save the report as an HTML file (*.htm or *.html) or as a text file (*.txt) click **Save**. Enter a name for the file and select the folder into which you want to save it.
- To update the report with the current values, click **Refresh**.
- To print the report, click **Print**.

**Compare** View Statistics in the Dialogs of the Objects

**Compare** Show Statistics in a Chart

**Compare** Show Statistics and Other Values in a Report

**Compare** Show Values During the Simulation with the Display

**Compare** Accessing Statistics with SimTalk

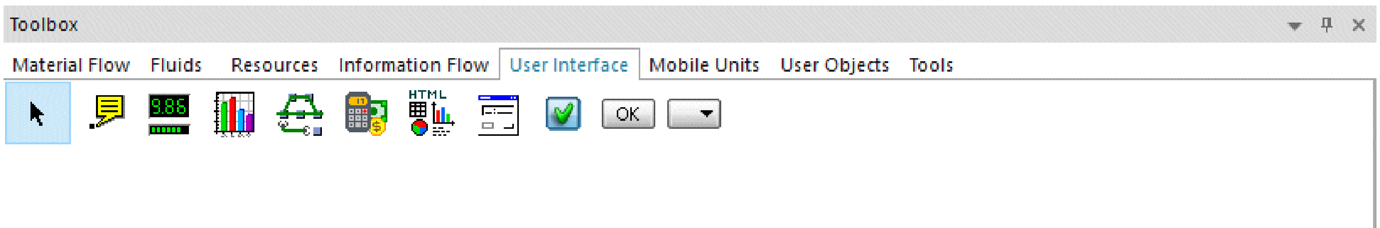
**Back to** View and Visualize Statistics

# Showing Statistics in a Chart

## Show Statistics in a Chart

To present the results of your simulation runs to colleagues and management, you will use the *Chart*.

You can insert the **Chart** into your simulation model from the folder **UserInterface** in the *Class Library* or from the toolbar **User Interface** in the *Toolbox*.




The *Chart* graphically displays the data sets that *Plant Simulation* recorded during a simulation run.

You can select to show:

- Data from a table, into which you saved the simulation results, for example.
- Data from input channels you define, that dynamically record the values of attributes of objects that your are interested in.

  To show statistics values of an object in the *Chart*, which you inserted into a *Frame*, you can use drag-and-drop:

| To | Drag a | To a |
|---|---|---|
| Show the statistics of a material flow object | material flow object, *Portioner*, *DePortioner* | *Chart* |
| Show the frequency distribution of the number of *MUs* | *PlaceBuffer*, *Buffer*, *Store* | *Chart* |
| Show the contents of a table | *DataTable*, *TimeSequence*, *MaterialsTable* | *Chart* |
| Show the value of the *Variable* you inserted into a *Frame* | *Variable* | *Chart* |

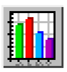
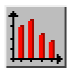
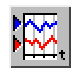
- To apply the settings you selected in the dialog and to show the records, click **Apply**. When you resize the *display* window, *Plant Simulation* also resizes the shown graph.
- To show the data the *Chart* collected in the *display* window, click **Show Chart**.
- To open the dialog of the *Chart*, double-click anywhere in the *display* window.
- To make a *Chart* of type **Histogram** or **Plotter** collect data during the simulation run, select **Collect data** in the dialog. To deactivate data collection, clear the check box.

- To delete all values that the *Chart* collected, select **Tools > Reset Values**.

The *Chart* lets you select settings in a number of ways. You can:

- **Select Settings in the Statistics Wizard**
- **Select Where the Data Comes From**
- **Select How the Chart Shows the Data**
- **Add Labels and a Legend**
- **Add Annotations**

When you select another **Category**, the *Chart* automatically shows the icon of this category in the *Frame* window.

| Chart | Histogram | Plotter |
|---|---|---|
|  |  |  |

**Compare Viewing the Statistics Report**

**Compare View Statistics in the Dialogs of the Objects**

**Compare Show Statistics and Other Values in a Report**

**Compare Show Values During the Simulation with the Display**

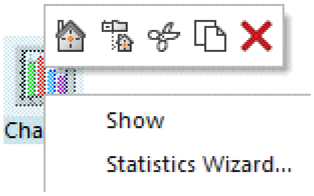
**Compare Accessing Statistics with SimTalk**

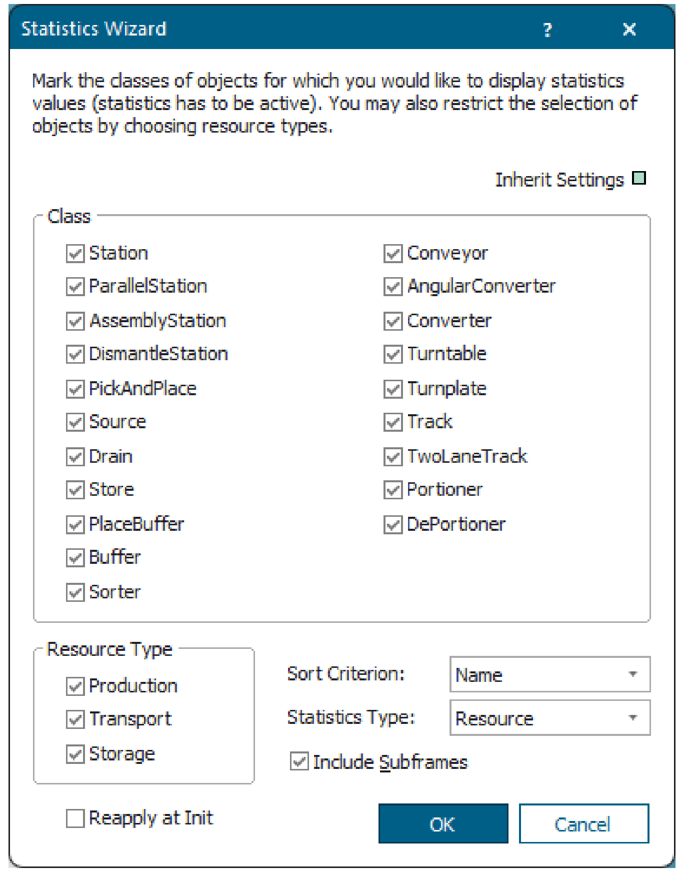
**Back to View and Visualize Statistics**

## Select Settings in the Statistics Wizard

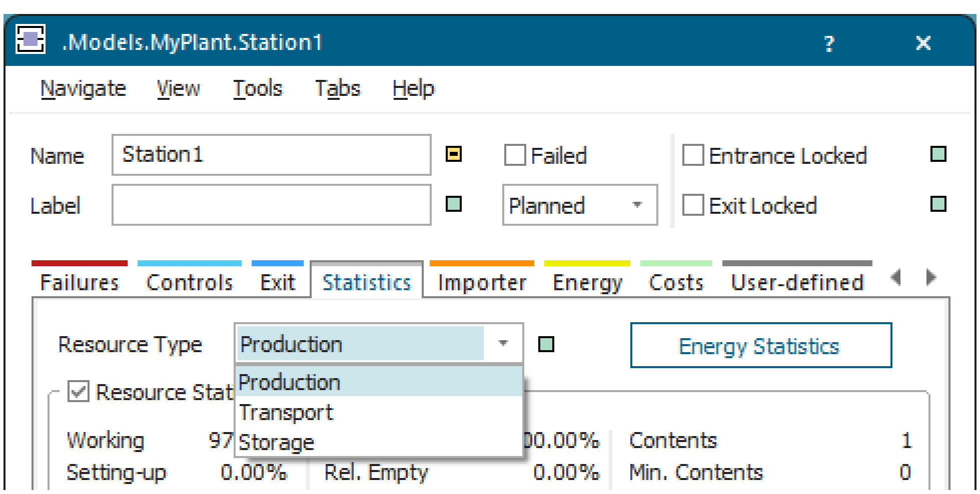
A *Chart*, which you insert into a *Frame*, provides the **Statistics Wizard**.

Right-click the *Chart* and select **Statistics Wizard** on the context menu.




- Select the class(es) of object(s) for which you would like to show statistical data in the *Chart* under the group box **Class**. The *Chart* only adds material flow objects to the display for which you selected the check box **Resource Statistics** on the **Tab Statistics**.
- Select the type(s) of resource(s) for which you would like to show statistics of the selected object(s) in the *Chart* from the drop-down list box **Resource type**: **Production**, **Transport** or **Storage**. Select the **Resource Type** on the tab **Statistics** of an object collecting statistical data. Use the check boxes in the **Statistics Wizard** to restrict which objects *Plant Simulation* shows.



- If you would like *Plant Simulation* to also show statistical data of objects within *Frames* that you inserted into the *Frame* in which the *Chart* is located, select the check box **Include Subframes**.
- Select the criterion which *Plant Simulation* uses when sorting statistical data from the drop-down list **Sort Criterion**: Either the **Name** of the object, or one of the states it is in: **Working**, **Setting-Up**, **Waiting**, **Blocked**, **Powering up/down**, **Failed**, **Stopped**, or **Paused**.

  To sort the stations by their **Name** in alphabetical order from A to Z, select **Sort Criterion > Name**. To sort the stations in the order that meets your needs, drag the stations onto the *Chart* in the order it is to show them. To change the order at a later point in time, edit the **table** of the **Input Channels** by cutting the respective columns of the objects and pasting them in the desired order.
- Finally, click **OK** to open the *display* window of the *Chart* with the settings you selected in the dialog.

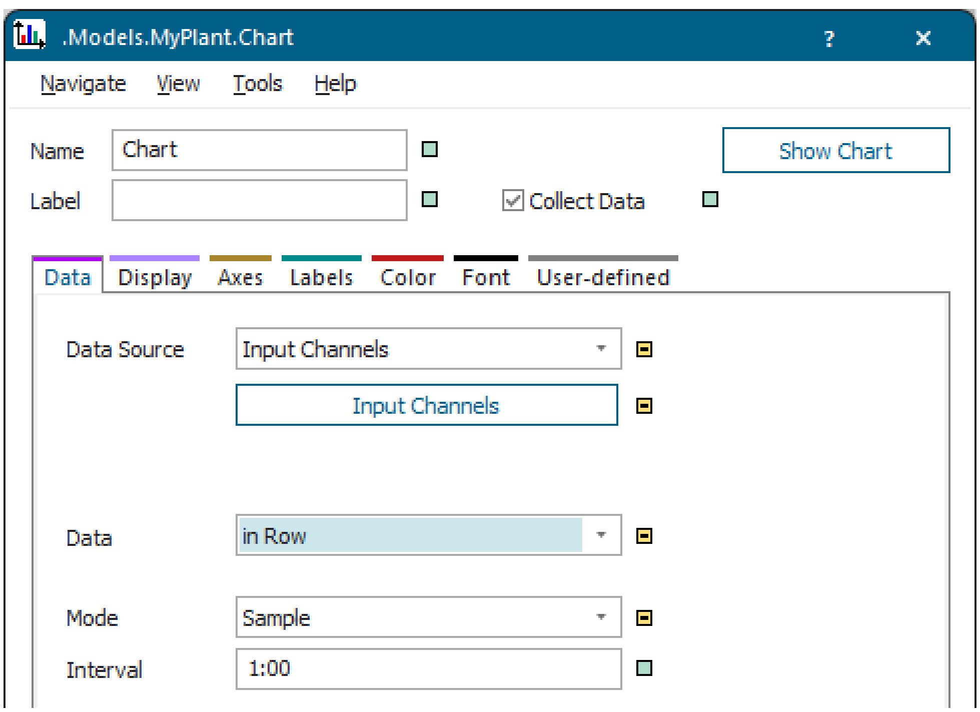
**Back to Show Statistics in a Chart**

### Select Where the Data Comes From

Before you start your simulation run, you can tell the *Chart* what to use as the source of the data it displays and in which mode it collects data.

Proceed as follows:

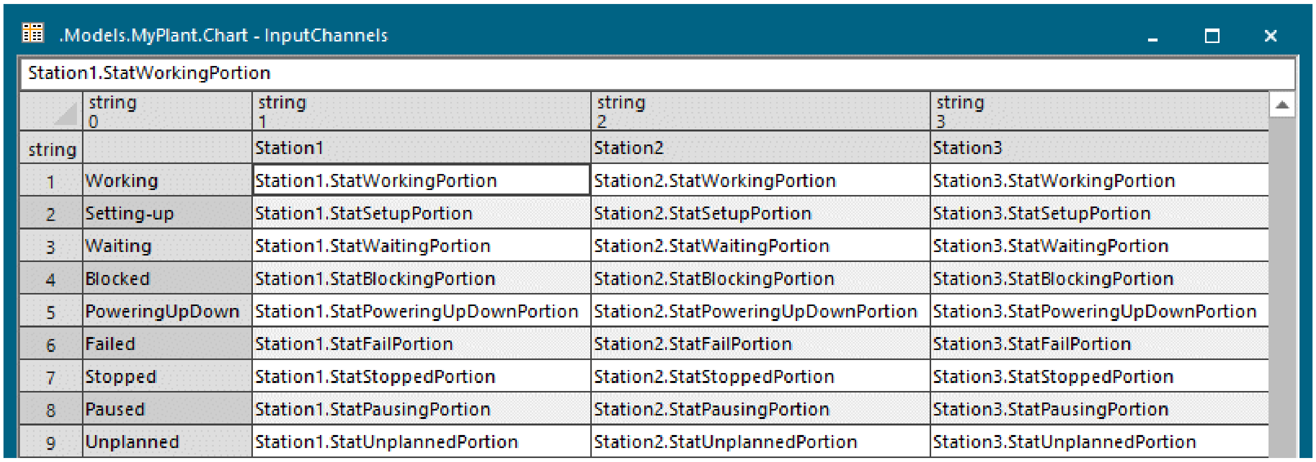
- Select the **Data Source**, i.e., whether the *Chart* takes the data you want to display from a set of input channels or from a table.
  - To show values of channels you define in a data table, select **Input Channels** from the drop-down list.



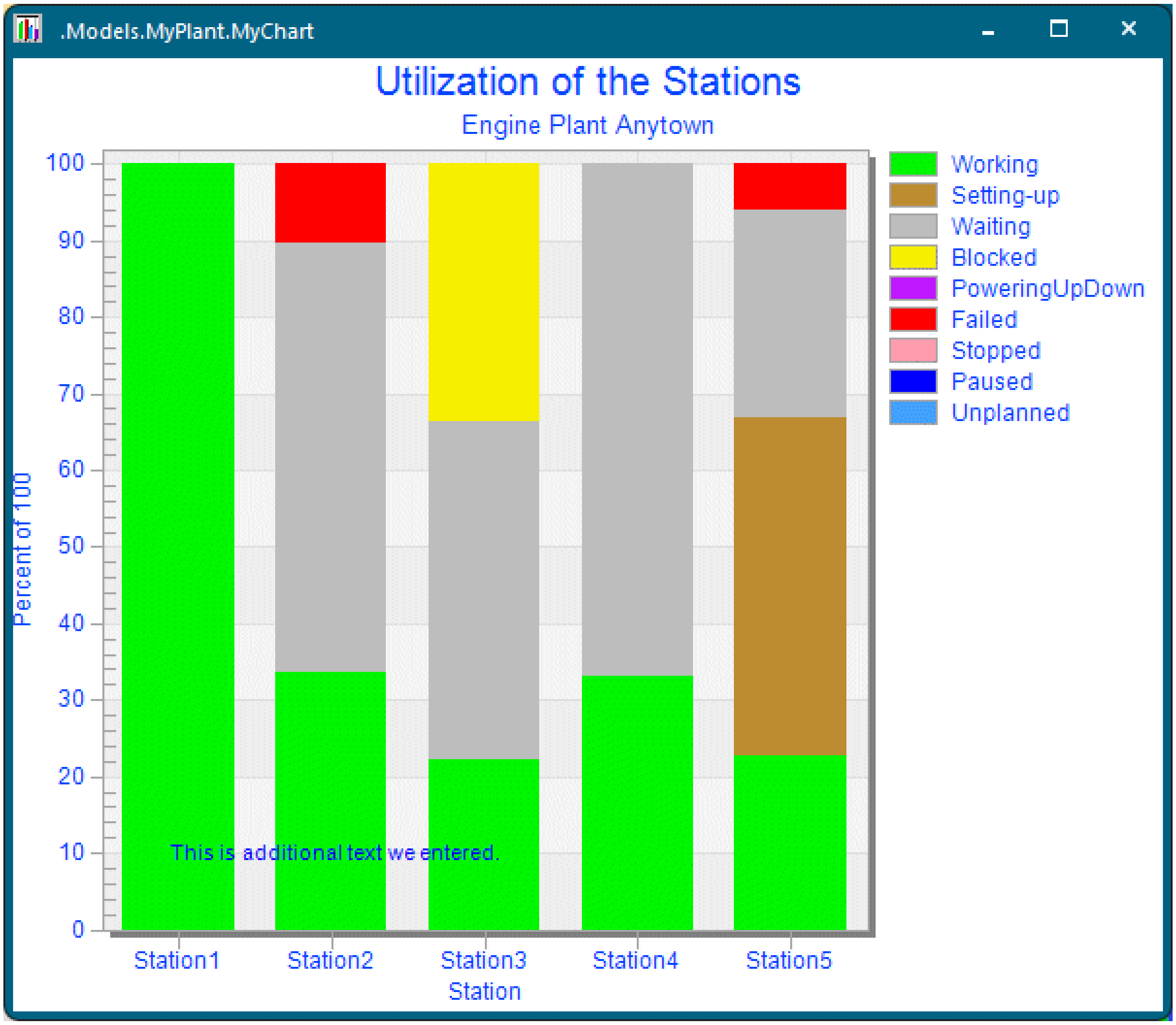
To open a table into which you can type the input channels, click **Data Table**. These channels are, as a rule, the paths to *attributes*, *read-only attributes*, or *methods* of objects, whose values you want to display. You can also type in complex expressions, such as the sum of two attributes, etc.

Type in any text you want to add as the **legend** to the *Chart* into the row index.

For the **Statistics type > Resource** we pre-configured the following entries in the *Statistics Wizard*:




| | string 0 | string 1 | string 2 | string 3 |
|---|---|---|---|---|
| string | | Station1 | Station2 | Station3 |
| 1 | Working | Station1.StatWorkingPortion | Station2.StatWorkingPortion | Station3.StatWorkingPortion |
| 2 | Setting-up | Station1.StatSetupPortion | Station2.StatSetupPortion | Station3.StatSetupPortion |
| 3 | Waiting | Station1.StatWaitingPortion | Station2.StatWaitingPortion | Station3.StatWaitingPortion |
| 4 | Blocked | Station1.StatBlockingPortion | Station2.StatBlockingPortion | Station3.StatBlockingPortion |
| 5 | PoweringUpDown | Station1.StatPoweringUpDownPortion | Station2.StatPoweringUpDownPortion | Station3.StatPoweringUpDownPortion |
| 6 | Failed | Station1.StatFailPortion | Station2.StatFailPortion | Station3.StatFailPortion |
| 7 | Stopped | Station1.StatStoppedPortion | Station2.StatStoppedPortion | Station3.StatStoppedPortion |
| 8 | Paused | Station1.StatPausingPortion | Station2.StatPausingPortion | Station3.StatPausingPortion |
| 9 | Unplanned | Station1.StatUnplannedPortion | Station2.StatUnplannedPortion | Station3.StatUnplannedPortion |

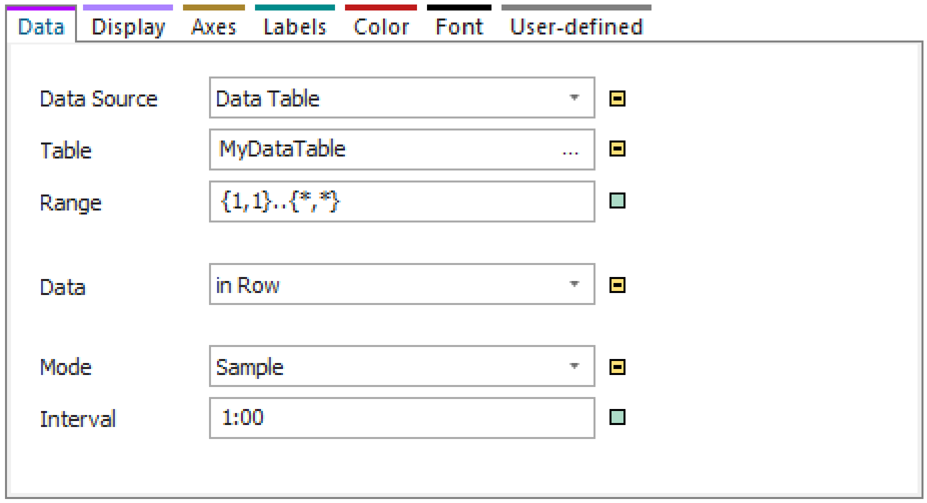



- To show the contents of a table as a diagram, select **Data Table**. This might, for example, be a table into which you saved the results of a simulation run.

  Type in the name and the path of the data table containing the data, or click the button … and select the table in the dialog **Select Object**.

  Type in the **Range** of the table cells you want to show.

  - To show all cells in all columns and rows, type in `{*,*}..{*,*}`.
  - To show the first four cells of column one, type in `{1,1}..{1,4}`.
  - To show all cells in column one, type in `{1,1}..{1,*}`.




You can also type in negative numbers:

- To show the one but last column of the table, type in `{-2,*}..{-2,*}`.
- To show the one but last and the last column, type in `{-2,*}..{*,*}`.

- Select which mode the *Chart* uses to update the data it shows.
  - **Sample** mode updates the *Chart* periodically and takes the next sample after the time span which you type into the text box **Interval**.
  - **Watch** mode updates the *Chart*, whenever the value of a **watchable** *read-only attribute*, for example *NumMU*, *NumMUParts*, *StatMaxNumMU*, *StatNumIn*, *StatNumOut*, etc. or of a watchable *attribute* changes, for example *Pause*, *Speed*, *Unplanned*, etc.

    The column **Watchable** in the window **Show Attributes and Methods** of the individual objects shows all *attributes*, *read-only attribute*, and *methods* that *Plant Simulation* can watch.
  - **Plot** mode updates the *Chart*, whenever a simulation event takes place.

**Back to Show Statistics in a Chart**

## Selecting How the Chart Shows the Data

### Select How the Chart Shows the Data

After you have selected where the *Chart* takes the data from that it displays, you will select the settings for displaying this data.

You can:

- Show a number of values or several sets of a number of values, see **Show Values in a Chart**.
- Show a number of values or several sets of a number of values in a three-dimensional *Chart*, see **Show Values in a 3D Chart**.
- Show the frequency distribution for one or several input channels, see **Show Values in a Histogram**.

- Show the frequency distribution for one or several input channels in a three-dimensional *Chart*, see **Show Values in a 3D Histogram**.
- Show the progression of one or more values over time, see **Show Values as the Chart Plots Them**.

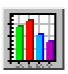
> **Note**
>
> When you select another **Category**, the *Chart* automatically shows the icon of this category in the *Frame* into which you inserted it.

- Show a number of x-y pairs of values or several sets of x-y pairs of values, see **Show Values as an XY Graph**.

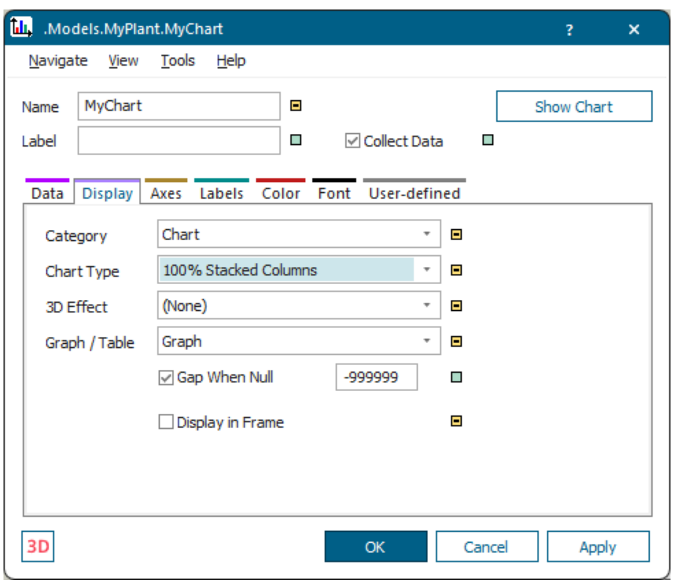
**Back to Show Statistics in a Chart**

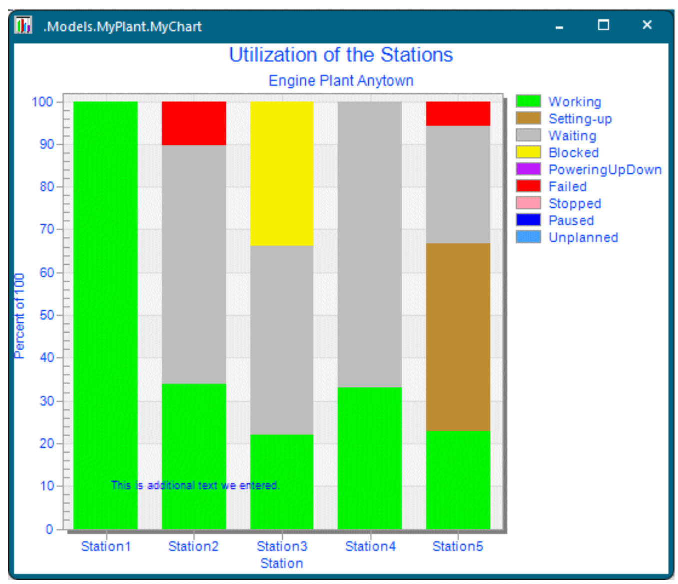
## Show Values in a Chart

To show a number of values or several sets of a number of values as a chart, click the tab **Display** and select **Chart** from the drop-down list **Category**.



To make the *Chart* collect data during the simulation run, select **Collect Data** in the dialog. To deactivate data collection, clear the check box.



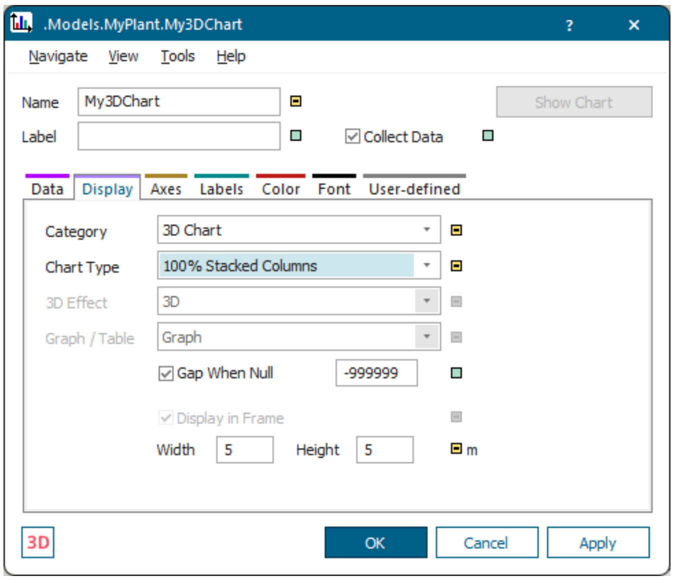



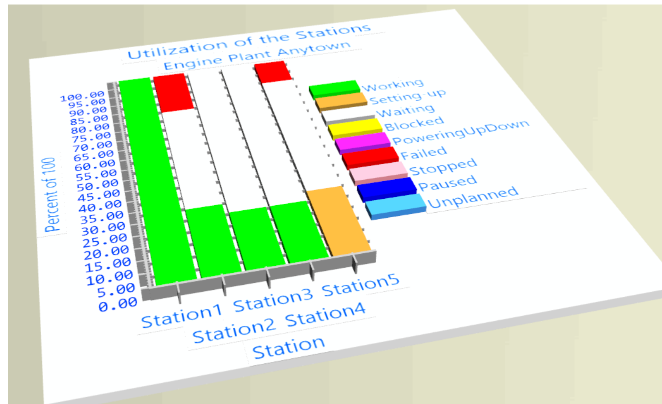

**Back to Select How the Chart Shows the Data**

## Show Values in a 3D Chart

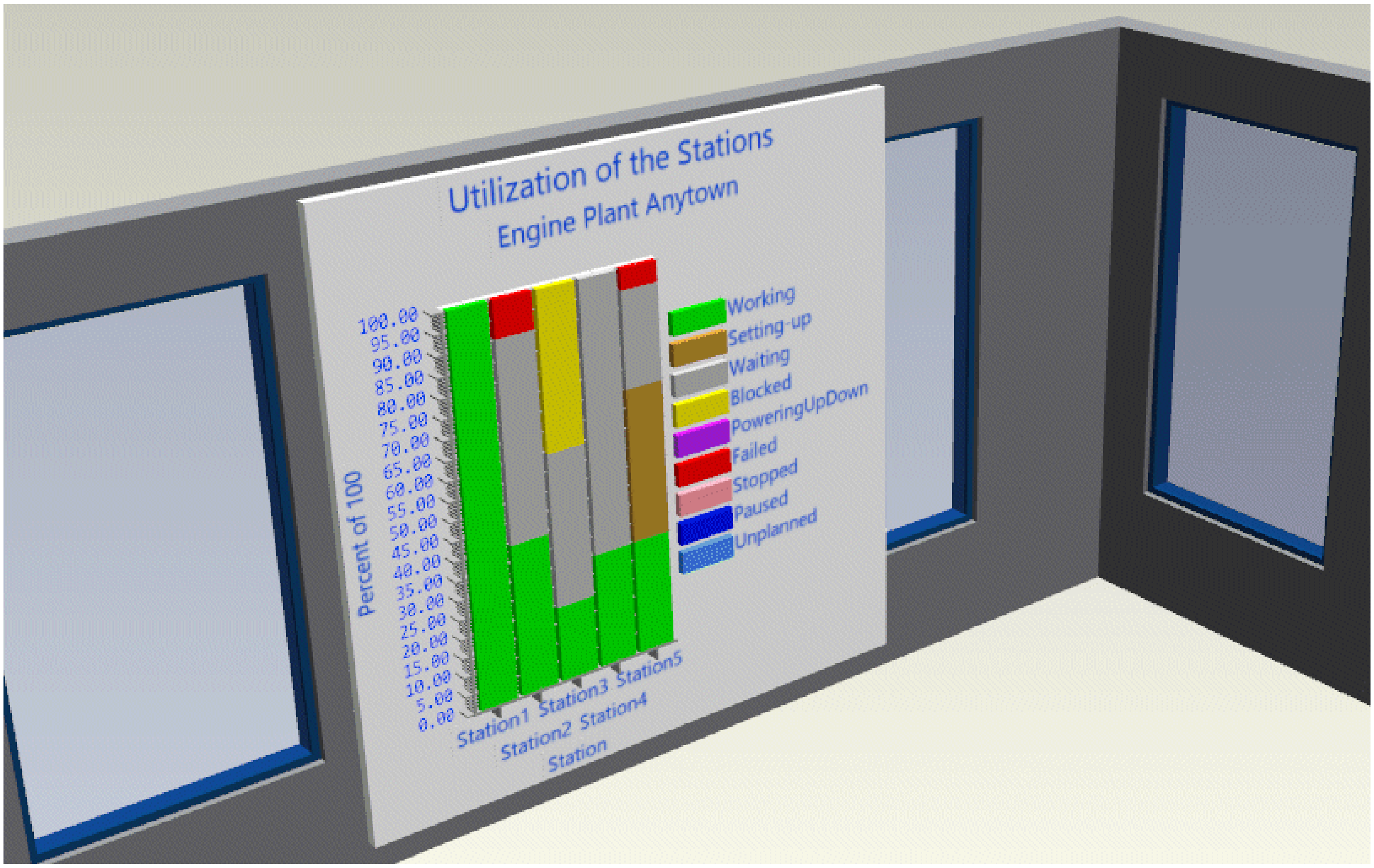
You can show a number of values or several sets of a number of values as a 3D chart. To do so, click the tab **Display** and select **3D Chart** from the drop-down list **Category**.

To make the *Chart* collect data during the simulation run, select **Collect Data** in the dialog. To deactivate data collection, clear the check box.



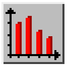

You can, for example hang the **3D Chart** onto the wall of your factory, thus vastly improving visibility and showing the operators the utilization of the machines at a glance.



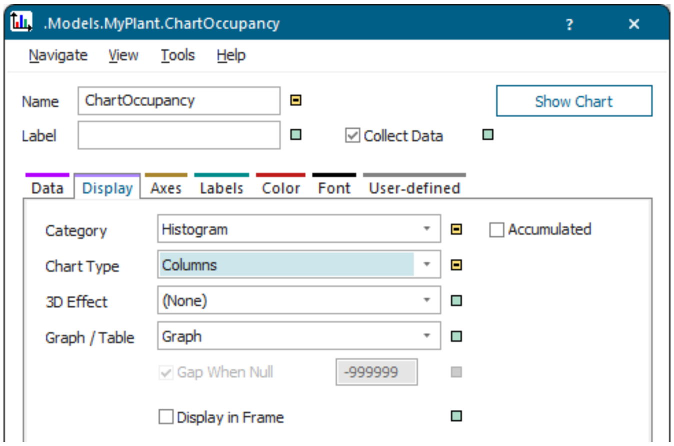

**Back to** **Select How the Chart Shows the Data**

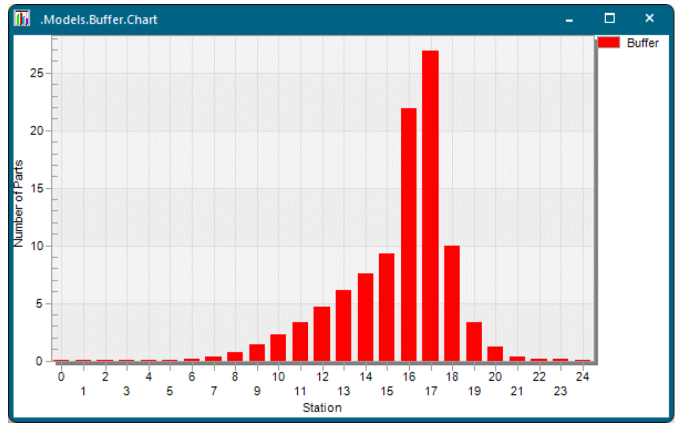
## Show Values in a Histogram

You can show the frequency distribution for one or several input channels as a histogram. To do so, click the tab **Display** and select **Histogram** from the drop-down list **Category**. A histogram graphically summarizes the distribution of a data set.



To make the *Chart* collect data during the simulation run, select **Collect Data** in the dialog. To deactivate data collection, clear the check box.



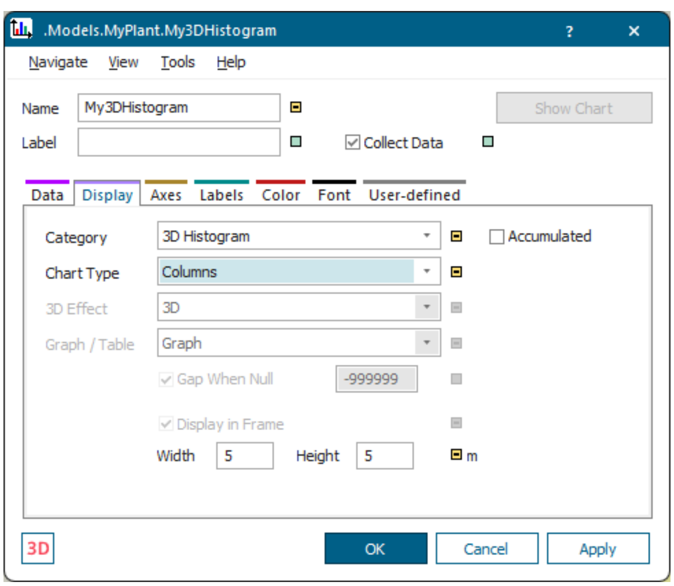



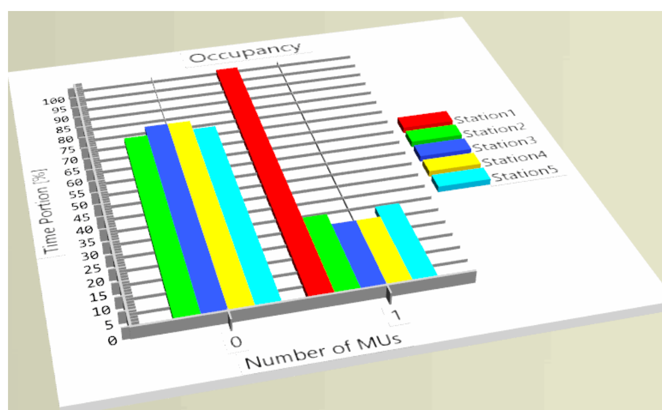

Back to Select How the Chart Shows the Data

## Show Values in a 3D Histogram

To show the frequency distribution for one or several input channels as a 3D histogram, click the tab **Display** and select **3D Histogram** from the drop-down list **Category**. A histogram graphically summarizes the distribution of a data set.

To make the *Chart* collect data during the simulation run, select **Collect Data** in the dialog. To deactivate data collection, clear the check box.



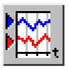




See also 3D Histogram

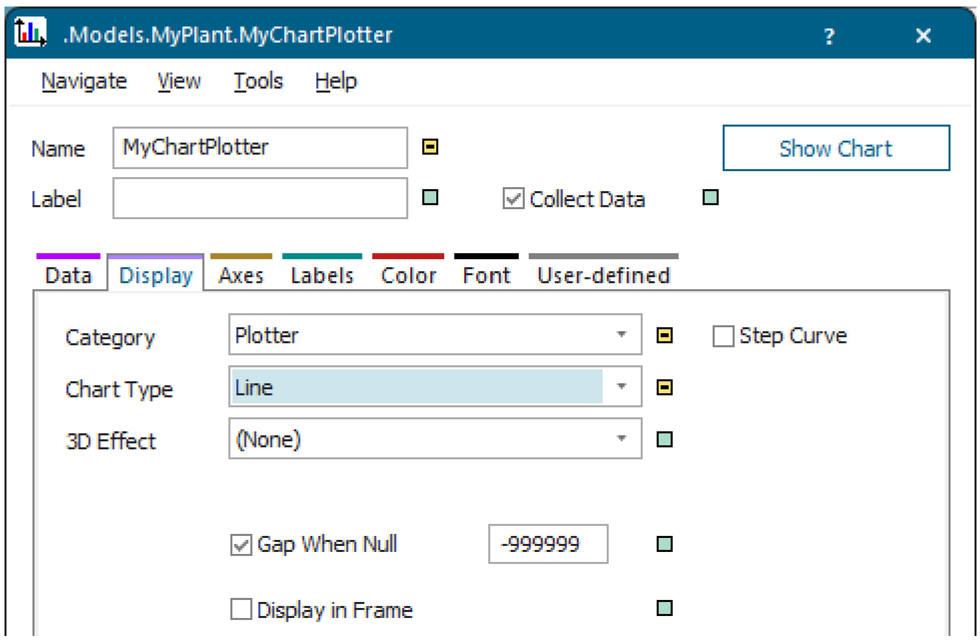
Back to Select How the Chart Shows the Data

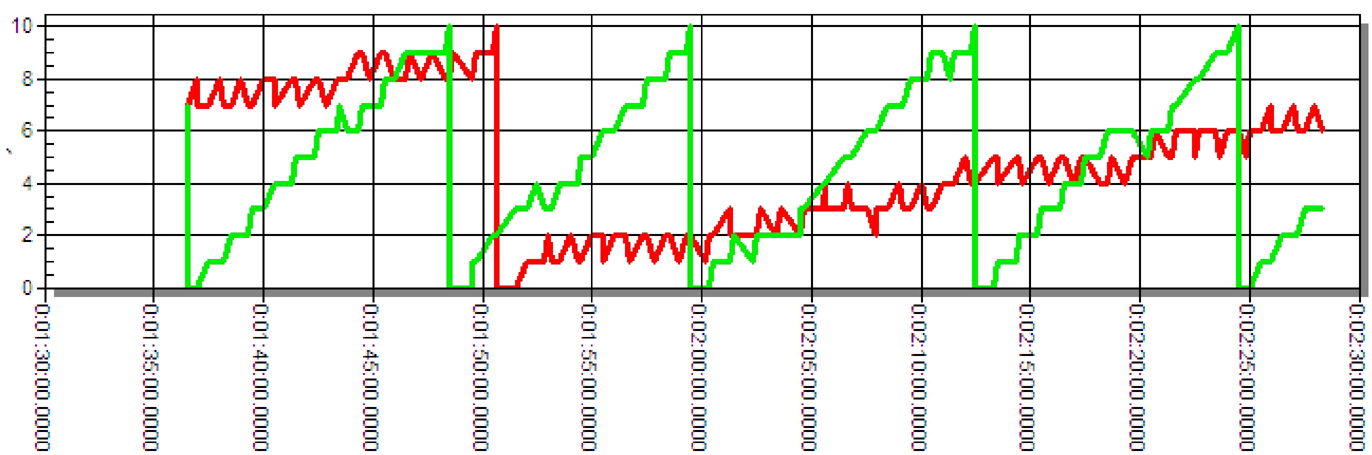
## Show Values as the Chart Plots Them

You can show the progression of one or more values over time. To do so, click the tab **Display** and select **Plotter** from the drop-down list **Category**.



To make the *Chart* collect data during the simulation run, select **Collect Data** in the dialog. To deactivate data collection, clear the check box.



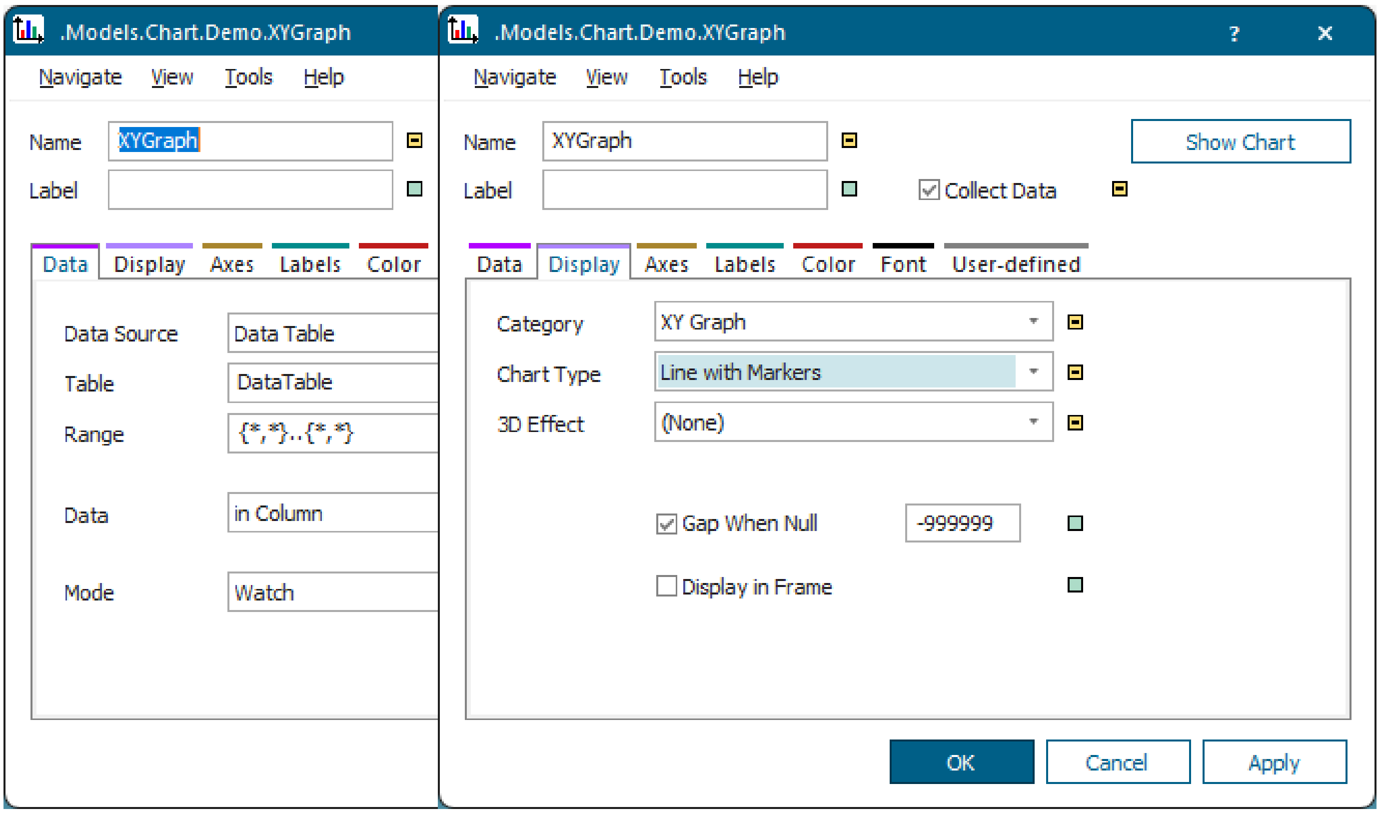




**Back to** **Select How the Chart Shows the Data**

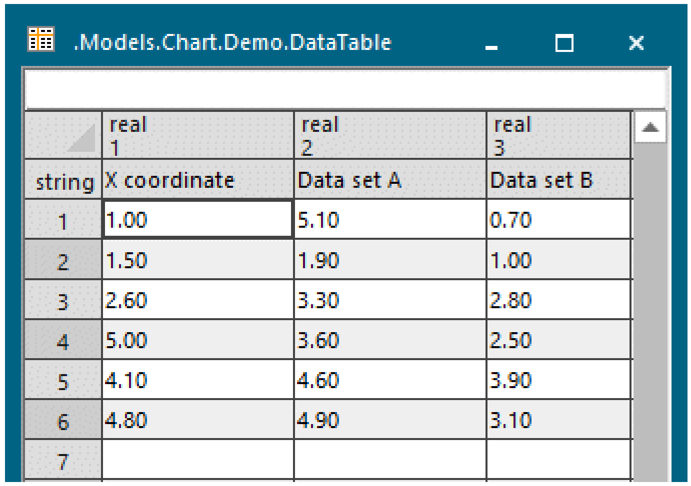
## Show Values as an XY Graph

You can show a number of x-y pairs of values or several sets of x-y pairs of values. To do so, click the tab **Display** and select **XY Graph** from the drop-down list **Category**.

> **Note**
> You can only show values as an XY graph, when you select **Data > Data Source > Data Table**.



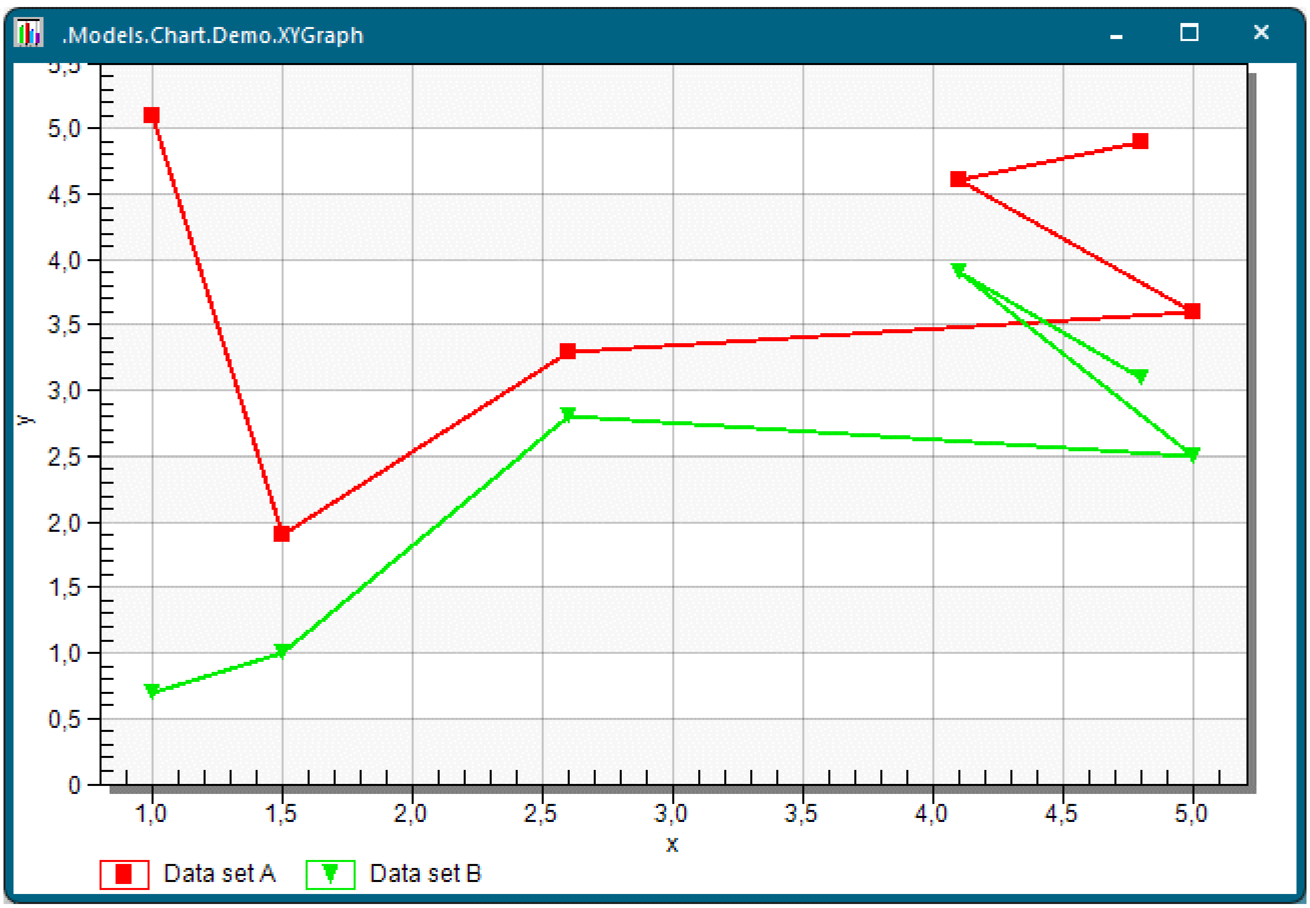

Our table containing the data that we want to show, looks like this:



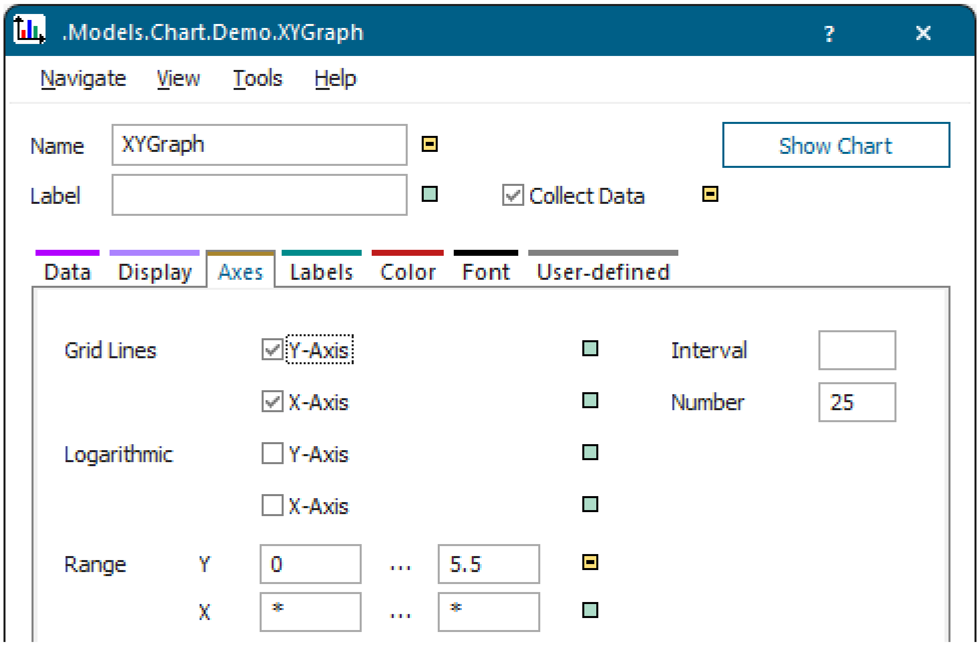
.Models.Chart.Demo.DataTable

| | real 1 | real 2 | real 3 |
|---|---|---|---|
| string | X coordinate | Data set A | Data set B |
| 1 | 1.00 | 5.10 | 0.70 |
| 2 | 1.50 | 1.90 | 1.00 |
| 3 | 2.60 | 3.30 | 2.80 |
| 4 | 5.00 | 3.60 | 2.50 |
| 5 | 4.10 | 4.60 | 3.90 |
| 6 | 4.80 | 4.90 | 3.10 |
| 7 | | | |

- When we select **Data in Column**, the XY graph looks as shown below. The XY graph shows the content of the row index as the **Legend**.

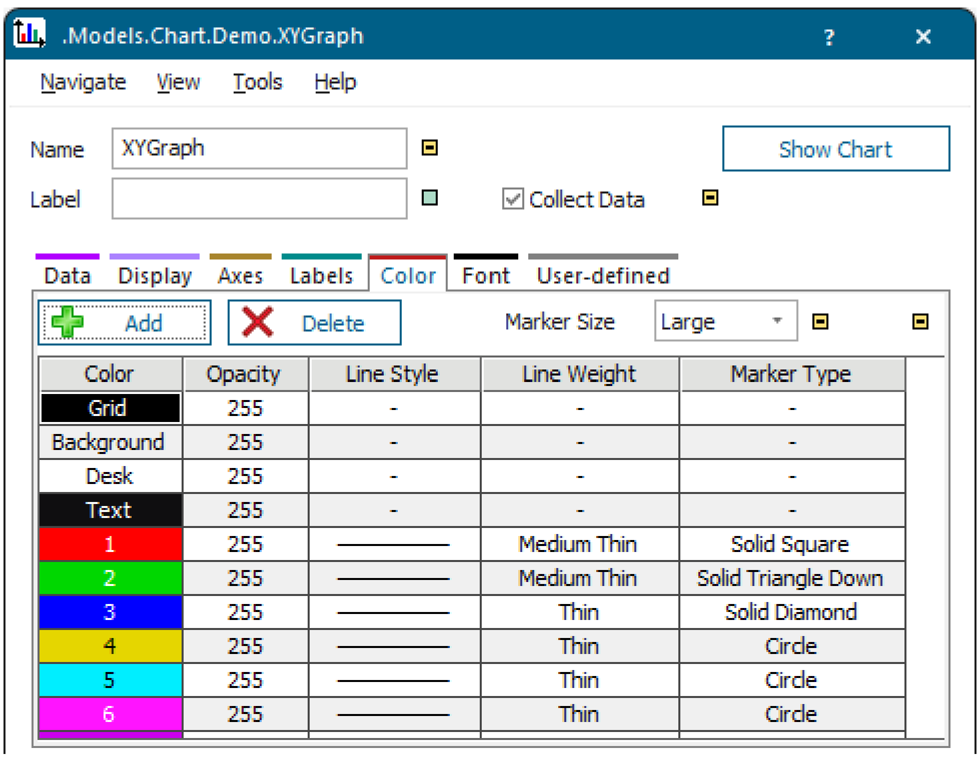



We selected the **Range** from 0 to 5.5, which is shown on the y axis, on the tab **Axes**.




On the tab **Labels** we selected to show the **Legend** below the chart.

We set the **Color**, the **Line Weight**, and the **Marker Type** of the **Line With Markers** on the tab **Color**.

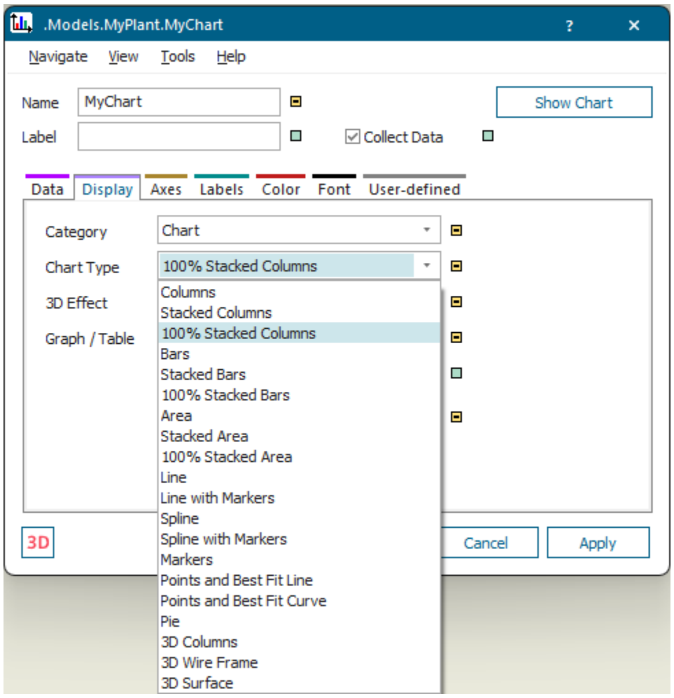


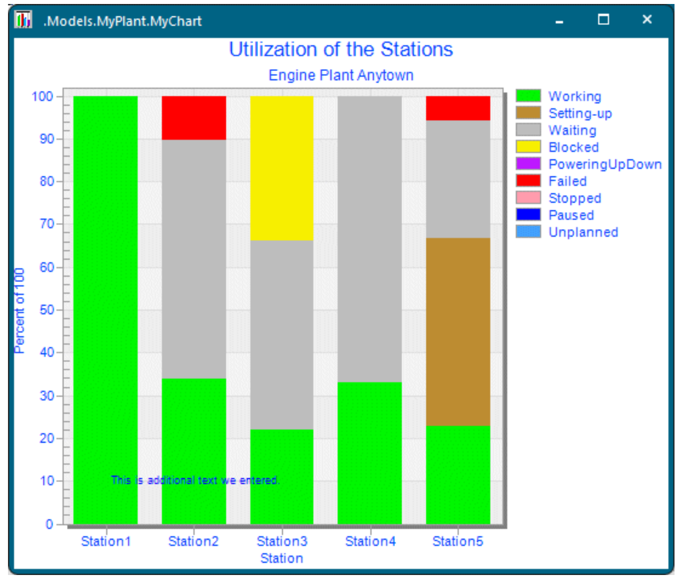

- If we select **Data in Row**, we have to transpose the respective data in the table.

**Back to** Select How the Chart Shows the Data

## Select the Chart Type

Select the type of *Chart* for displaying the recorded data. A record consists of the entries of a row or a column, depending on the setting you selected under **Display > Data**.




Depending on the **Category** you selected (**Chart**, **Histogram**, or **Plotter**), the *Chart* provides some or all of the following chart types:

| Category | Description |
|---|---|
| **Columns** | Shows each record as a column. When displaying several records, the *Chart* places the columns next to each other. Use this to compare values across categories. |
| **Stacked Columns** | Shows each record as a column. When displaying several records, the *Chart* places the columns of the individual records on top of each other. It stacks positive values upward, negative values downward. |
| **100% Stacked Columns** | Shows each record as a column, stacked one on top of the other, showing the respective percentage of the entire set of 100 percent. |
| **Bars** | Shows each record as bars positioned horizontally on top of each other. When displaying several records, the *Chart* places the bars of the individual records next to each other. Use this to compare values across categories. |
| **Stacked Bars** | Shows each record as a set of bars stacked horizontally next to each other. Positive values are stacked to the right, negative values are stacked to the left. |

| Category | Description |
|---|---|
| **100% Stacked Bars** | Shows each record as a set of bars, one stacked horizontally next to each other, showing its respective percentage of the entire set of 100 percent. |
| **Area** | Shows each record as areas. When displaying several records, the *Chart* stacks the areas from the individual records one behind the other. Use this to show the trend of the contribution of each value over time or categories. |
| **Stacked Area** | Shows each record as areas. When displaying several records, the *Chart* stacks the areas from the individual records one on top of the other. |
| **100% Stacked Area** | Shows each record as a set of areas, one stacked on top of the other, showing the respective percentage of the entire set of 100 percent. Displays the trend of the percentage each value contributes over time or categories. |
| **Line** | Shows each record as a line. |
| **Line with Markers** | Shows each record as a line with markers. |
| **Spline** | Shows each record as a curve. |
| **Spline with Markers** | Shows each record as a curve with markers. |
| **Markers** | Shows each record as a set of different markers. |
| **Points and Best Fit Conveyor** | Shows each record as a set of data points, and fits the best straight line to them. The line is calculated from least-squares-approximations of the data set. |
| **Points and Best Fit Curve** | Shows each record as a set of data points, and fits the best curved line to them. The curve is calculated from least-squares-approximations of the data set. |
| **Pie** | Shows each record as a slice of a pie, depicting its percentage of the whole. Enter a negative prefix, so that the *Chart* offsets this slice from the rest of the pie. Use the scrollbar to scroll from record to record. Use this to show the contribution of each value to a total. |
| **XY Points** | Shows x-y pairs as data points. The first record contains the x-coordinates. The *Chart* interprets all additional records as the associated y-coordinates. Use this to compare pairs of values. |
| **Line** | Shows x-y pairs as data points and connects them with a line. The first record contains the x-coordinates. The *Chart* interprets all additional records as the associated y-coordinates. |
| **Sticks** | Shows a vertical bar for each x-y pair, starting from the zero line. The first record contains the x-coordinates. The *Chart* interprets all additional records as the associated y-coordinates. |
| **Area** | Shows the connected points belonging to the x-y pairs and fills the areas thus created. The first record contains the x-coordinates. The *Chart* interprets all additional records as the associated y-coordinates. |
| **3D Columns** | Shows each record as columns stacked in three-dimensional space. To turn the chart left or right you can enter an angle of **Rotation** or you can also drag the |

| Category | Description |
|---|---|
| | horizontal slider in the display window. To turn the chart up or down, you can enter the **Height** or you can drag the vertical slider. |
| **3D Wire Frame** | Shows each record as wire-frames stacked in three-dimensional space. To turn the chart left or right you can enter an angle of **Rotation** or you can also drag the horizontal slider in the display window. To turn the chart up or down, you can enter and the **Height** or you can drag the vertical slider. |
| **3D Surface** | Shows each record as solid areas stacked in three-dimensional space. To turn the chart left or right you can enter an angle of **Rotation** or you can also drag the horizontal slider in the display window. To turn the chart up or down, you can enter the **Height** or you can drag the vertical slider. |

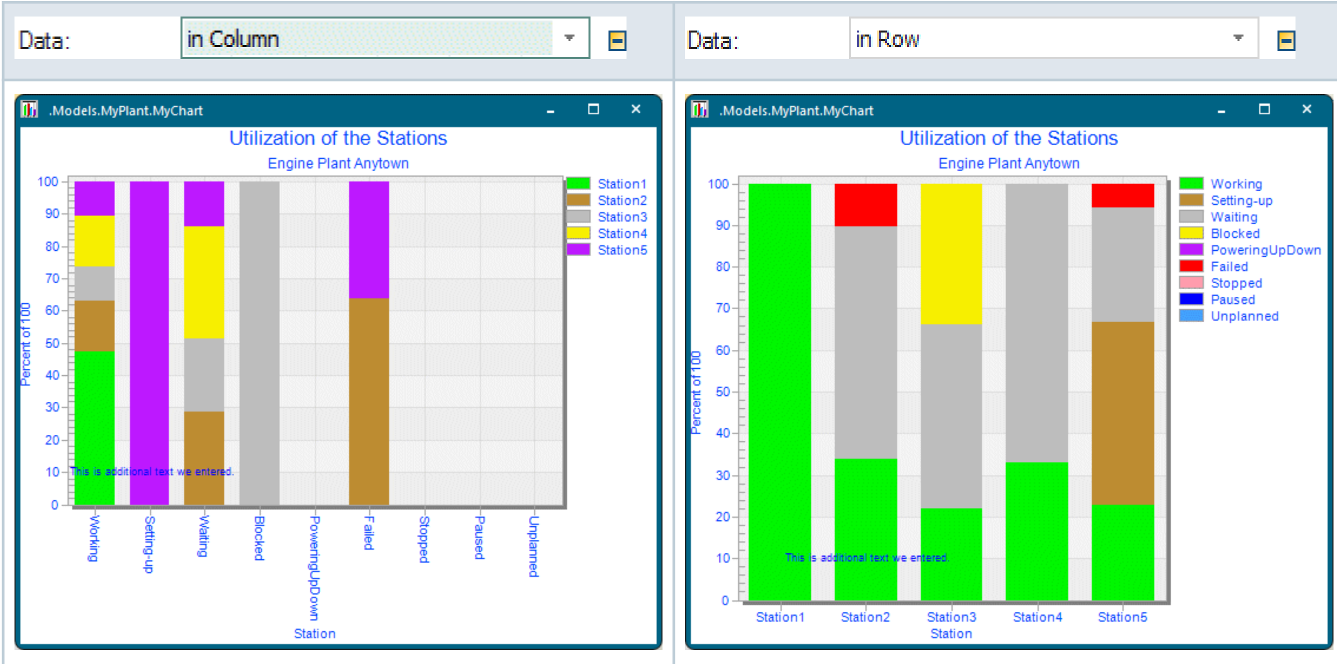
**Back to Select How the Chart Shows the Data**

## Select Additional Display Options

You can select additional display options for the *Chart*.

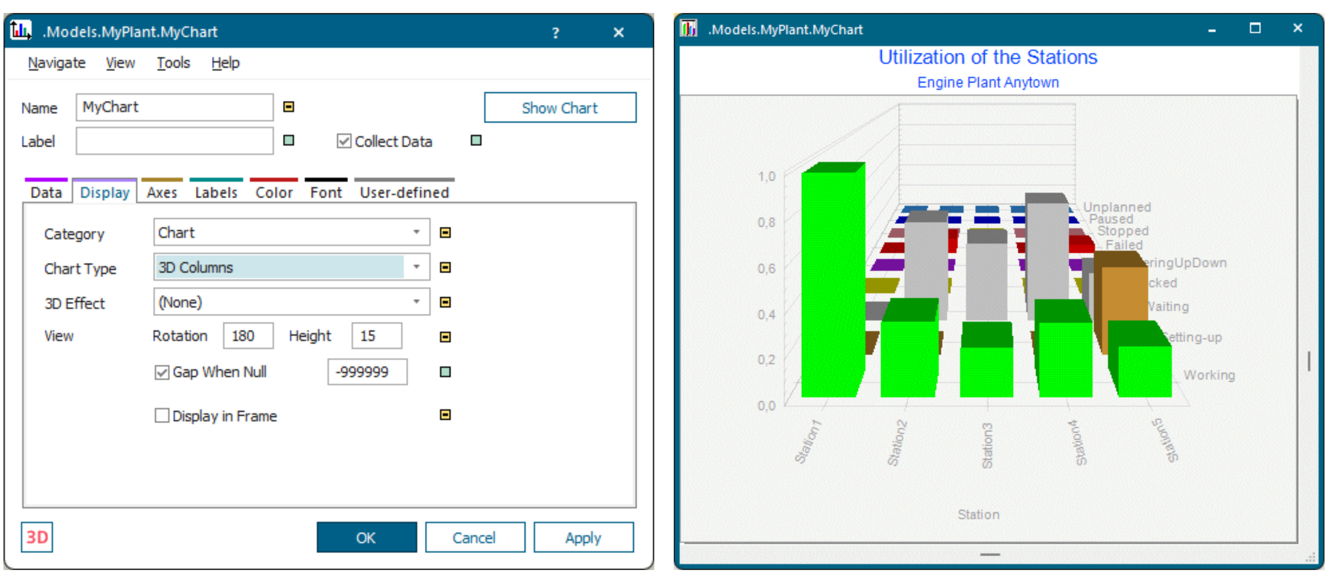
- Select how the *Chart* show the **records** i.e., the entries of a row or of a column.
  - **Graph** show the records as a graph only in the *display* window.
  - **Table** show the records as a table, i.e., as text only in the *display* window. Select a **font** and a relative font size on the tab **Font**.


  - **Graph and Table** show the records as a graph and as a table in the *display* window.
- Select how the *Chart* show the **data**. This way you can present different views of the same record.



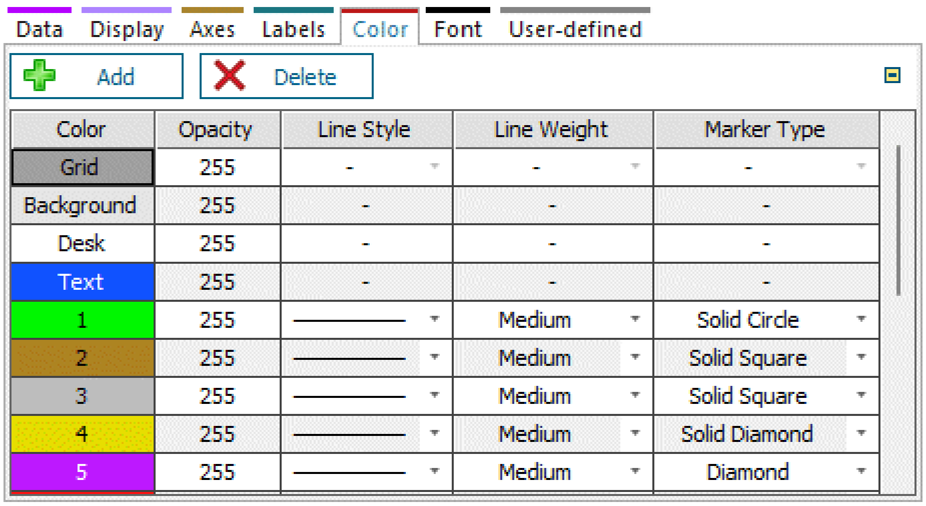

- If you want to, you can also select a **3D Effect** for some of the **Chart Types**:
  - **None** adds no 3D effect to the **Chart Type** you selected.
  - **Shadow** adds a black shadow to the chart type you selected.
  - **3D** adds three-dimensional depth to the chart type you selected.
  - **Gradient Bars** adds an effect between a gradient and a 3D look to the chart type you selected.
  - **Contoured** for the chart types **3D Columns** and **3D Surface**.

    You can also type in the **Rotation** angle and the **Height** of the columns/surface. To turn the chart left or right, drag the horizontal slider in the display window. To turn it up or down, drag the vertical slider.



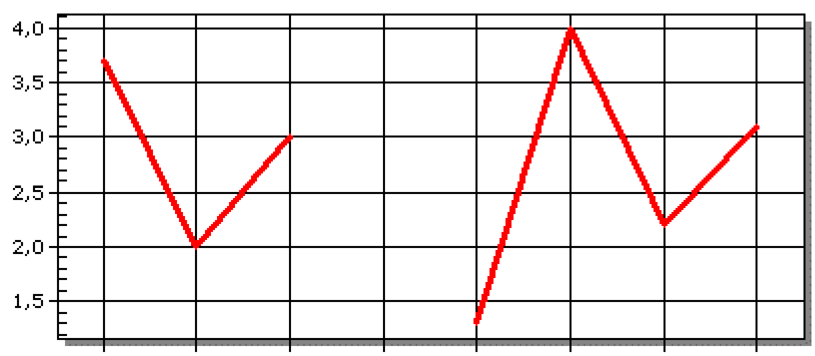
> **Note**
>
> Some chart types do not provide **3D effects**.

- To change the background color of the *Chart* proper in the display window, click the tab **Color**. Double-click the cell labeled **Background** and select another color. Click **Apply** to change the color.




- To change the color around the *Chart* proper in display window, click the tab **Color**. Double-click the color field next to **Desk** and select another color. Click **Apply** to change the color.
- To display the *Chart* in the *Frame*, instead of the icon of the object, select **Display in Frame** on the tab **Display**.

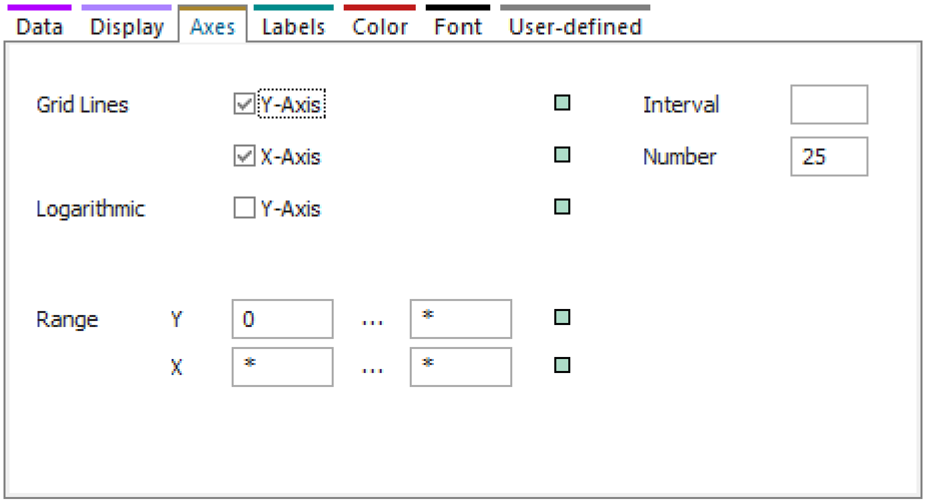
  We advise to increase the size of the icon of the *Chart* in the *Frame*: Hold down **Shift + Ctrl**, grab one of the four corners of the icon with the mouse and drag the icon to a convenient size.
- To interrupt the line the *Chart* draws for values that are not defined, select **Gap When Null** on the tab **Display**. To continue drawing the line, clear the check box.



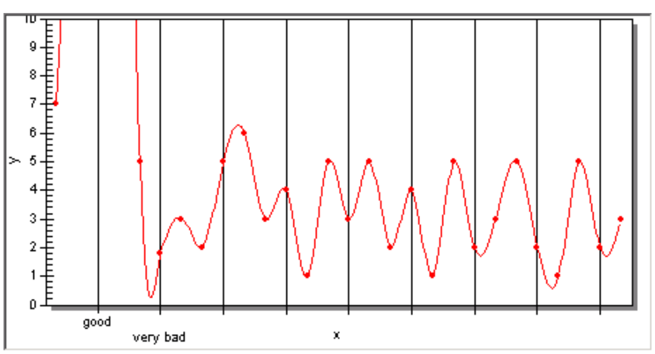
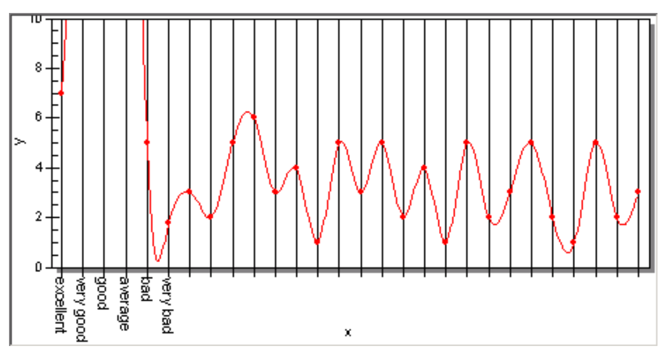

- To print the *Chart*, select **Tools > Print**.
- To copy the contents of the display window to the *Clipboard* is a bitmap, select **Tools > Copy to Clipboard**. You can then paste it into another application with **Ctrl+V** or **Home > Paste**.
- To show **Grid Lines** in the x-axis in the *display* window, select **X-axis** on the tab **Axes**. Type in the maximum number of grid lines on the x-axis. This feature is especially handy, when the *Chart* displays

numerous points within a limited space, and you want to regulate how many grid lines the display window shows.

To hide them, clear the check box.



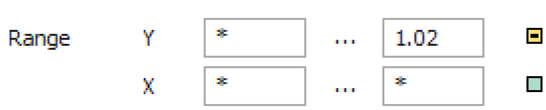

Compare these screenshots:

| 10 grid lines on the x-axis | 20 grid lines on the x-axis |
| --- | --- |
|   |   |

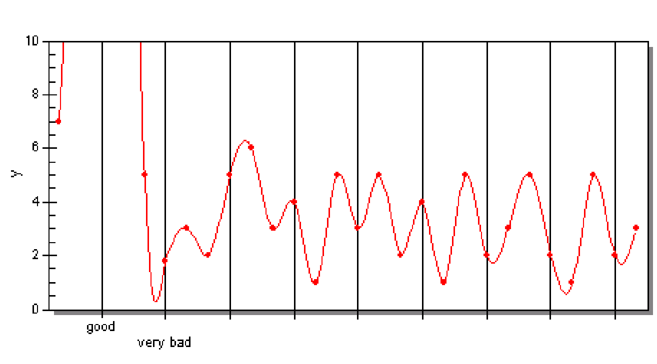
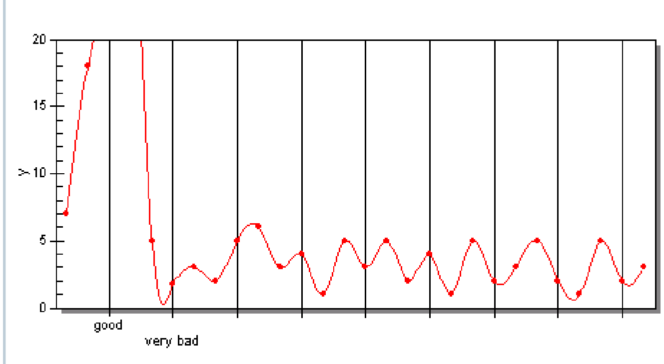
- To show grid lines on the y-axis in the *display* window, select **Y-axis**  . To hide the grid lines, clear the check box.
- To set the range the *Chart* is to display scaled on the y-axis, type the first value of into the left text box. The default setting `0 … 0` means that *Plant Simulation* scales automatically.

  Type the last value of the range the *Chart* is to display scaled on the y-axis into the right text box.

> **Note**
>
> When you type in neither **Title** for **Subtitle**, the *Chart* might, under certain circumstances, cut off the upper part of the **y Range**. To prevent this, you can type a blank space into the text box **Subtitle**.



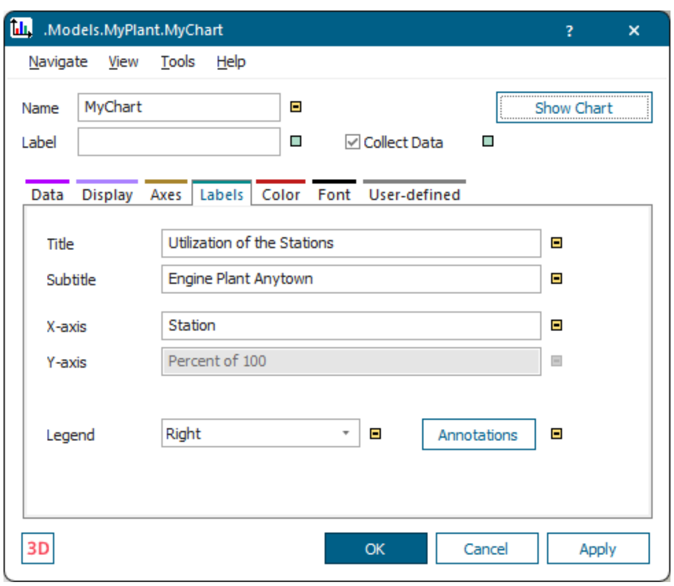
Compare these screenshots:

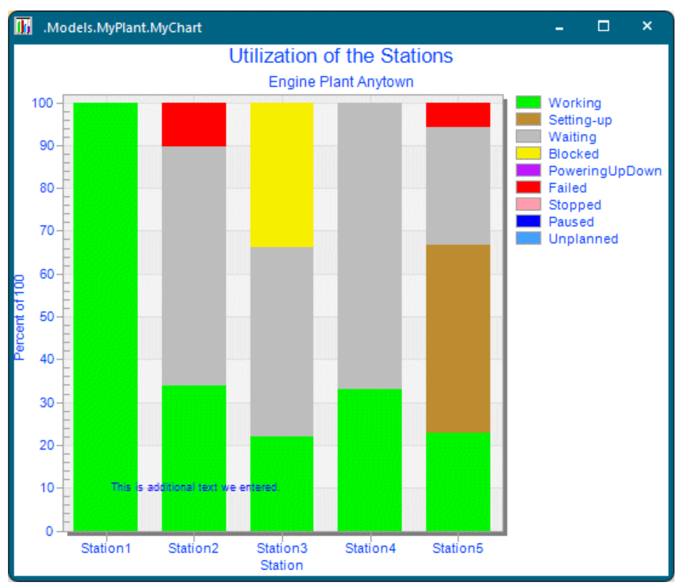
| Scaling on the y-axis, 10 units | Scaling on the y-axis, 20 units |
|---|---|
|  |  |

**Back to** **Select How the Chart Shows the Data**

## Add Labels and a Legend

You can add a title and a subtitle to your chart, and format them, add text to the x-axis and to the y-axis, and add a legend that tells what the values mean.

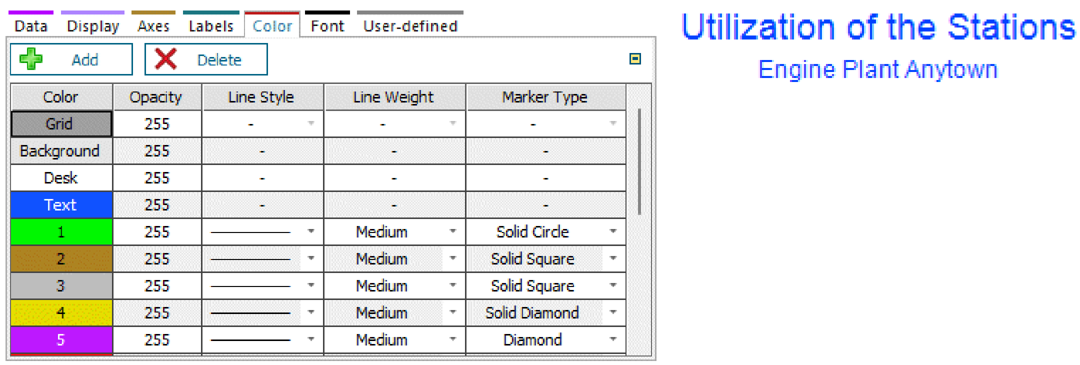



- Enter any text that *Plant Simulation* shows as the title of the *Chart* in the *display* window into the text box **Title**. In our example we entered `Utilization of the Stations`.

  Select a font, if you want to **Bold** face or **Italicize** the text and a relative font size on the tab **Font**.


Select a color for all text in the *display* window on the tab **Color**. Double-click the color field next to **Font** and select another color. Click **Apply** to change the color.



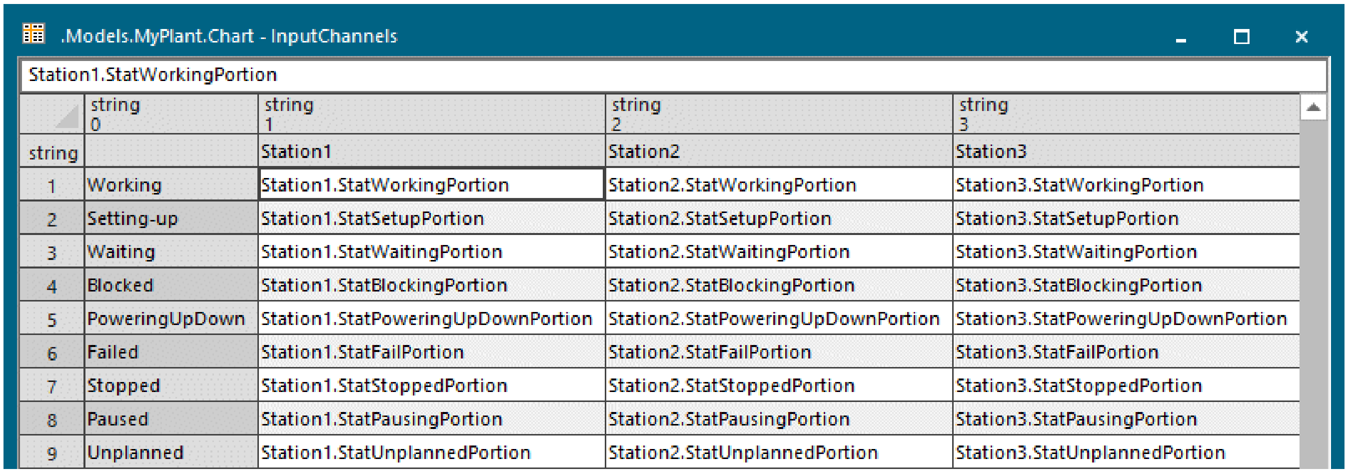

- Enter any text that *Plant Simulation* shows as the subtitle of the *Chart* in the *display* window into the text box **Subtitle**. In our example we entered `Engine Plant Anytown`.
- Enter any text that *Plant Simulation* shows in the *display* window on the x-axis. In our example we entered `Station`.

  Select a font, if you want to **Bold** face or **Italicize** the text and a relative font size on the tab **Font**.

> **Note**
>
> This font applies to all labels, except for the **Title**, the **Subtitle** and data displayed in **Table** format in the *display* window

- Enter any text that *Plant Simulation* shows in the *display* window on the y-axis. In our example we entered `Percent of 100`.
- Select if and on which side of the *display* window *Plant Simulation* shows the legend for the values the *Chart* displays.

  As you remember, you enter the text you want to show as the **legend** into the row index of the table when you select **Data Source > Input Channels > Table File**.



.Models.MyPlant.Chart - InputChannels

Station1.StatWorkingPortion

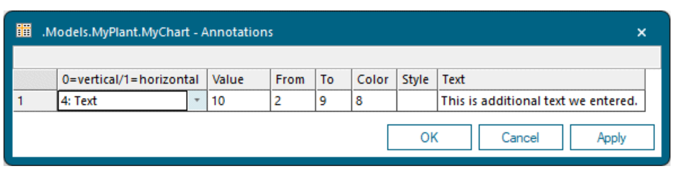
| string | string 0 | string 1 | string 2 | string 3 |
|---|---|---|---|---|
| string | | Station1 | Station2 | Station3 |
| 1 | Working | Station1.StatWorkingPortion | Station2.StatWorkingPortion | Station3.StatWorkingPortion |
| 2 | Setting-up | Station1.StatSetupPortion | Station2.StatSetupPortion | Station3.StatSetupPortion |
| 3 | Waiting | Station1.StatWaitingPortion | Station2.StatWaitingPortion | Station3.StatWaitingPortion |
| 4 | Blocked | Station1.StatBlockingPortion | Station2.StatBlockingPortion | Station3.StatBlockingPortion |
| 5 | PoweringUpDown | Station1.StatPoweringUpDownPortion | Station2.StatPoweringUpDownPortion | Station3.StatPoweringUpDownPortion |
| 6 | Failed | Station1.StatFailPortion | Station2.StatFailPortion | Station3.StatFailPortion |
| 7 | Stopped | Station1.StatStoppedPortion | Station2.StatStoppedPortion | Station3.StatStoppedPortion |
| 8 | Paused | Station1.StatPausingPortion | Station2.StatPausingPortion | Station3.StatPausingPortion |
| 9 | Unplanned | Station1.StatUnplannedPortion | Station2.StatUnplannedPortion | Station3.StatUnplannedPortion |

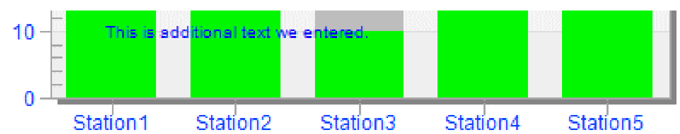
**Back to Show Statistics in a Chart**

## Add Annotations

You can also additional lines and/or text to the display window to the *Chart*

Click **Annotations** and fill in the table that opens.



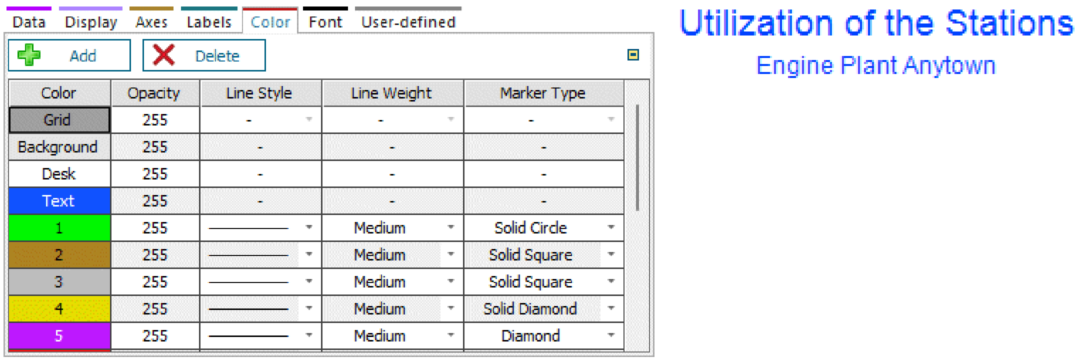




- Enter any text that *Plant Simulation* shows as the title of the *Chart* in the *display* window into the text box **Title**. In our example we entered `Utilization of the Stations`.

  Select a font, if you want to **Bold** face or **Italicize** the text and a relative font size on the tab **Font**.

  Select a color for all text in the *display* window on the tab **Color**. Double-click the color field next to **Font** and select another color. Click **Apply** to change the color.




- Enter any text that *Plant Simulation* shows as the subtitle of the *Chart* in the *display* window into the text box **Subtitle**. In our example we entered `Engine Plant Anytown`.
- Enter any text that *Plant Simulation* shows in the *display* window on the x-axis. In our example we entered `Station`.

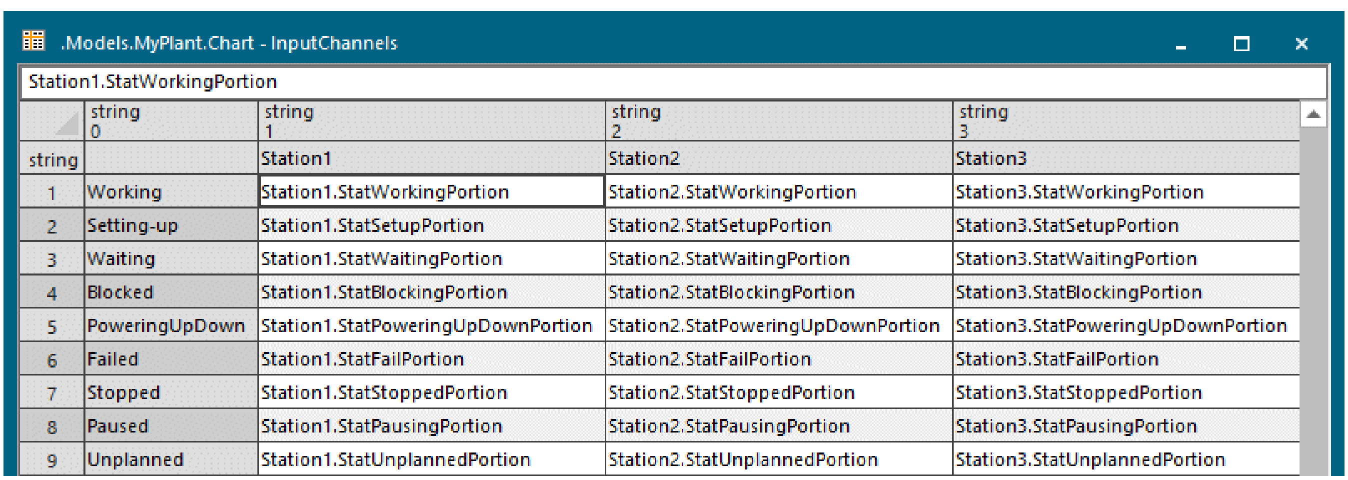
  Select a font, if you want to **Bold** face or **Italicize** the text and a relative font size on the tab **Font**.

  **Note**

  This font applies to all labels, except for the **Title**, the **Subtitle** and data displayed in **Table** format in the *display* window

- Enter any text that *Plant Simulation* shows in the *display* window on the y-axis. In our example we entered `Percent of 100`.
- Select if and on which side of the *display* window *Plant Simulation* shows the legend for the values the *Chart* displays.

  As you remember, you enter the text you want to show as the **legend** into the row index of the table when you select **Data Source > Input Channels > Table File**.



.Models.MyPlant.Chart - InputChannels

Station1.StatWorkingPortion

| | string 0 | string 1 | string 2 | string 3 |
|---|---|---|---|---|
| string | | Station1 | Station2 | Station3 |
| 1 | Working | Station1.StatWorkingPortion | Station2.StatWorkingPortion | Station3.StatWorkingPortion |
| 2 | Setting-up | Station1.StatSetupPortion | Station2.StatSetupPortion | Station3.StatSetupPortion |
| 3 | Waiting | Station1.StatWaitingPortion | Station2.StatWaitingPortion | Station3.StatWaitingPortion |
| 4 | Blocked | Station1.StatBlockingPortion | Station2.StatBlockingPortion | Station3.StatBlockingPortion |
| 5 | PoweringUpDown | Station1.StatPoweringUpDownPortion | Station2.StatPoweringUpDownPortion | Station3.StatPoweringUpDownPortion |
| 6 | Failed | Station1.StatFailPortion | Station2.StatFailPortion | Station3.StatFailPortion |
| 7 | Stopped | Station1.StatStoppedPortion | Station2.StatStoppedPortion | Station3.StatStoppedPortion |
| 8 | Paused | Station1.StatPausingPortion | Station2.StatPausingPortion | Station3.StatPausingPortion |
| 9 | Unplanned | Station1.StatUnplannedPortion | Station2.StatUnplannedPortion | Station3.StatUnplannedPortion |

- Select the **Type** of the annotation:

  Select `0: Vertical Line` for displaying a vertical line.

  Select `1: Horizonal Line` for a displaying a horizontal line.

Select `2: X-axis Label` for labeling the x-axis.

Select `3: Y-axis Label` for labeling the y-axis.

Select `4: Text` for any text.

- **Value**: Enter the value where the *Chart* displays the line in the *display* window. For a horizontal line or for an annotation of the Y axis or for this is the Y value. For a vertical line or for an annotation of the X axis this is X value.

  Y values are specified in the Y scale of the *Chart*.

  X values are specified in the X scale of the *Chart*. For a bar chart the value 3.5 means that the X value is placed between the third and the fourth bar.

- **From**: Enter the starting point of the line here. If you do not enter a value, the *Chart* extends an existing line. This way you can not only create horizontal and vertical lines but also diagonal lines.

  For a horizontal line or for text this is an X value. For a vertical line this is a Y value.

- **To**: Enter the end point of the line here. If you do not enter a value in the cells **From** and **To**, the *Chart* draws the line from the left to right, or from the top to the bottom of the *display* window.

  For a horizontal line this is a X value. For a vertical line this is a Y value.

- **Color**: Enter the number of the color of the line here. This is the number of one of the lines you defined on the tab **Color**.

- **Style**: Enter the style of the line here, such as dotted, dashed, etc.

  For lines of **Type** `0` or `1`, you can enter the line style. For text of **Type** `4`, you can enter the marker style.

| Value | Line style | Marker style |
|---|---|---|
| `0` | a thin solid line | text only without a marker |
| `1` | a dashed line | a plus sign |
| `2` | a dotted line | a cross |
| `3` | a dash-dot line | a circle |
| `4` | a dash-dot-dot line | a filled circle |
| `5` | a medium thin solid line | a square |
| `6` | a thick solid line | a filled square |
| `7` | a grid tick | a diamond |
| `8` | a grid line | a filled diamond |
| `9` | none | upward triangle |
| `10` | a medium thick solid line | solid upward triangle |
| `11` | an extra thick solid line | downward triangle |
| `12` | none | solid downward triangle |

| Value | Line style | Marker style |
|---|---|---|
| 13 | none | small plus |
| 14 | none | small cross |
| 15 | none | small circle |
| 16 | none | small solid circle |
| 17 | none | small square |
| 18 | none | small solid square |
| 19 | none | small diamond |
| 20 | none | small solid diamond |
| 21 | none | small upward triangle |
| 22 | none | small solid upward triangle |
| 23 | none | small downward triangle |
| 24 | none | small solid downward triangle |
| 25 | none | large plus |
| 26 | none | large cross |
| 27 | none | large circle |
| 28 | none | large solid circle |
| 29 | none | large square |
| 30 | none | large solid square |
| 31 | none | large diamond |
| 32 | none | large solid diamond |
| 33 | none | large upward triangle |
| 34 | none | large solid upward triangle |
| 35 | none | large downward triangle |
| 36 | none | large solid downward triangle |
| 92 | none | arrow north |
| 93 | none | arrow north east |
| 94 | none | arrow east |
| 95 | none | arrow south east |
| 96 | none | arrow south |

| Value | Line style | Marker style |
|---|---|---|
| 97 | none | arrow south west |
| 98 | none | arrow west |
| 99 | none | arrow north west |

- **Text**: Enter any text that describes the line.

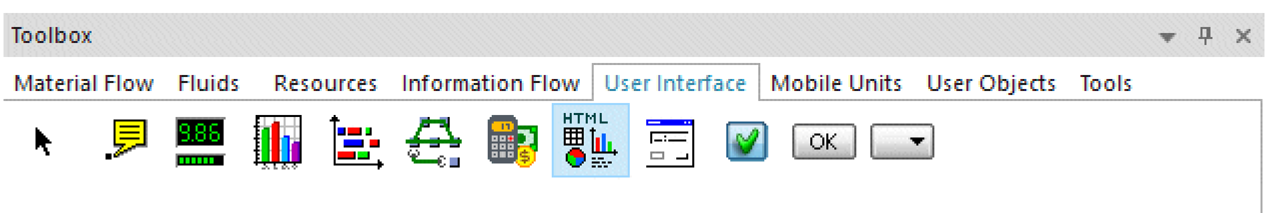
| Prefix | Position |
|---|---|
| `|l` | left inside edge of the graph |
| `|L` | left outside edge of the graph |
| `|r` | right inside edge of the graph |
| `|R` | right outside edge of the graph |
| `|c` | centered inside of the graph |

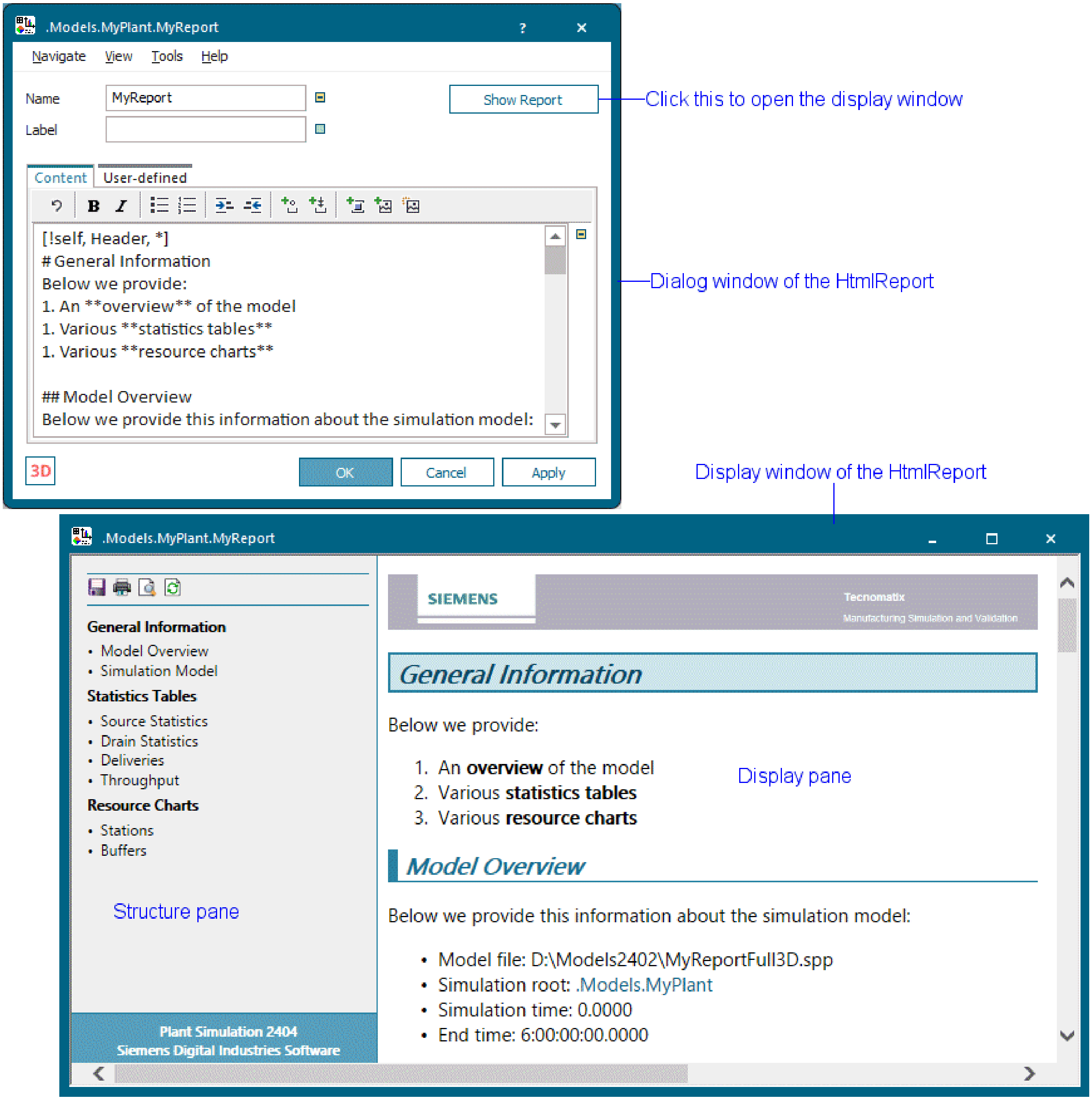
**Back to** Show Statistics in a Chart

## Show Statistics and Other Values in a Report

If you want to present all of the results of a simulation run, which *Plant Simulation* outputs to tables and diagrams, on an HTML page on screen, you will use the object *HtmlReport*. You can also print, and save, this HTML page for sharing it with your colleagues or for displaying it on a website.

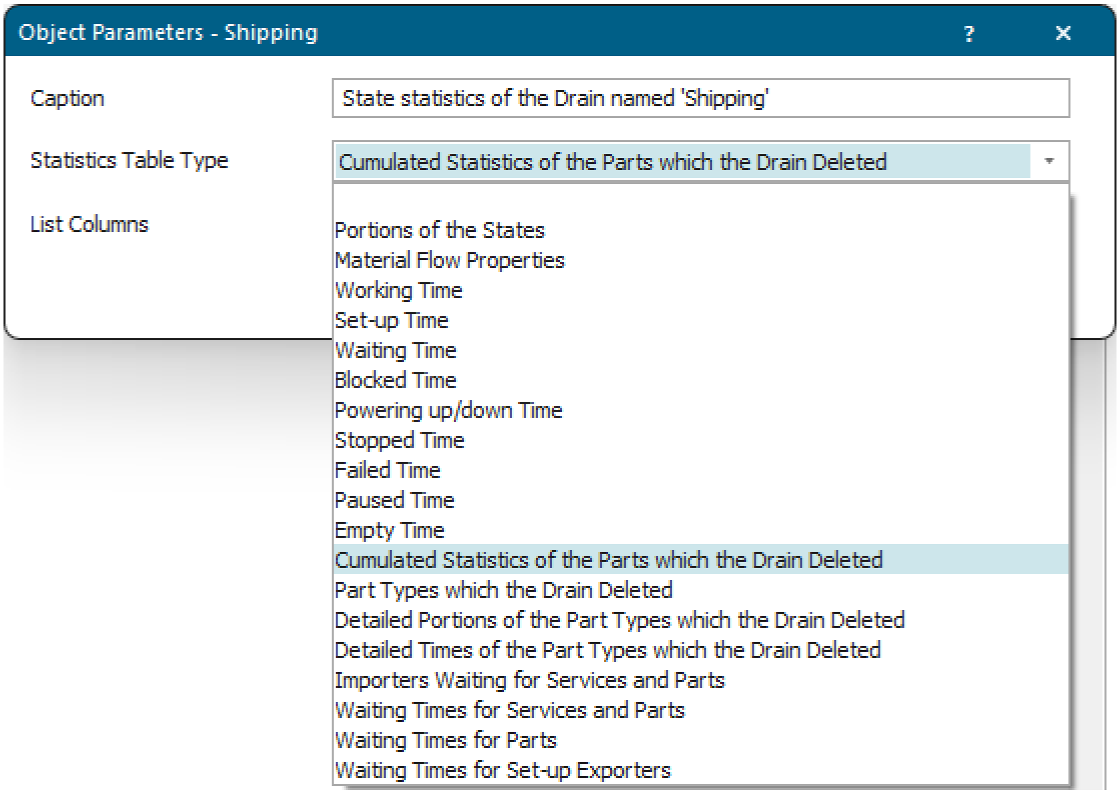
You can insert the **HtmlReport**  into your simulation model from the folder **UserInterface** in the *Class Library* or from the toolbar **User Interface** in the *Toolbox*.



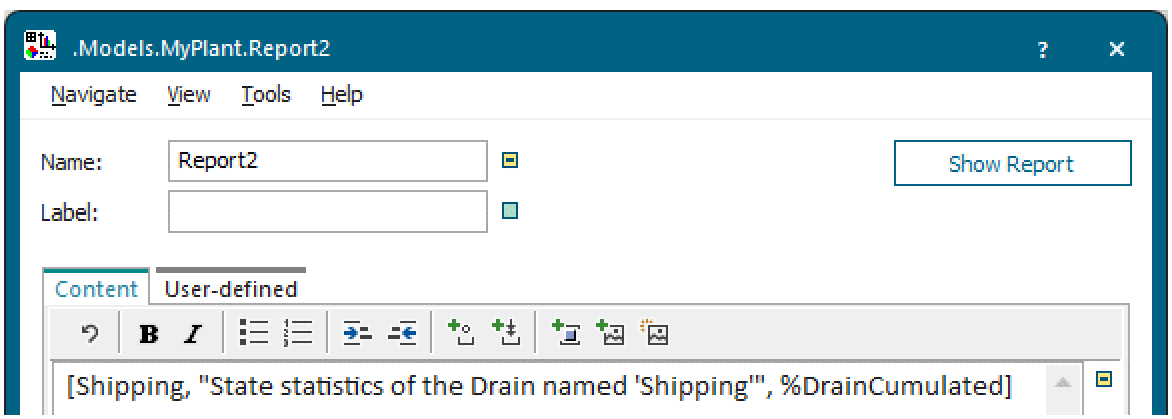



Creating a report is really quick and comfortable:

- Type in the name of the object in brackets on the tab **Content** and highlight the name. Click on the toolbar and enter the required settings into the dialog **Object Parameters**. The dialog offers different settings for the different objects.




To open the dialog **Object Parameters** for an object, which you added to the tab **Content** with **drag-and-drop**, click the mouse between the name of the object and the closing bracket, and click .




Then, select the settings you need in the dialog **Select Object**.

- Enter the content, which the *HtmlReport* shows, as text on the Tab Content of the object. As you will see, the syntax is easy to understand allowing you to quickly type in the required instructions.

  To open the dialog **Object Parameters** for an object, which you added to the tab **Content** with **drag-and-drop**, click the mouse between the name of the object and the closing bracket, and click .

  Then, select the settings you need in the dialog **Select Object**.

In our example below, we'll introduce the most important items which you can add to your report.

- To address an object, we entered its name within brackets, `[MyStation]` in our example.
- To show the **icon** named **logo** of the object *HtmlReport* and to zoom the icon when zooming the report window, we entered an **exclamation mark** followed by the **name of the icon** enclosed in **brackets** followed by an asterisk. The asterisk denotes the icon will be zoomed in/out as well.

```
[!self, logo, *]
```

To show the picture of your company, for example its logo, add it as an icon to the object *HtmlReport* and name it **logo**. In our example below, we used the Siemens logo.


- To show a **heading of level 1**, we entered a number sign # followed by the title of the heading:

```
# General Information
```

- To show **body text**, we entered:

```
Below we provide:
```

- To show a **numbered list**, we entered `1.` followed by the text we want to display.

```
1. An **overview** of the model
1. Various **statistics tables**
1. A **resource chart**
```

To **bold** text, we enclosed it with two asterisks `**`.

- To show a **heading of level 2**, we entered two number signs ## followed by the title of the heading:

```
## Model Overview
```

- To show **body text**, followed by a **bulleted list**, we entered the body text, followed by a * denoting the bullet, and the text we want to display.

```
Below we provide this information about the simulation model:
* Model file: [=modelFile]
* Simulation root: [EventController.Location]
* Simulation time: [EventController.SimTime]
* End time: [EventController.EndTime]
```

- To show a **formula**, we entered the **equal sign** = followed by the **name of the function** enclosed **in brackets**:

```
* Model file: [=modelFile]
```

- To show the **value of the attribute** *SimTime* of the *EventController*, we entered the **name of the object** followed by the **name of the attribute** enclosed **in brackets**:

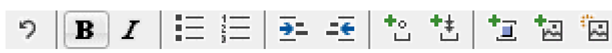
```
* Simulation time: [EventController.SimTime]
```

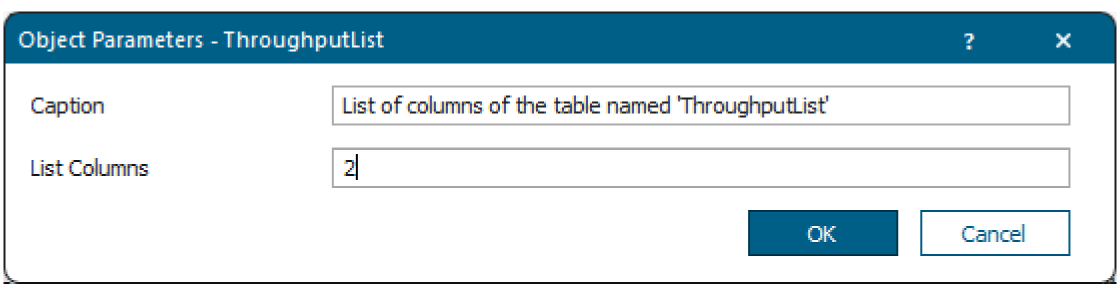
- To show a **heading of level 2**, followed by a screenshot of the *Frame* containing our model, we entered:

```
## Simulation Model
The simulation model looks like this:
><**[current, "Test Simulation"]**
```

  The expression `Test Simulation` is the **caption** that is shown below the screenshot.

  The greater than sign followed by a less-than sign >< centers the screenshot.

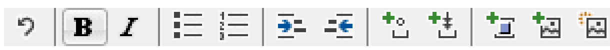
- To show the **values of objects of type DataTable** from our model:
  - We clicked  on the toolbar 
  - We selected the *DataTable/List* whose values you would like to show. We then clicked **OK**.
  - We entered the **caption** which the *HtmlReport* shows below the *DataTable*.
  - We entered the identifiers of the **columns** of the *DataTable* which the *HtmlReport* shows on the HTML page.

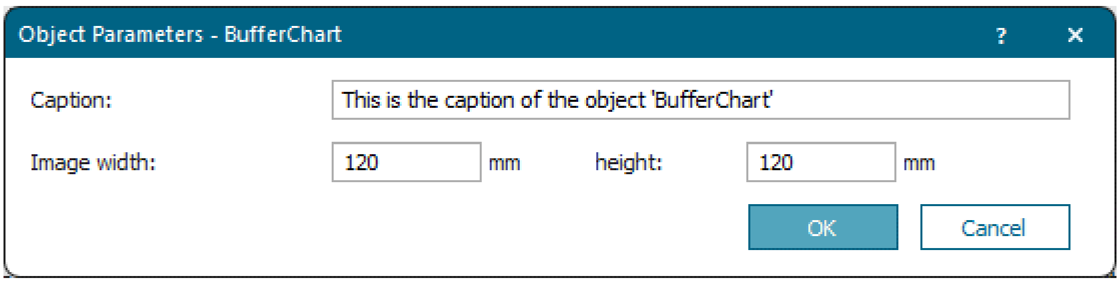



  - Instead, we can also directly type in the respective instructions, compare the example below:

```
# Statistics Tables
These **statistics values** are of general interest. Feel free to add
more.
## Drain Statistics
Drain statistics shows the contents of the *summary report* table.
[.MaterialFlow.Drain*]
## Deliveries
```

```
Delivieries shows the contents of the table object *DeliveryList.*
[DeliveryList]
```

- To show the **contents of objects of type Chart** from our model as screenshots:
  - We clicked  on the toolbar 
  - We selected the *Chart* which you would like to show. We then clicked **OK**.
  - We entered the **Caption** which the *HtmlReport* shows below the *Chart*.
  - We entered the **Image Width** with which the *HtmlReport* shows the picture of the object. You can also enter an asterisk (*) for either the **Width** or the **Height** to ensure that the image will not be distorted.
  - We entered the **Image Height** with which the *HtmlReport* shows the picture of the object. You can also enter an asterisk (*) for either the **Width** or the **Height** to ensure that the image will not be distorted. .




We clicked **OK** to apply our settings.

Instead, we can also directly type in the respective instructions, compare the example below:

```
# Resource Chart
The *resource chart* for the *Stations* look like this. Feel free to
add more.
## Stations
The utilization of the processing  *Stations* is as follows:
><[StationChart, 150, 100]
```

The numbers `150` and `100` designate the **Width** and the **Height** of the screenshot.

The greater than sign followed by a less-than sign >< centers the screenshot.

- To show a **solid horizontal line** between the contents of the *Report* proper and the footer, we entered:

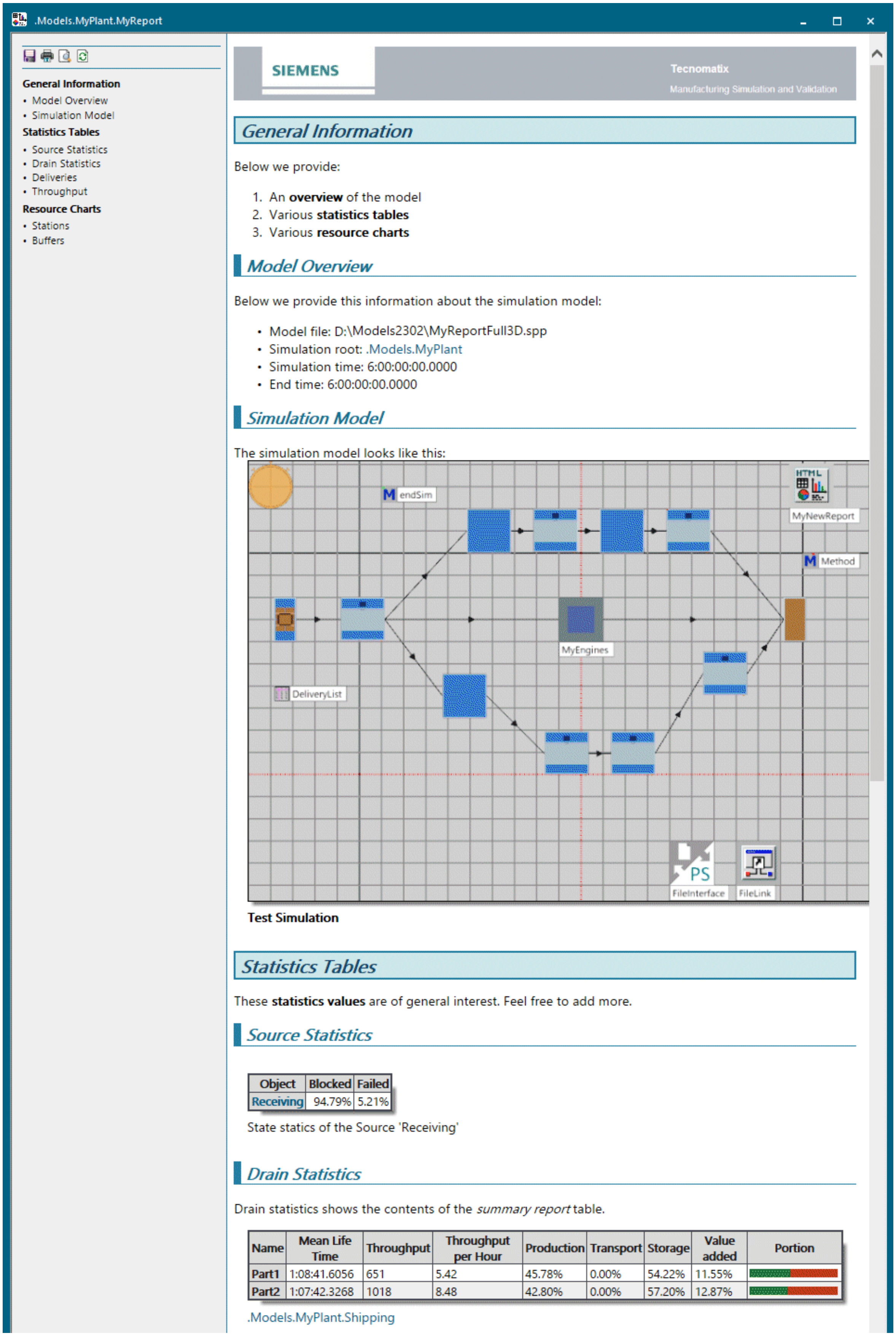
```
---
```

- To show a footer containing indented *text*, designated by the greater sign >, the *current date* and a *signature*, we entered the following **HTML tag**:

```
> Cedar Rapids, [=day(sysdate)].[=month(sysdate)].[=year(sysdate)+1900]
Best regards,
```

```
<span style="font-family: Segoe Script; font-size:16pt">H. Thompson</
span>
```

Our finished report with the settings above looks like this:


.Models.MyPlant.MyReport

General Information
- Model Overview
- Simulation Model

Statistics Tables
- Source Statistics
- Drain Statistics
- Deliveries
- Throughput

Resource Charts
- Stations
- Buffers

SIEMENS

Tecnomatix

Manufacturing Simulation and Validation

## *General Information*

Below we provide:

1. An **overview** of the model
2. Various **statistics tables**
3. Various **resource charts**

### *Model Overview*

Below we provide this information about the simulation model:

- Model file: D:\Models2302\MyReportFull3D.spp
- Simulation root: .Models.MyPlant
- Simulation time: 6:00:00:00.0000
- End time: 6:00:00:00.0000

### *Simulation Model*

The simulation model looks like this:



**Test Simulation**

## *Statistics Tables*

These **statistics values** are of general interest. Feel free to add more.

### *Source Statistics*

| Object | Blocked | Failed |
|---|---|---|
| Receiving | 94.79% | 5.21% |

State statics of the Source 'Receiving'

### *Drain Statistics*

Drain statistics shows the contents of the *summary report* table.

| Name | Mean Life Time | Throughput | Throughput per Hour | Production | Transport | Storage | Value added | Portion |
|---|---|---|---|---|---|---|---|---|
| Part1 | 1:08:41.6056 | 651 | 5.42 | 45.78% | 0.00% | 54.22% | 11.55% | |
| Part2 | 1:07:42.3268 | 1018 | 8.48 | 42.80% | 0.00% | 57.20% | 12.87% | |

.Models.MyPlant.Shipping

## Deliveries

Delivieries shows the contents of the table object *DeliveryList*.

| MU | Number | Name | Attributes |
|---|---|---|---|
| .MUs.Entity | 2 | Part1 | |
| .MUs.Entity | 3 | Part2 | |

## Throughput

Throughput shows the contents of the table object *ThroughputList*.

| Part | Total Throughput | Throughput per Hour |
|---|---|---|
| Part1 | 651 | 5.42 |
| Part2 | 1018 | 8.48 |

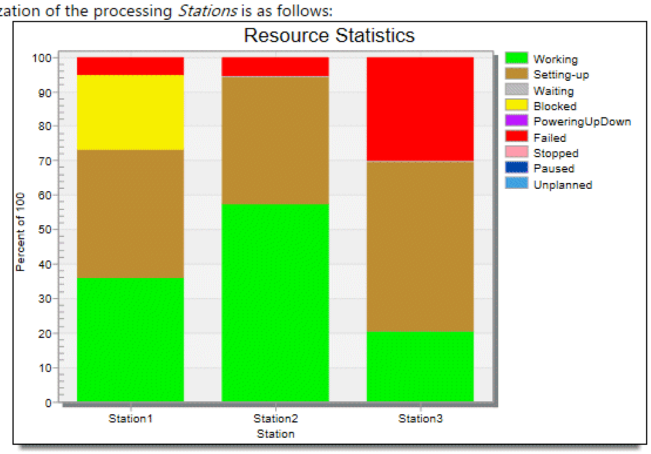
Figure 1: Mean value of the throughput times

# Resource Charts

The *resource charts* for the *Buffers* and *Stations* look like this. Feel free to add more.

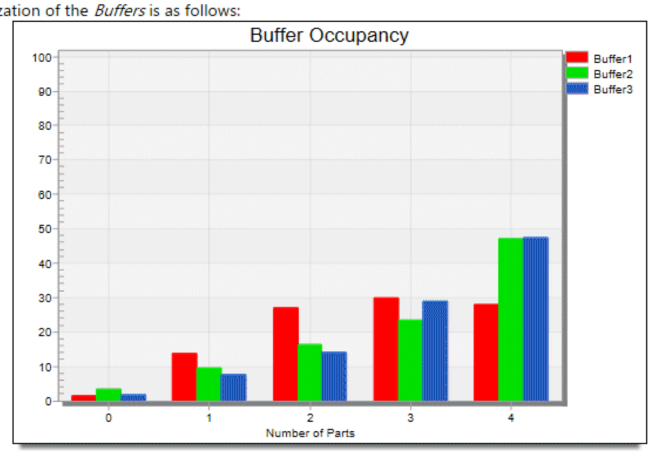
## Stations

The utilization of the processing *Stations* is as follows:




## Buffers

The utilization of the *Buffers* is as follows:




Cedar Rapids, 31.1.2023
Best regards,
H. Thompson

Tecnomatix Plant Simulation 2302
Siemens Digital Industries Software

**Compare** Viewing the Statistics Report

**Compare** Show Reports in a Hierarchically Structured Model

**Compare** View Statistics in the Dialogs of the Objects

**Compare** Show Statistics in a Chart

**Compare** Show Values During the Simulation with the Display

**Go to** Accessing Statistics with SimTalk

**Back to** View and Visualize Statistics

## Showing Reports in a Hierarchically Structured Model

### Show Reports in a Hierarchically Structured Model

In this example we show how to clearly arrange and model a complex production facility and generate reports with objects of type *HtmlReport*. We also demonstrate inheritance of our user objects which we created in the folder **UserObjects**.

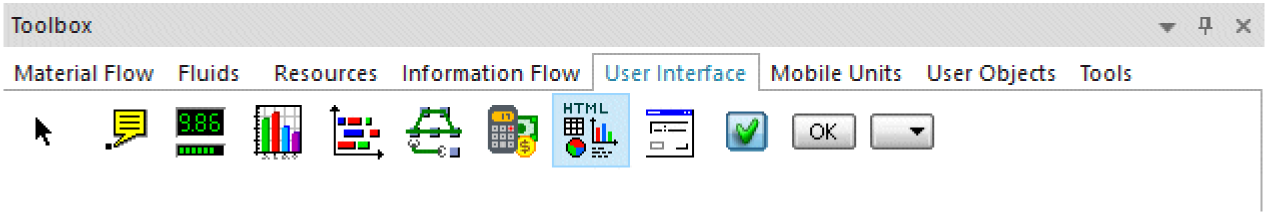
We insert an **HtmlReport** into each of our *sub-Frames* and rename it to `ReportObject`. The expression *ReportObject* is a special name that the programmers built in so that *Plant Simulation* then uses this object when we reference the *Frame* in the *HtmlReport* in brackets.

To do so, we entered these instructions into the overall *HtmlReport* of the car assembly model, i.e., into **.Models.MyCarAssembly.HtmlReport**:

```
# Production Areas
The car assembly encompasses three production areas:
* CarBody
* PaintShop
* FinalAssembly
[CarBody, #]        // Show the HtmlReport of the Frame ‘CarBody’ instead
of the Frame itself
[PaintShop, #]      // Show the HtmlReport of the Frame ‘PaintShop’ instead
of the Frame itself
[FinalAssembly, #]  // Show the HtmlReport of the Frame ‘FinalAssembly’
instead of the Frame itself
```

The overall report of our car assembly gathers all objects named *ReportObject* during the simulation and then shows them.

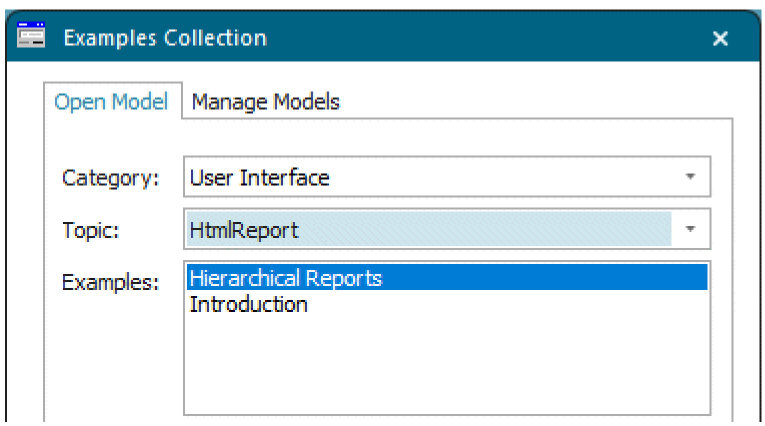
To add the *HtmlReport* to your simulation model, click **Manage Class Library > Basic Objects > UserInterface > HtmlReport** on the **Home** ribbon tab.




We will demonstrate how to:

- Create the Classes of the Cells and the Lines
- Model the Installation and Configure the Components
- Run the Simulation and Show the Report
- Study the HtmlReport

If you do not want to re-create the model yourself, you can also open and study the sample model **UserInterface > HtmlReport > Hierarchical Reports**. Click **Start Page** on the **Window** ribbon tab. Then, click **Getting Started > Example Models > Small Examples**.




**Compare** Show Statistics and Other Values in a Report

**Back to** View and Visualize Statistics

**Compare HtmlReport** > Tab Content

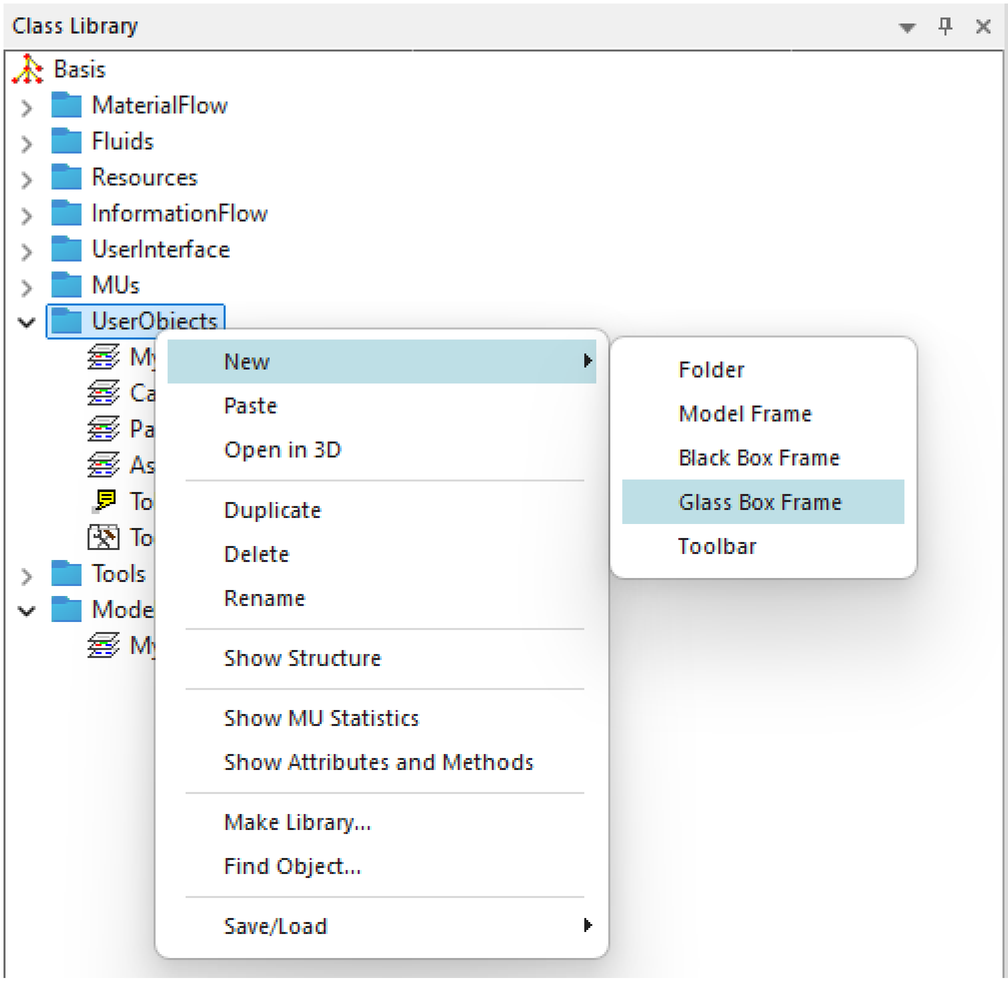
## Creating the Classes of the Cells and the Lines

### Create the Classes of the Cells and the Lines

As we do not want to change the default settings of the built-in objects, we duplicate the built-in *Frames* and objects in the *Class Library* and then work with these duplicated classes in the folder **UserObjects**.

Proceed as follows:

- First, we create the classes of our cells and lines. To do so, click the folder **UserObjects** with the right mouse button and select **New > Glass Box Frame**. This creates a *Frame* that shows its content toward the outside.

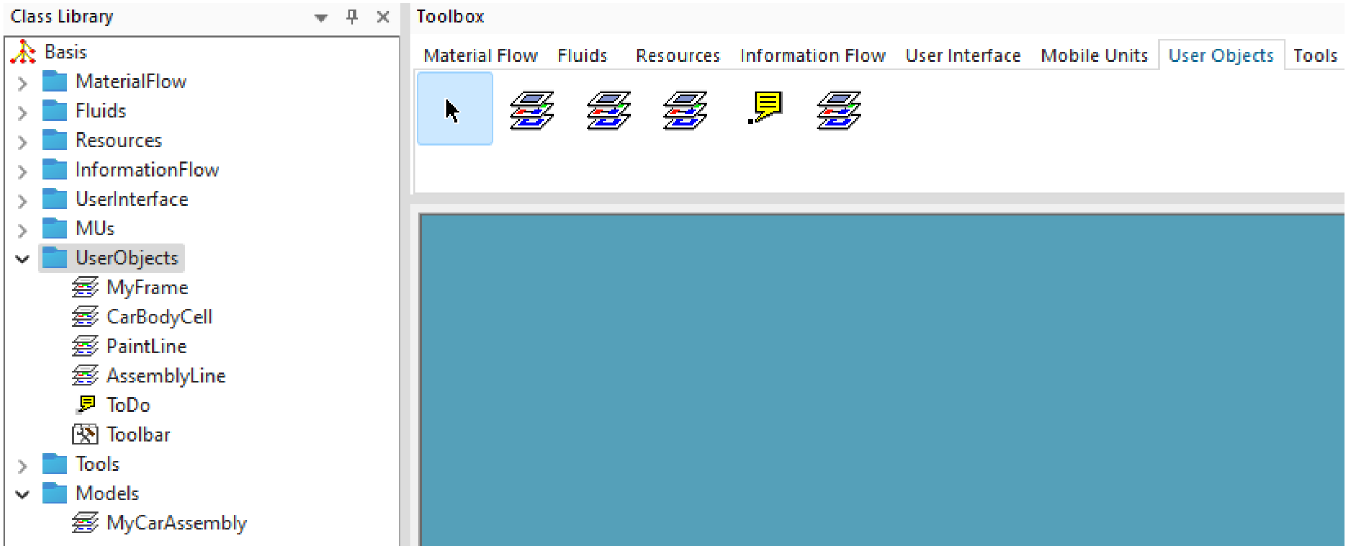



  Repeat this twice and rename our *Frames* that represent our cells and lines. We renamed our *Frames* to `CarBodyCell`, `PaintShop`, and `AssemblyLine`.

- Then, we duplicate the object *Comment* in the folder **UserInterface**. *Plant Simulation* automatically places the duplicated object into the folder **UserObjects**. We renamed our *Comment* to `ToDo`.

  We need this independent class of the *Comment* so that *Plant Simulation* gathers the instances with the name *ToDo* during the simulation run and displays them. The report in our plant as a whole shows a HTML page/a topic with the tasks that are still to be done. When you click an individual task, *Plant Simulation* opens the respective object of type *Comment*. Then, click in the dialog on **Navigate > Open Location** to open the location/*Frame*, in which this *Comment* is inserted.

- In addition we add the objects to the toolbar **UserObjects** in the *Toolbox*. *Class Library* and *Toolbox* then look like this:




Then we will:

- **Model the Car Body Cell**
- **Model the Paint Line**
- **Model the Assembly Line**

To display the data of the instances of the **UserObjects** consistently and in a structured way, we inserted an object of type *HtmlReport* each into the classes of our *Frames* and then renamed it to `ReportObject`. Then *Plant Simulation* shows this object instead of the *Frame* on the HTML page in our overall report of the car assembly.

To do so, we entered these instructions into the overall *HtmlReport* of the car assembly model, i.e., in **.Models.MyCarAssembly.HtmlReport**:

```
# Production Areas
The car assembly encompasses three production areas:
* CarBody
* PaintShop
* FinalAssembly
[CarBody, #]         // Show the HtmlReport of the Frame ‘CarBody’ instead
of the Frame itself
[PaintShop, #]       // Show the HtmlReport of the Frame ‘PaintShop’ instead
of the Frame itself
[FinalAssembly, #]  // Show the HtmlReport of the Frame ‘FinalAssembly’
instead of the Frame itself
```

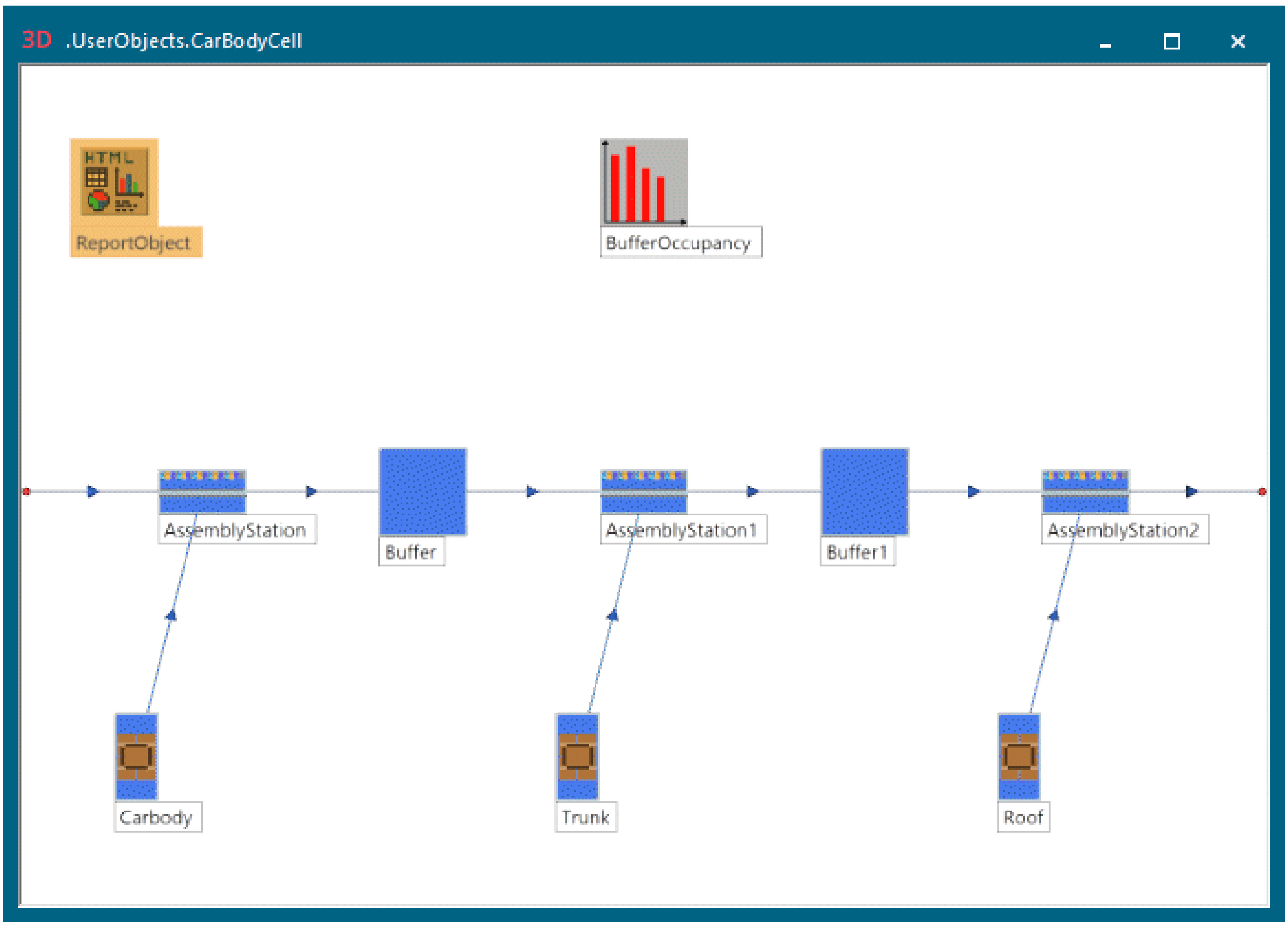
**Back to Show Reports in a Hierarchically Structured Model**

# Modeling the Car Body Cell

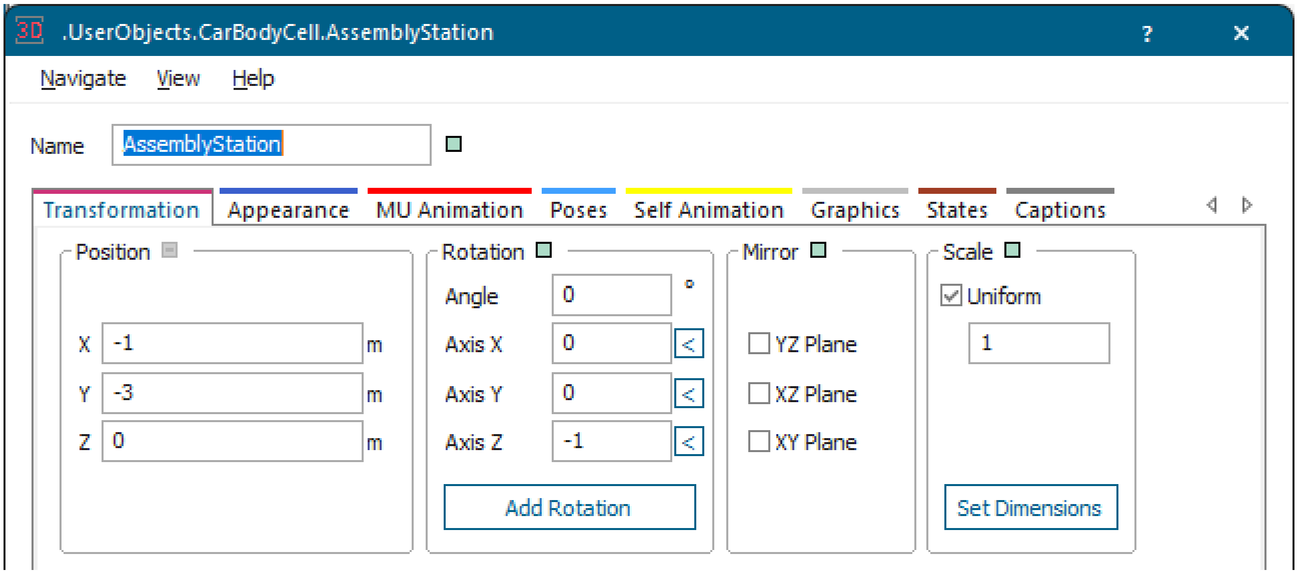
## Model the Car Body Cell

Our *CarBodyCell* consists of the material flow objects *Interface/Entrance*, two *AssemblyStations* between which a *Buffer* each is located, and an *Interface /Exit*. The three *Sources* named `CarBody`, `Trunk`, and `Roof` provide the *AssemblyStations* with parts.

We also inserted an *HtmlReport* which we renamed to `ReportObject` and a *Chart* as a **Histogram** which we renamed to `BufferOccupancy`.




As we want to show the object names in our window, we select the respective object, press the spacebar, and change to the tab **Captions** in the dialog. There we activate the check box **Name/Label Activated**. We repeated this for all objects.




In our *CarBodyCell* we will:

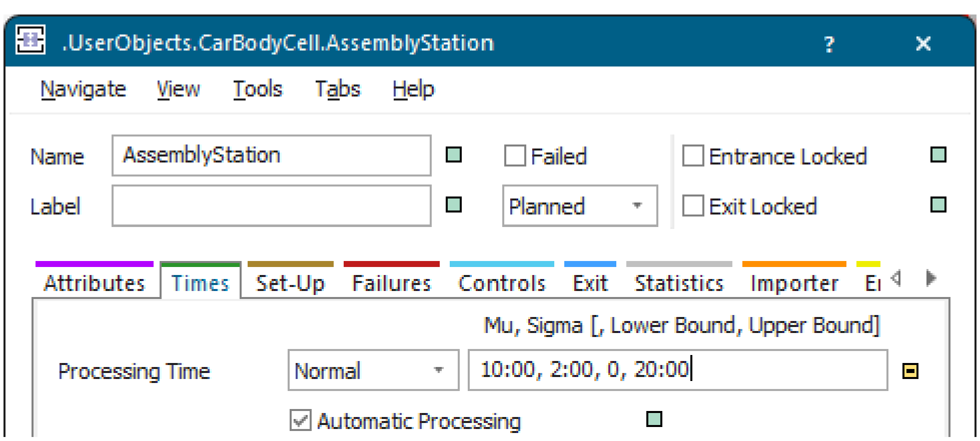
- Configure the Assembly Stations That Assemble the Parts
- Configure the Sources That Provide the Assembly Stations With Parts
- Configure the Buffers between the Assembly Stations
- Configure the Chart as Histogram
- Configure the ReportObject of the Car Body Cell

**Back to** Show Reports in a Hierarchically Structured Model

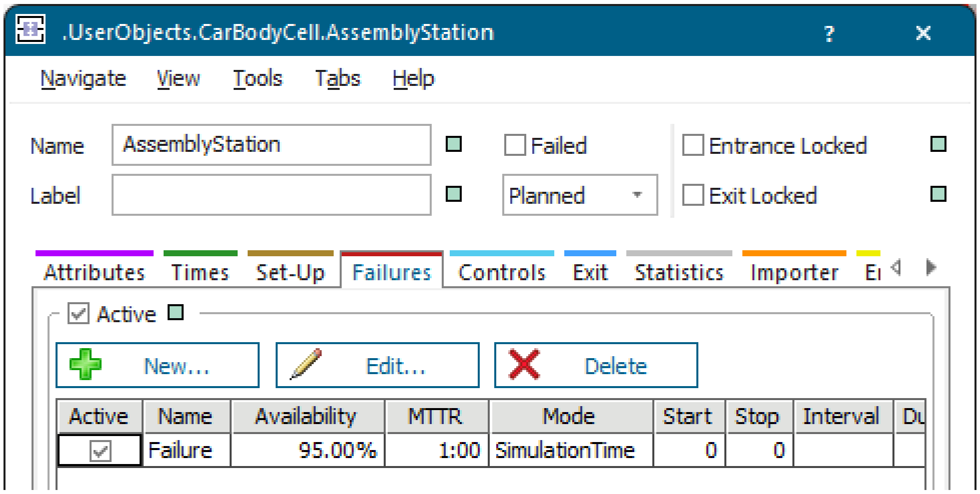
## Configure the Assembly Stations That Assemble the Parts

You can configure the *AssemblyStations* which assemble the delivered parts.

- Select a **normally distributed processing time**. We typed in these parameters: `10:00, 2:00, 0, 20:00`.

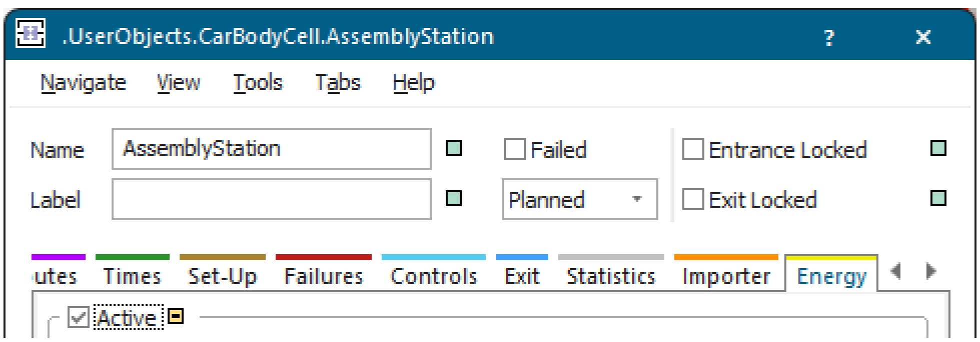


- Create a **failure** with these default values:



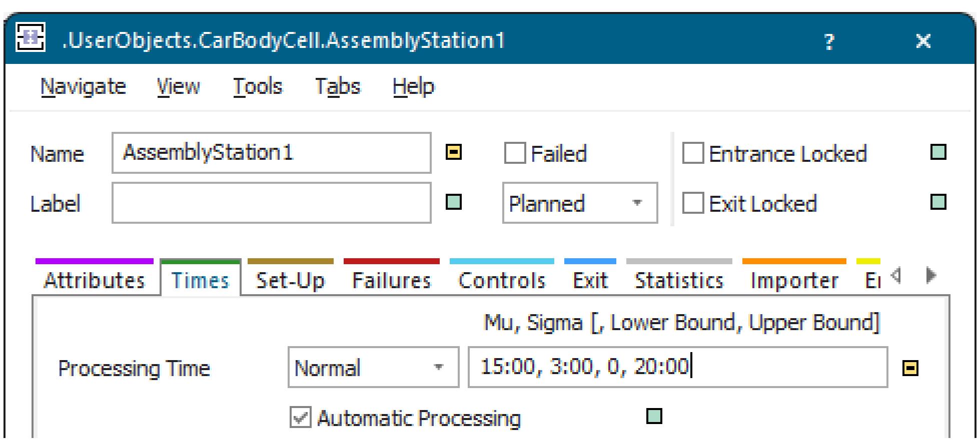

- Activate the energy settings on the tab **Energy**.

  We did not change any other settings.

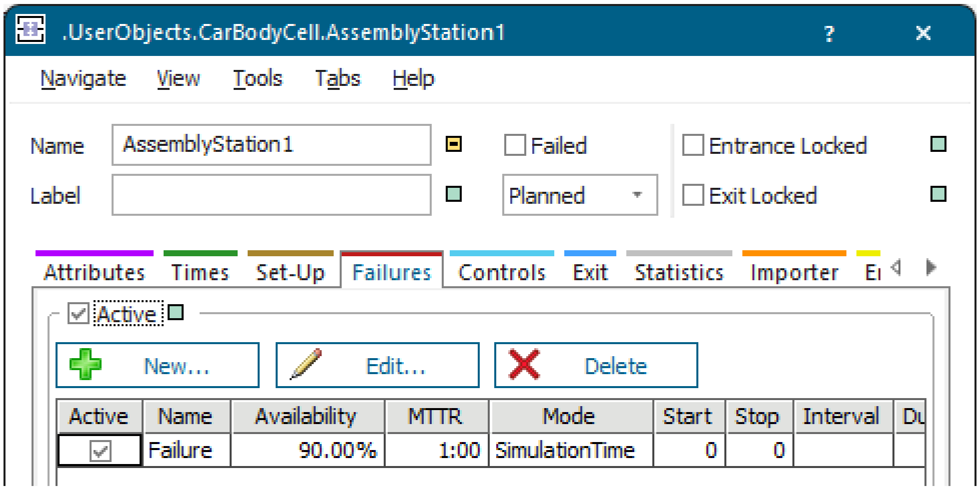



To configure *AssemblyStation1*:

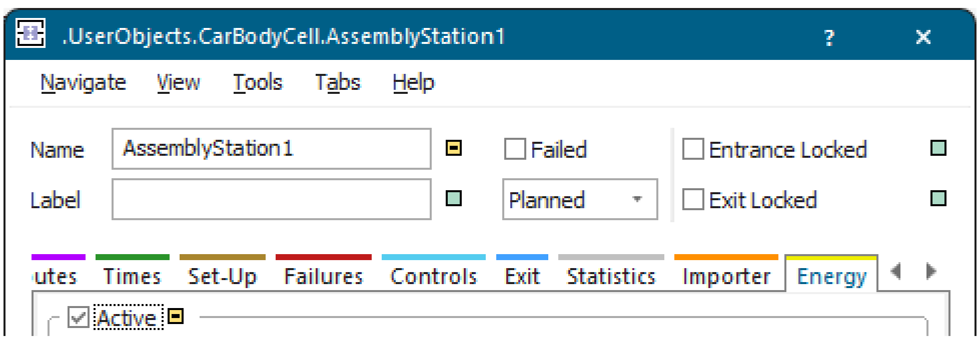
- Select a **normally distributed processing time**. We typed in these parameters: `15:00, 3:00, 0, 20:00`.




- Create a **failure** with these values:



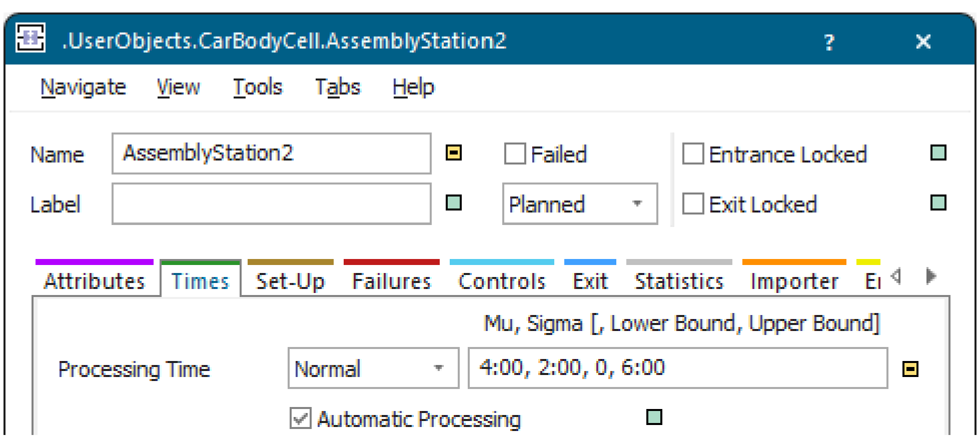

- Activate the energy settings on the tab **Energy**.



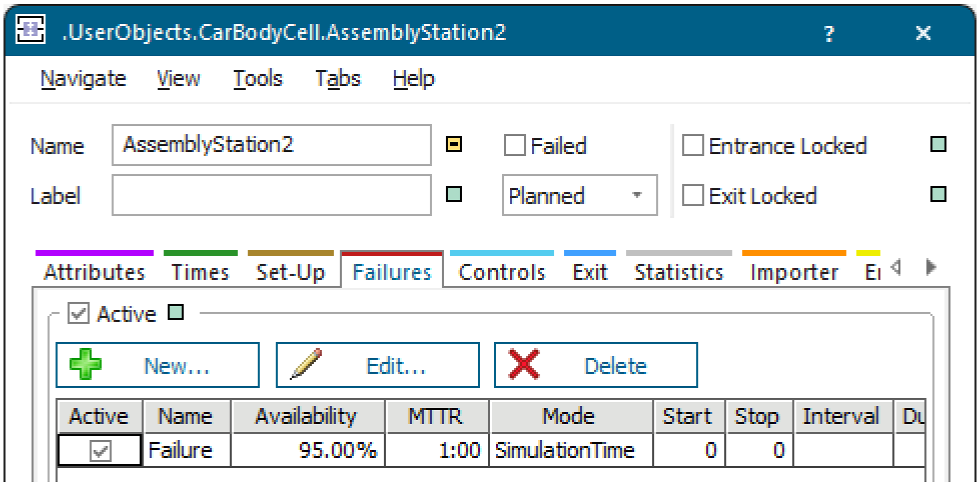

  We did not change any other settings.

To configure *AssemblyStation2*:

- Select a **normally distributed processing time**. We typed in these parameters: `4:00, 2:00, 0, 6:00`.

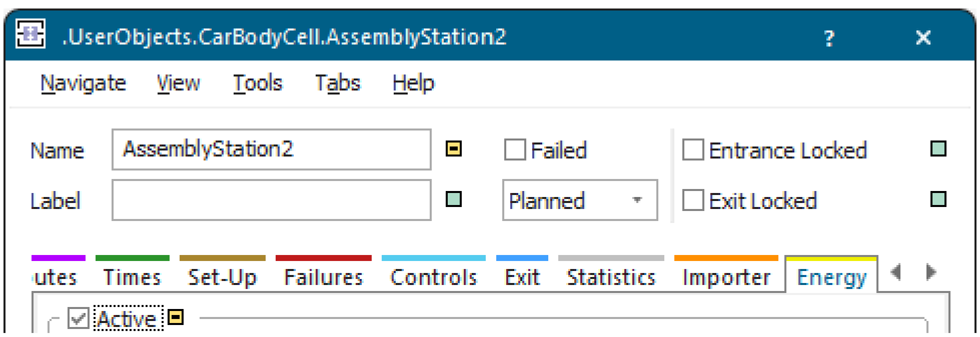



- Create a **failure** with these values:




- Activate the energy settings on the tab **Energy**.

  We did not change any other settings.



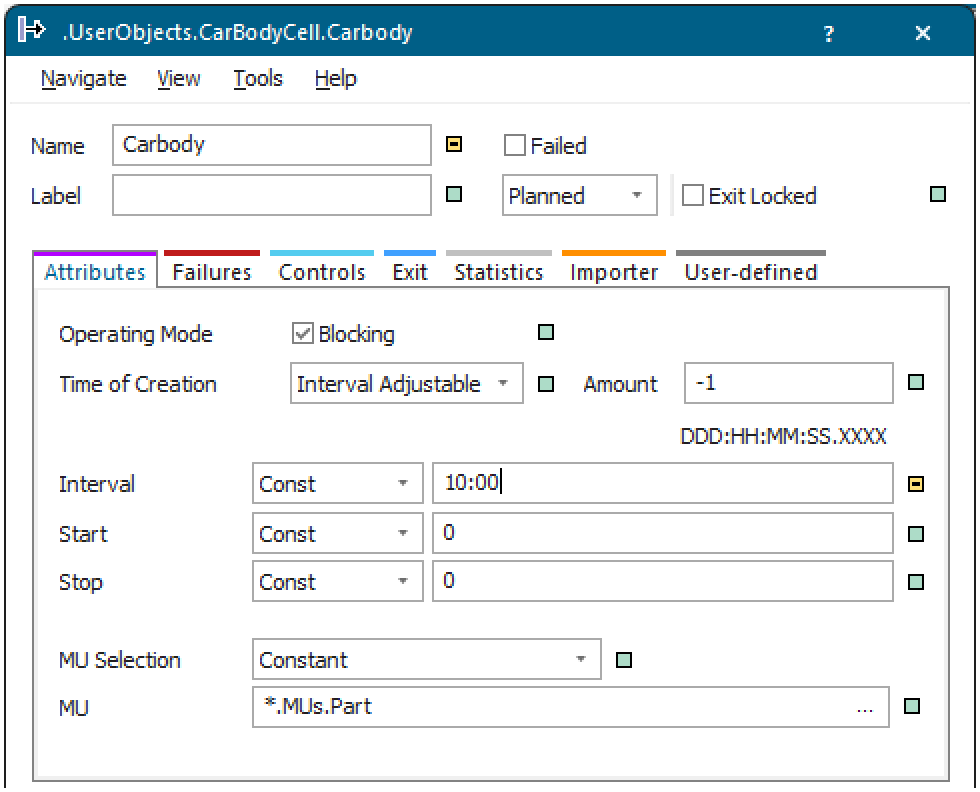

**Back to Model the Car Body Cell**

### Configure the Sources That Provide the Assembly Stations With Parts

You can configure the three *Sources* that provide the three *AssemblyStations* with parts.

Proceed as follows:

- For the *Source* named `CarBody` we entered a **constant Interval** of `10:00` minutes between the creation events.




- We use the *Sources* named `Trunk` and `Roof` with the default settings.

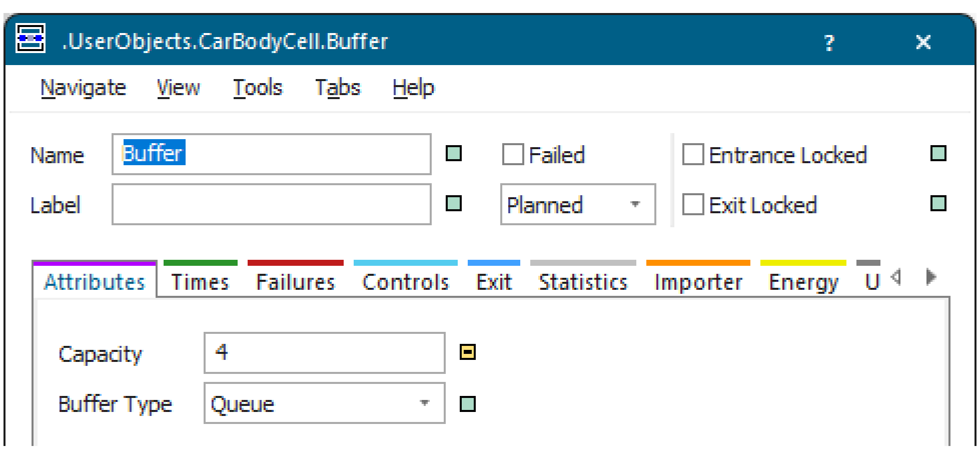
**Back to Model the Car Body Cell**

## Configure the Buffers between the Assembly Stations

Next, configure the two *Buffers*, named *Buffer* and *Buffer1*, which we inserted between the *AssemblyStations*.

Proceed as follows:

- Enter their **Capacity**. We typed in 4 each.



- We did not change any other settings.

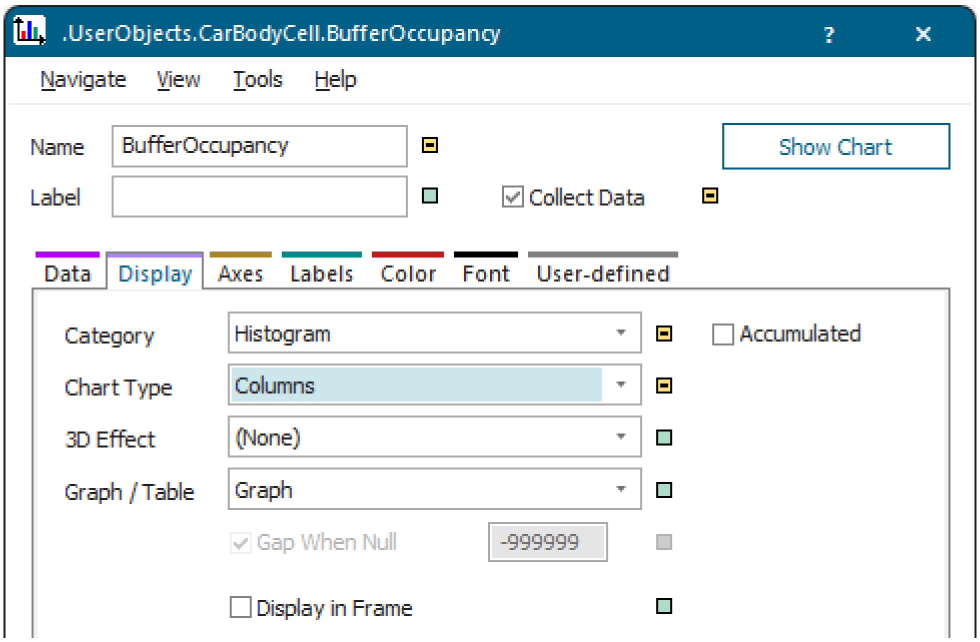
**Back to Model the Car Body Cell**

### Configure the Chart as Histogram

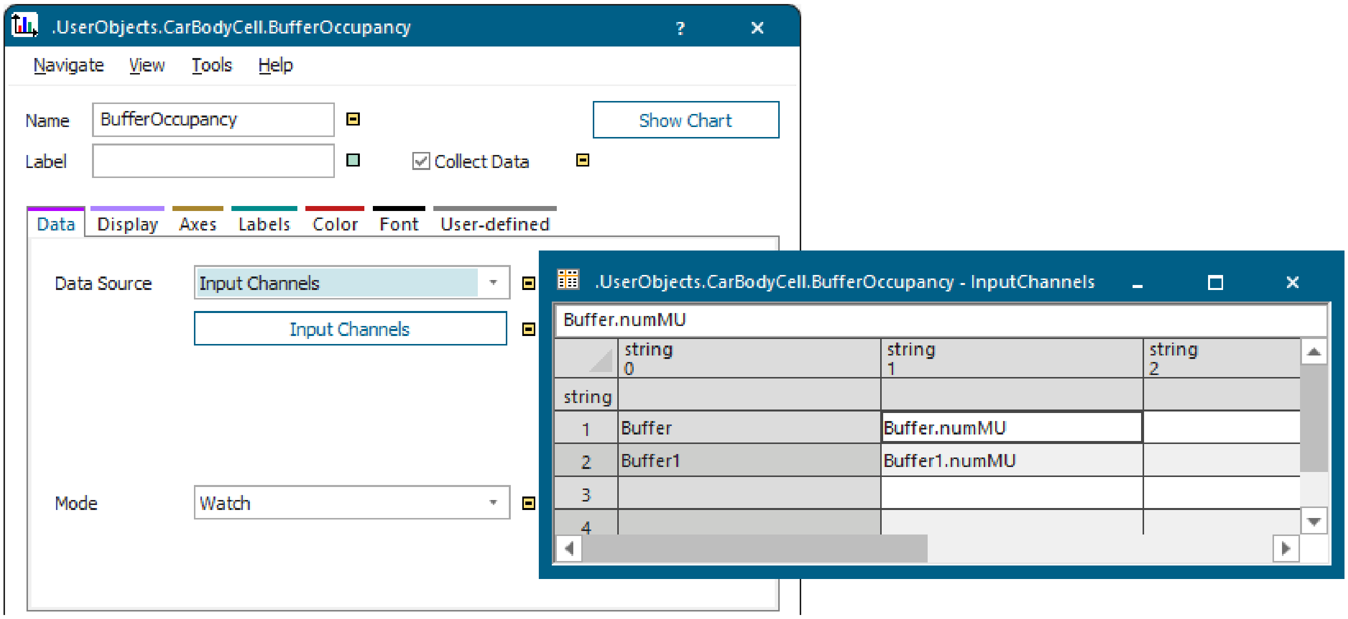
Double-click the icon of the *Chart*, configure our *Chart* named `BufferOccupancy` as a **histogram**.

Proceed as follows:

- Change to the tab **Display** and select the **Category > Histogram**.




- Drag the two *Buffers*, named *Buffer* and *Buffer1*, one after the other to the icon of the *Chart* and drop them there. This sets the **Data Source > Input Channels** and enters the *Buffers* into the table of the input channels.




We did not change any other settings.

**Back to Model the Car Body Cell**

## Configure the ReportObject of the Car Body Cell

When configuring the *ReportObject* of the `CarBodyCell` we have to decide which data we want to show.

We entered these instructions into the *Method*. The comments after the instructions explain what the code executes.

```
# [current.name]                // name of the Frame as heading on the first
level
The above car body cell looks like this: // descriptive text
## Layout                       // second level heading
><[current, *]                  // centered screenshot, adjusted to maximum
window width
## Buffer Occupancy             // second level heading
The occupancy of the individual Buffers looks like this: // descriptive text
><[BufferOccupancy, 150, 100]   // centered screenshot of the buffer
occupancy,
                                // size of the screenshots is 150 mm times
100 mm
[Buffer]                        // show statistics values of the object
'Buffer' as a table
[Buffer1]                       // show statistics values of the object
'Buffer1' as a table
## Statistics                   // second level heading
Statistics of the AssemblyStations and the Buffers looks like this: //
descriptive text
[AssemblyStation, "Assembly statistics"]    // show statistics values of
the object 'AssemblyStation' as
```

```
                                            // a table, "Assembly
statistics" is the caption of the table
[AssemblyStation1]            // show statistics values of the object
'AssemblyStation1' as a table
[AssemblyStation2]            // show statistics values of the object
'AssemblyStation2' as a table
[Buffer, "Buffer statistics ", %MatFlowProperties]    // show statistics
values of the object
                              // 'Buffer' as a table
                              // "Buffer statistics" is the caption of the
table
                              // % MatFlowProperties shows the material
flow properties
[Buffer1]                     // show statistics values of the object
'Buffer1' as a table
```

When we run the simulation after we finished modeling, *Plant Simulation* gathers the individual *HtmlReports* named *ReportObject* from all *sub-Frames* and shows them in the overall report of the car assembly model.

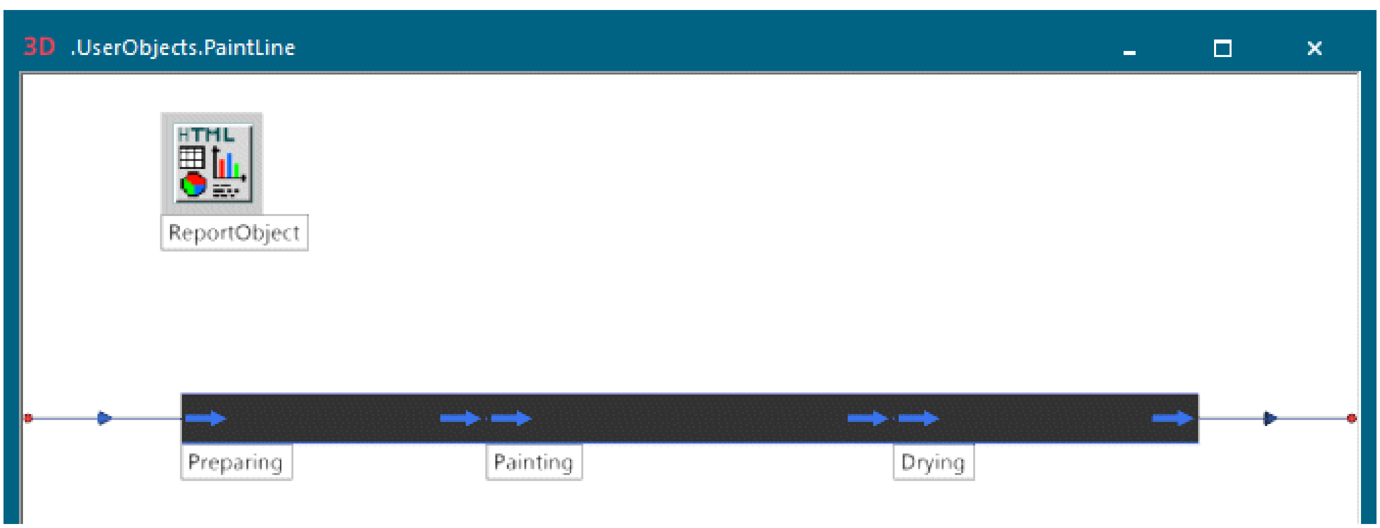
**Back to Model the Car Body Cell**

## Modeling the Paint Line

### Model the Paint Line

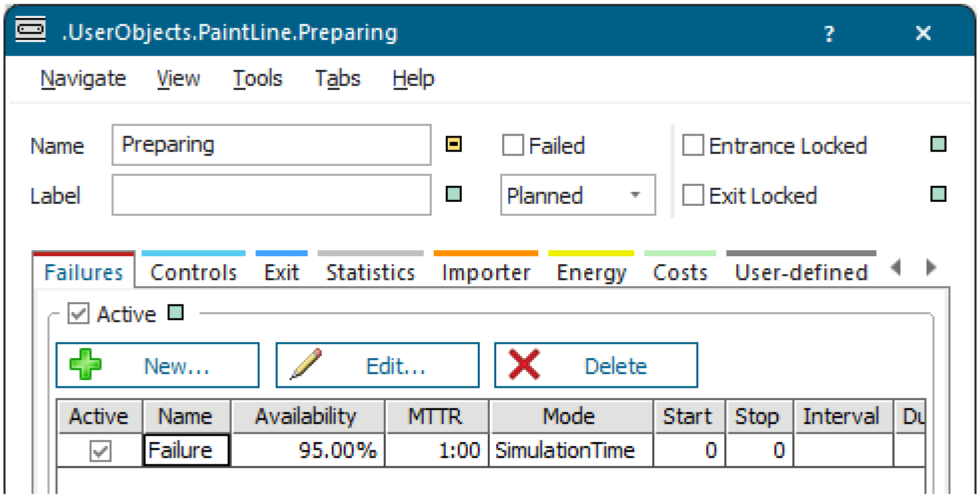
Our *PaintLine* consists of an *Interface/Entrance*, three *Conveyors* named `Preparing`, `Painting`, and `Drying`, and of an *Interface/Exit*.

We also inserted an *HtmlReport* and renamed it to `ReportObject`.




To configure the *Conveyor* named *Preparing*:

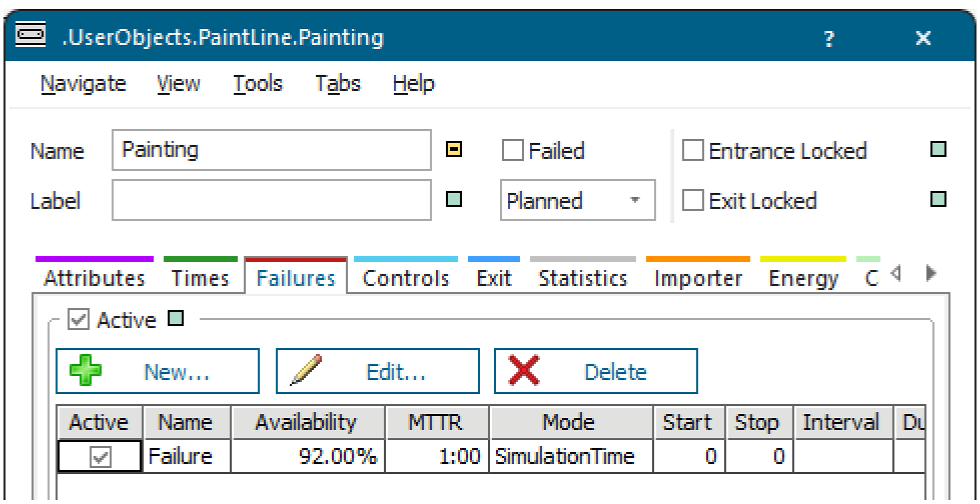
- Insert it with a **Length** of 6 meters.
- Create a **failure** with these default values:




We did not change any other settings.

To configure the *Conveyor* named *Painting*:

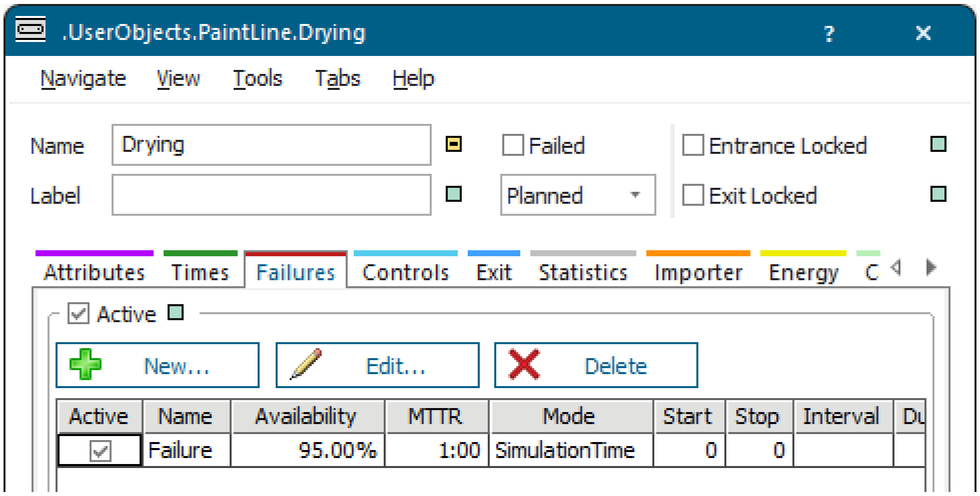
- Insert it with a **Length** of 8 meters.
- Create a **failure** with these values:




We did not change any other settings.

To configure the *Conveyor* named *Drying*:

- Insert it with a **Length** of 6 meters.
- Create a **failure** with these default values:




We did not change any other settings.

To define the *HtmlReport* named *ReportObject* of the *PaintLine*, we typed in these instructions:

```
# [current.name]
The above paint line of the car assembly looks like this:
## Layout
><[current, *]
## Statistics
Statistics of our paint lines look like this:
><[Painting, "Flow statistics", %MatFlowProperties]
[Preparing]
[Drying]
```

When we run the simulation after we finished modeling, *Plant Simulation* gathers the individual *HtmlReports* named *ReportObject* from all *subframes* and shows them in the overall report of the car assembly model.

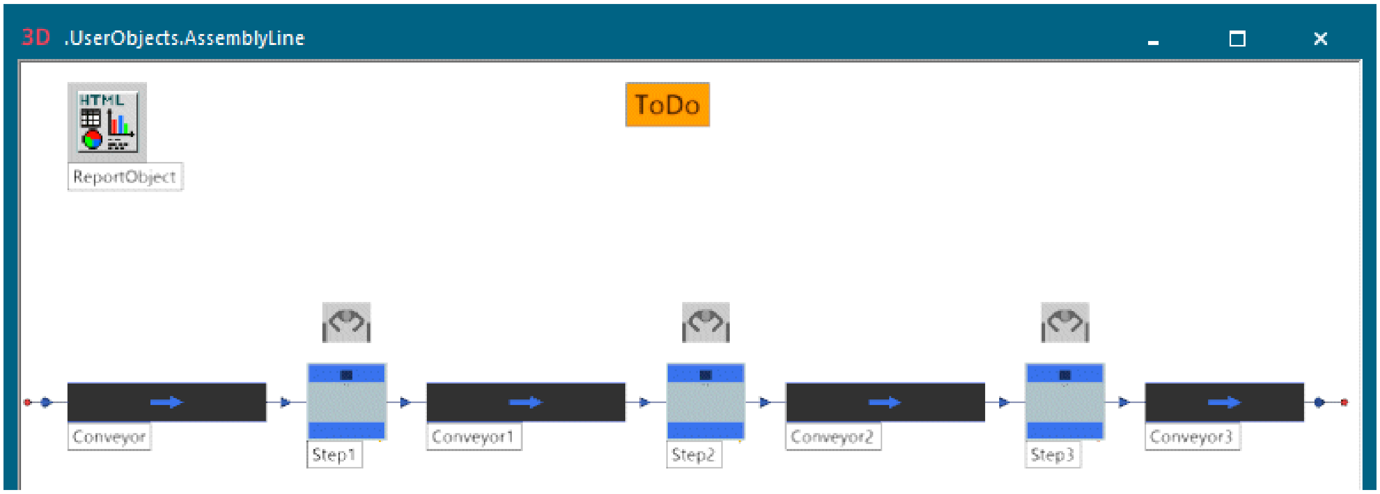
**Back to Show Reports in a Hierarchically Structured Model**

## Modeling the Assembly Line

### Model the Assembly Line

Our *AssemblyLine* consists of an *Interface/Entrance*, four *Conveyors*, three *Stations* named `Step1`, `Step2`, and `Step3`, and of an *Interface/Exit*.

We attached a *Workplace* each to each of the *Stations*. We also inserted an *HtmlReport* which we renamed to `ReportObject` and our *Comment* named *ToDo*.




We will demonstrate how to:

- Configure the Conveyors of the Assembly Line
- Configure the Processing Stations and the Workplaces
- Configure the ReportObject of the Assembly Line
- Configure the ToDos of the Assembly Line

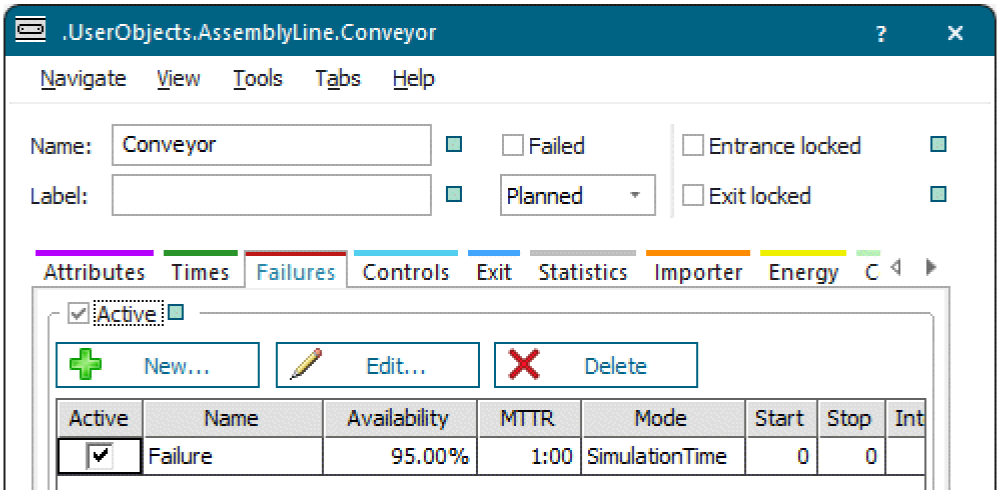
**Back to** Show Reports in a Hierarchically Structured Model

### Configure the Conveyors of the Assembly Line

Configure the four *Conveyors* between the *processing stations*.

Proceed as follows:

- Insert the first three *Conveyors* with a length of 5 meters each.
- For the fourth *Conveyor*, named *Conveyor3*, we typed in a length of 4 meters.
- Create a **failure** each for the *Conveyors* with these default settings:




We did not change any other settings.

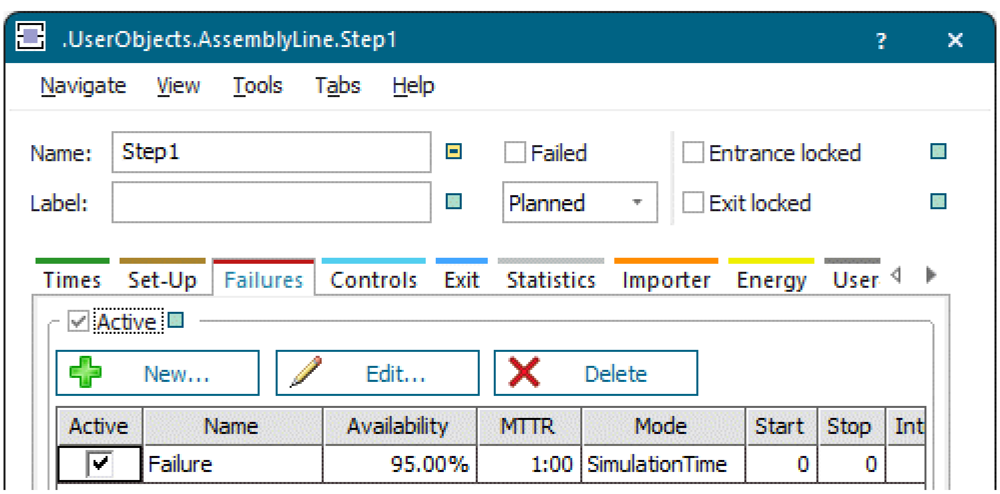
**Back to Model the Assembly Line**

## Configure the Processing Stations and the Workplaces

Configure the three processing *Stations* named *Step1*, *Step2*, and *Step3*.

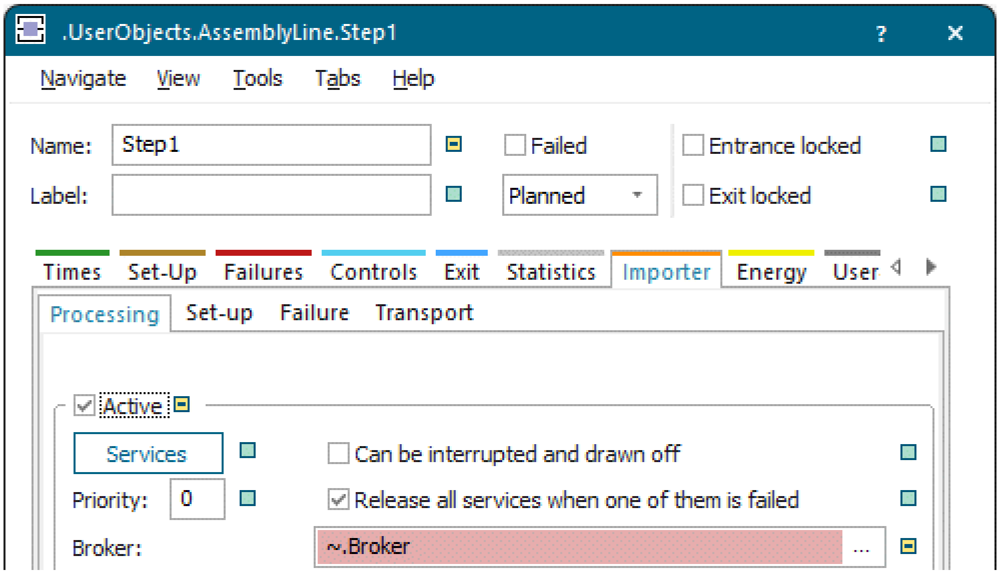
Proceed as follows:

- Create a **failure** each with these default values:




- Change to the tab **Importer** and activate the **processing importer**.

  Type `~.Broker` into the text box **Broker**. This points to the location of the *Broker* which controls the *Importers*. In our case this is the *Frame*, in which we are going to model the *FinalAssembly* in our overall model.




We did not change any other settings.

- Insert three *Workplaces* and attach one each to one of the *Stations*. We use them with their default settings.

**Compare Model the Assembly Line**

**Back to Configure the Final Assembly**

## Configure the ReportObject of the Assembly Line

Configure the *HtmlReport* named *ReportObject* of the *AssemblyLine* in a *Method*.

We typed in these instructions. The comments describe what the source code does.

```
# [current.name]               // name of the Frame as first level heading
The above assembly line of the car assembly looks like this:    //
descriptive text
## Layout                      // second level heading
><[current, *]    // centered screenshot, adjusted to maximum width of the
window
## Statistics     // second level heading
Statistics of the stations Step1, Step2, and Step3              //
descriptive text
[Step1, "Station statistics"]  // show statistics values of the object
'Step1' as a table
                               // "Station statistics" is the caption of
the table
[Step2]                        // show statistics values of the object
'Step2' as a table
[Step3]                        // show statistics values of the object
'Step3' as a table
```

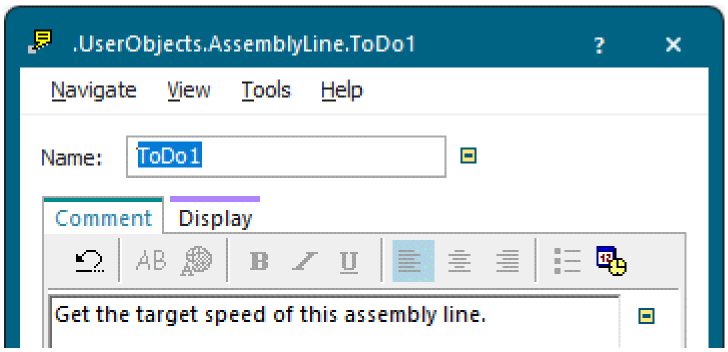
**Back to Model the Assembly Line**

### Configure the ToDos of the Assembly Line

Configure the *Comment* of the *AssemblyLine* named *ToDo*.

We typed in these instructions on the tab **Comment**:

```
Get the target speed of this AssemblyLine.
```




When we run the simulation after we finished modeling, *Plant Simulation* gathers the individual *Comments* named *ToDo* from all *sub-Frames* and shows them in the overall report of the car assembly model.

To do so, we entered these instructions into the overall *HtmlReport* of the car assembly model, i.e., in **.Models.MyCarAssembly.HtmlReport**:

```
# ToDo
These tasks still have to be completed:
[.UserObjects.ToDo*]
```

**Back to Model the Assembly Line**

### Modeling and Configuring the Installation

### Model the Installation and Configure the Components

Now that we created the classes of our *CarBodyCell*, of the *PaintShop*, and of the *AssemblyLine*, we can build our simulation model.

We create our car body assembly model in the *Frame* named **Model** intended for that in the folder *Models* in the *Class Library*. If you want to, you can also rename the *Frame* **Model**. We entered `MyCarAssembly`.

Our car assembly consist of these objects:

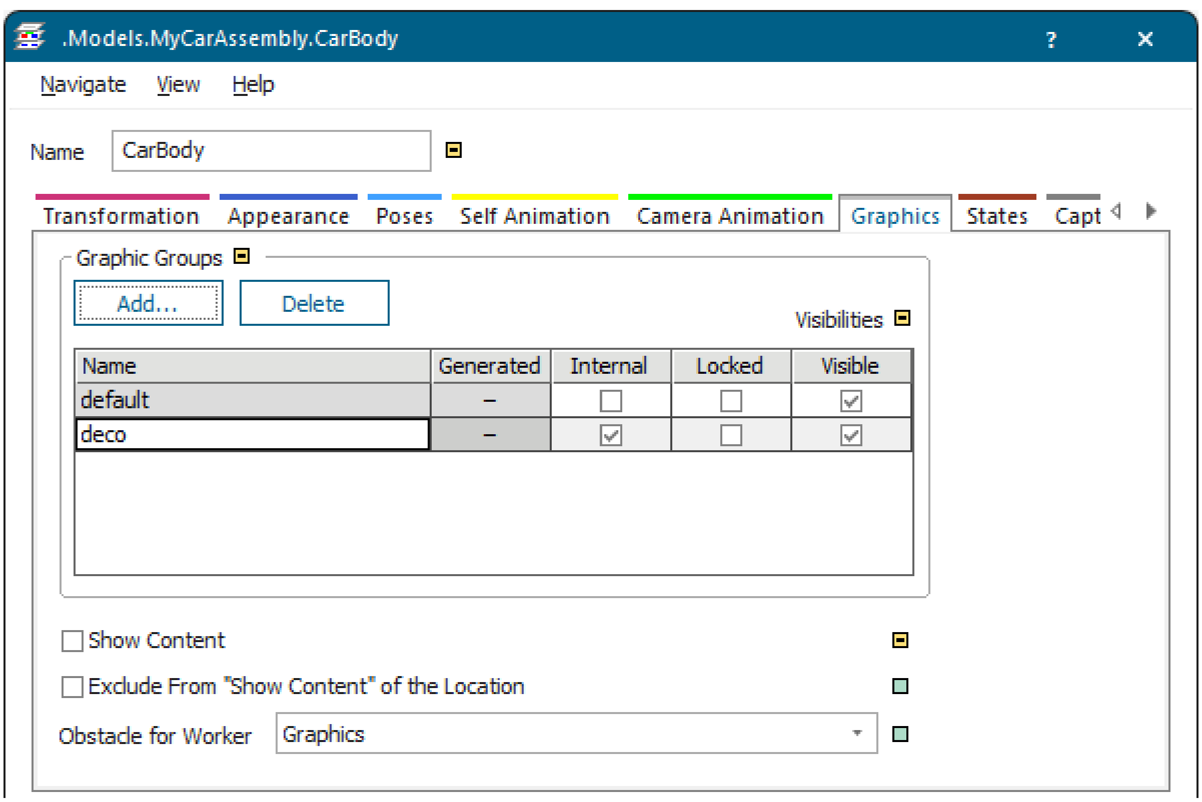
- A *Source*, representing the receiving department, and the *PartsTable*, defining the parts to be produced.
- An *HtmlReport* showing the collected data of the individual components of the car assembly.
- Three *Frames* representing the *CarBody*, the *PaintShop*, and the *FinalAssembly* departments of the assembly plant.

  Insert a *Frame* from the tab **MaterialFlow** into the *Frame MyCarAssembly*.

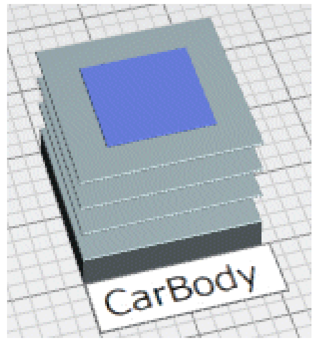
  By default *Plant Simulation* inserts a **Glass-Box-Frame** 

that shows its content toward the outside in the *Frame MyCarAssembly*. We want to show our *Frames* as a **Black-Box-Frames** though, which do not show their content in the *Frame MyCarAssembly*. For this reason we have to convert the **Glass-Box-Frame** to a **Black-Box-Frame**:

  - Click the *Frame* with the left mouse button and press the **spacebar**. Change to the tab **Graphics**.
  - Clear the check box **Show Content**.
  - Select the check box **Visible** for the graphic group named **default**.
  - Do this for each of the three *Frames*.
  - Rename our three *Frames*. We changed the names to `CarBody`, `PaintShop`, and `FinalAssembly`.




  The inserted **Black-Box-Frame** now looks like this.




In the next step we will:

- Configure the Receiving Department
- Configure the Body Assembly
- Configure the Paint Shop
- Configure the Final Assembly
- Configure the HtmlReport

**Back to** Show Reports in a Hierarchically Structured Model

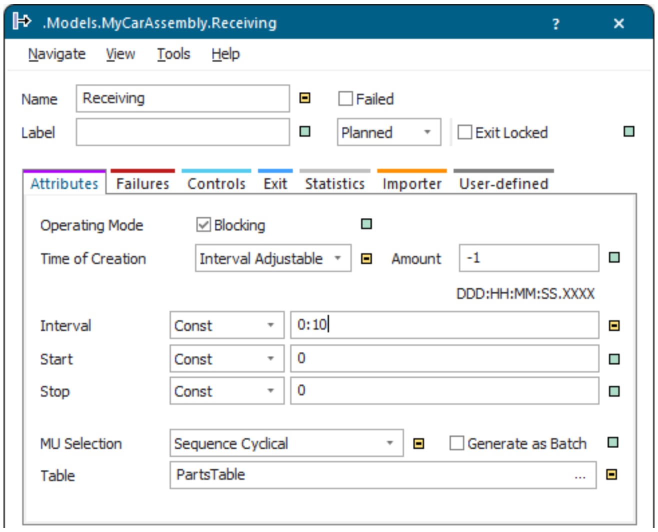
## Configure the Receiving Department

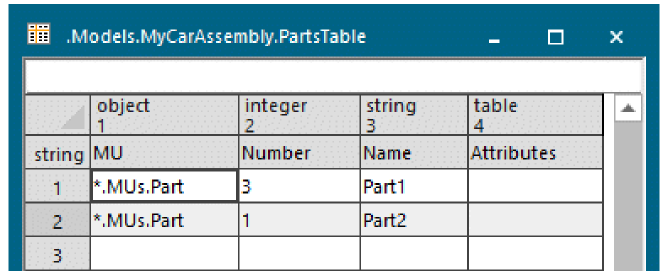
We model and configure the receiving department with a *Source*.

Proceed as follows:

- Insert a *Source* from the tab **MaterialFlow**. Rename it to `Receiving`.
- Select the **Mu Selection** > Sequence Cyclical.
- As the setting **Sequence Cyclical** requires a table, insert an object of type *DataTable*. We renamed it to `PartsTable` and entered the *PartsTable* into the text box **Table**.

  *Plant Simulation* automatically formats the *PartsTable*. We entered the data shown below.




We did not change any other settings.

**Back to Model the Installation and Configure the Components**

## Configure the Body Assembly

We model the *CarBody* section of our car assembly plant in a *Frame*, which we insert from the tab **MaterialFlow** into the *Frame MyCarAssembly*.

- Convert the **Glass-Box-Frame** to a **Black-Box-Frame** as described under **Model the Installation and Configure the Components**.
- Click the Frame *CarBody* with the right mouse button and select **Open in New Window**.
- Delete the default graphic.
- Insert an *Interface*.
- Insert our *CarBodyCell* twice, one above the other.
- Insert a second *Interface*.
- Connect the Interfaces along the material flow with *Connectors* with the *CarBodyCells*. *Plant Simulation* then renames the *Interfaces* to **Entrance** and **Exit**.
- Insert an *HtmlReport*.
  - Rename it to `ReportObject`.
  - To configure the *ReportObject*, type in these instructions:

```
# [current.name]
The car body section of our car assembly looks like this:
## Layout
><[current, *]
## CarBodyCells
```

```
The car body section encompasses several CarBodyCells:
[CarBodyCell.Origin*, ##]  // shows the heading two levels lower
[EnergyAnalyzer, #]        // shows the heading one level lower
```

When we run the simulation after we finished modeling, *Plant Simulation* gathers the individual *HtmlReports* named *ReportObject* from all *sub-Frames* and shows them in the overall report of the car assembly model.

- Insert our *Comment* named *ToDo* from the tab **UserObjects**.

  Type in this comment on the tab **Comment**:

  ```
  Question: Would adding a third CarBodyCell increase the throughput?
  ```

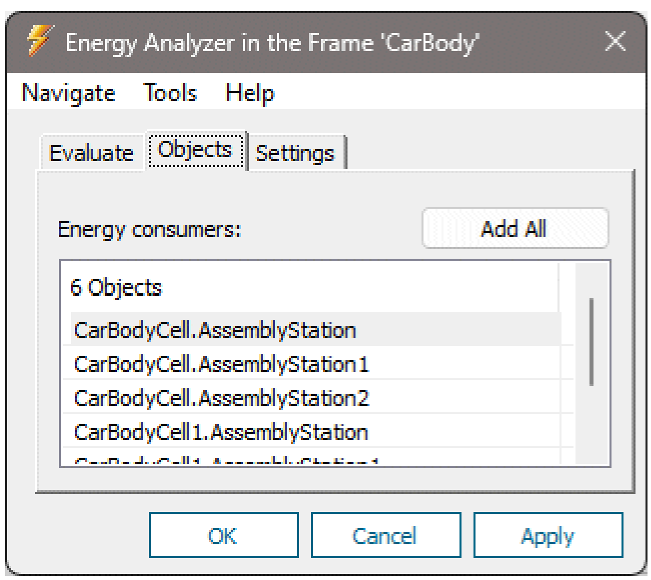
  When we run the simulation after we finished modeling, *Plant Simulation* gathers the individual *Comments* named *ToDo* from all *sub-Frames* and shows them in the overall report of the car assembly model.

- Insert the object *EnergyAnalyzer* from the tab **Tools**.

  Configure the *EnergyAnalyzer* for the *CarBodyCell*:

  - Click the *Frame CarBodyCell* with the right mouse button and select **Open in New Window**.
  - This opens the window with the path **.Models.MyCarAssembly.CarBody.CarBodyCell**. Hold down **Ctrl** and select the objects *AssemblyStation*, *AssemblyStation1*, and *AssemblyStation2*. Drag these objects to the object *EnergyAnalyzer* in the path **.Models.MyCarAssembly.CarBody** and drop them there.
  - Repeat this for *CarBodyCell1*.

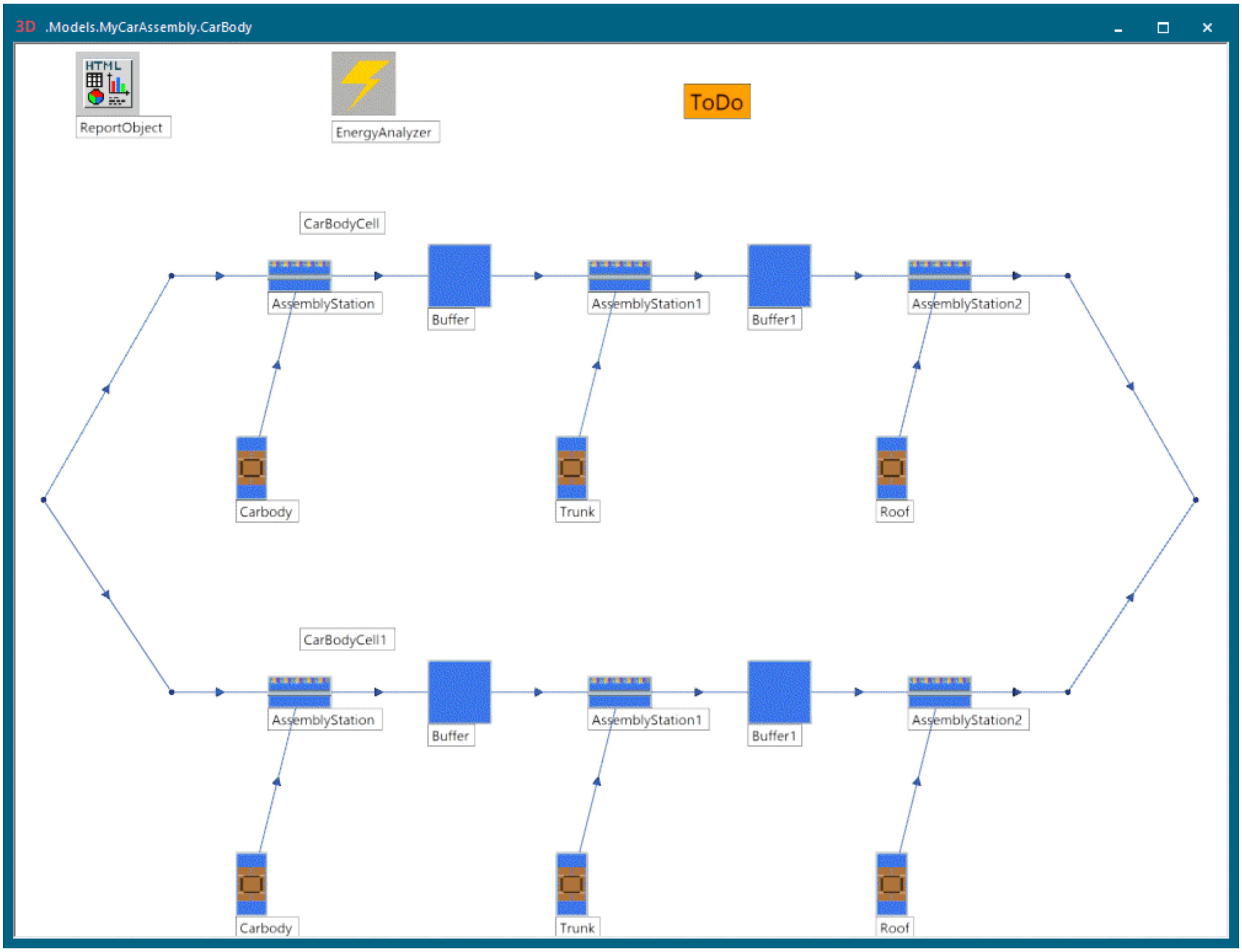
    The dialog of the *EnergyAnalyzer* then looks like this:




    We did not change any other settings.

Our *ReportObject* then shows the energy consumption values of the *AssemblyStations* after the simulation.

Our finished car body section with the two *CarBodyCells* looks like this:




**Go to Study the HtmlReport > Display the Energy Analysis**

**Back to Model the Installation and Configure the Components**

## Configure the Paint Shop

We model the *PaintShop* section of our car assembly plant in a *Frame*, which we insert from the tab **MaterialFlow** into the *Frame MyCarAssembly*.

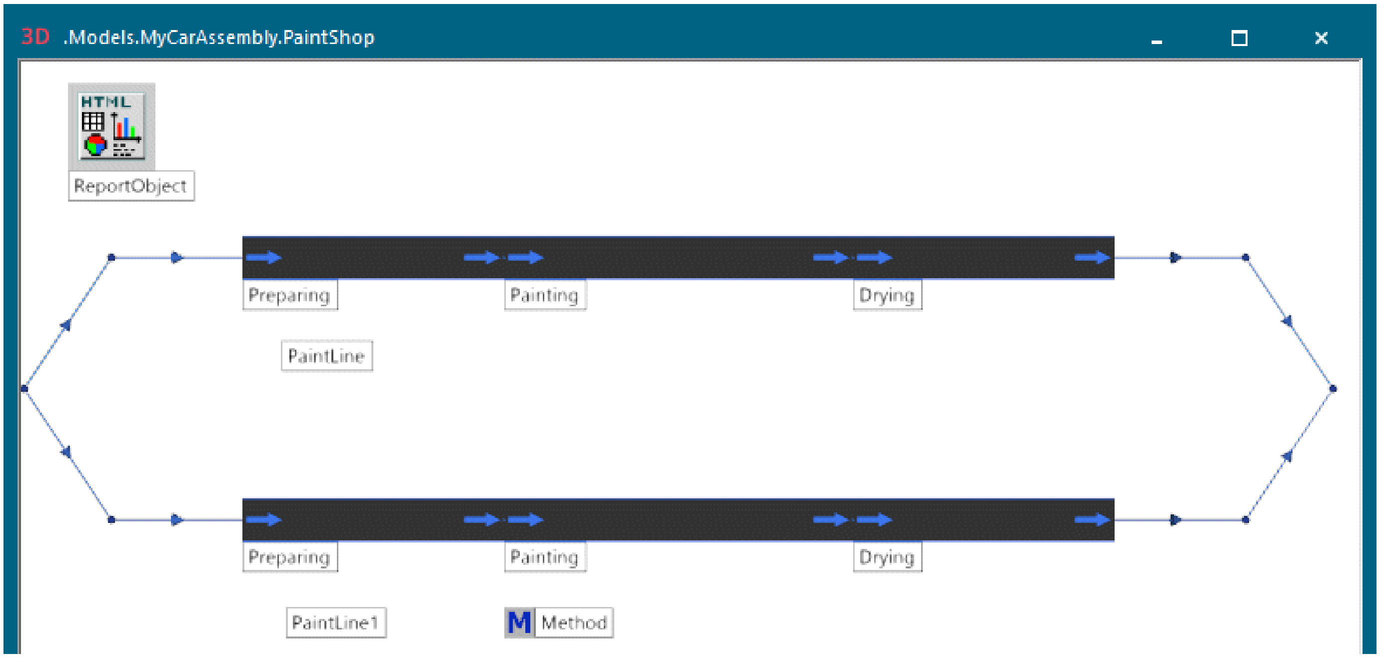
Proceed as follows:

- Convert the **Glass-Box-Frame** to a **Black-Box-Frame** as described under **Model the Installation and Configure the Components**.
- Click the *Frame PaintShop* with the right mouse button and select **Open in New Window**.
- Delete the default graphic.
- Insert an *Interface*.
- Insert our *PaintShop* twice, one above the other.
- Insert a second *Interface*.
- Connect the *Interfaces* along the material flow with *Connectors* with the *paint lines*. *Plant Simulation* then renames the *Interfaces* to **Entrance** and **Exit**.
- Insert an *HtmlReport*.
    - Rename it to `ReportObject`.
    - To configure the *ReportObject*, type in these instructions:

    ```
    # [current.name]
    The PaintShop section of the car assembly looks like this:
    ## Layout
    ><[current, *]
    ## PaintLines
    The PaintShop encompasses several PaintLines.
    [PaintLine.Origin*, ##]
    ```

    When we run the simulation after we finished modeling, *Plant Simulation* gathers the individual *HtmlReports* named *ReportObject* from all *sub-Frames* and shows them in the overall report of the car assembly model.

Our finished *PaintShop* section with the two *PaintLines* looks like this:



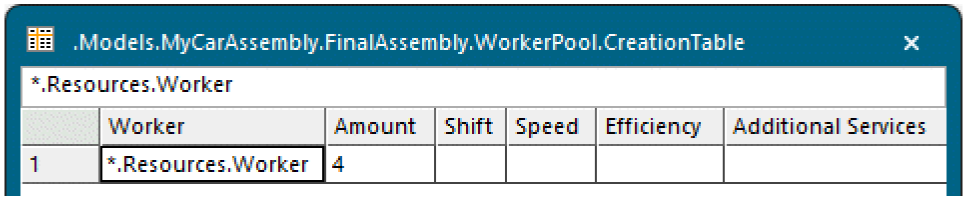

**Back to Model the Installation and Configure the Components**

## Configure the Final Assembly

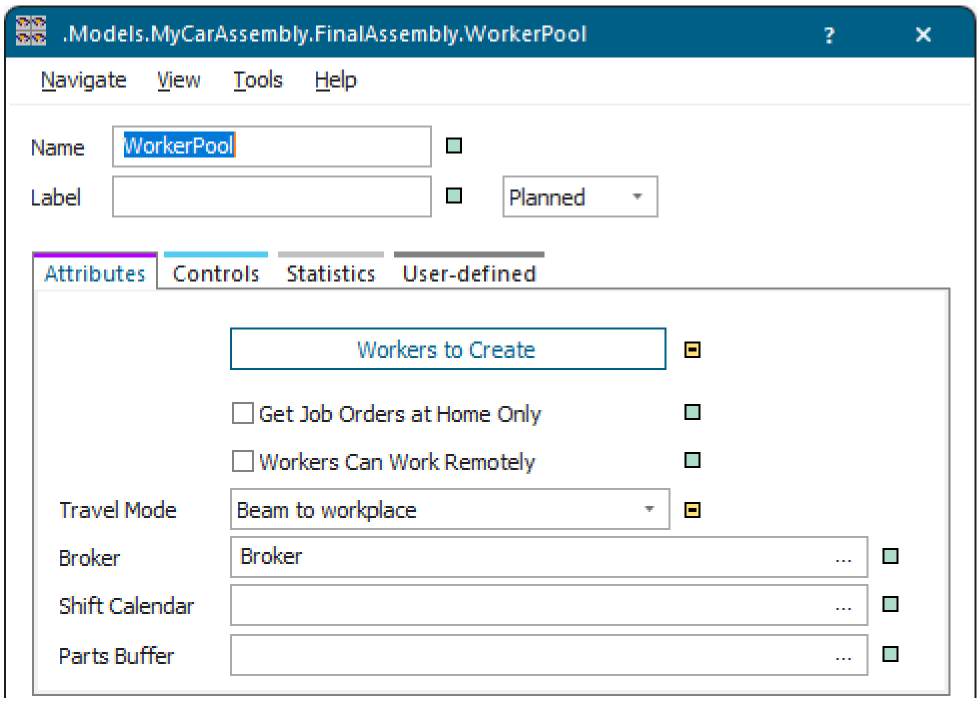
We model the *FinalAssembly* section of our car assembly plant in a *Frame*, which we insert from the tab **MaterialFlow** into the *Frame MyCarAssembly*.

Proceed as follows:

- Convert the **Glass-Box-Frame** to a **Black-Box-Frame** as described under **Model the Installation and Configure the Components**.
- Click the *Frame FinalAssembly* with the right mouse button and select **Open in New Window**.
- Delete the default graphic.
- Insert an *Interface*.
- Insert our *AssemblyLine* three times, one above the other.
- Insert a second *Interface*.
- Connect the *Interfaces* along the material flow with *Connectors* with the *assembly lines*. *Plant Simulation* then renames the *Interfaces* to **Entrance** and **Exit**.
- Insert the object *WorkerPool* from the tab **Resources**. Configure the *WorkerPool*:
  - Click the **Inheritance** check box for the **Creation Table** so that it looks like this . Then type 4 into the cell below **Amount** to create four *Workers*.




- Select the **Travel Mode [drop-down list] > Beam to Workplace**.




We did not change any other settings.

- Insert the *Broker* from the tab **Resources**. We use the *Broker* with its default settings.
- Insert the object *WorkerChart* from the tab **Tools**.

  Type `WorkerPool` into the text box below **Create charts of this WorkerPool**.
- Insert an *HtmlReport*.
  - Rename it to *ReportObject*.
  - To configure the *ReportObject*, type in these instructions:

```
# [current.name]
The final assembly section of the car assembly looks like this:
## Layout
><[current, *]
## Broker Statistics
The Broker collected these statistics values:
[Broker]
## Worker Chart
```

```
The worker chart shows statistics of all Workers that are staying in
the WorkerPool.
[WorkerChart]
## Assembly Lines
The final assembly section encompasses several assembly lines.
[AssemblyLine.Origin*, ##]
```

When we run the simulation after we finished modeling, *Plant Simulation* gathers the individual *HtmlReports* named *ReportObject* from all *sub-Frames* and shows them in the overall report of the car assembly model.

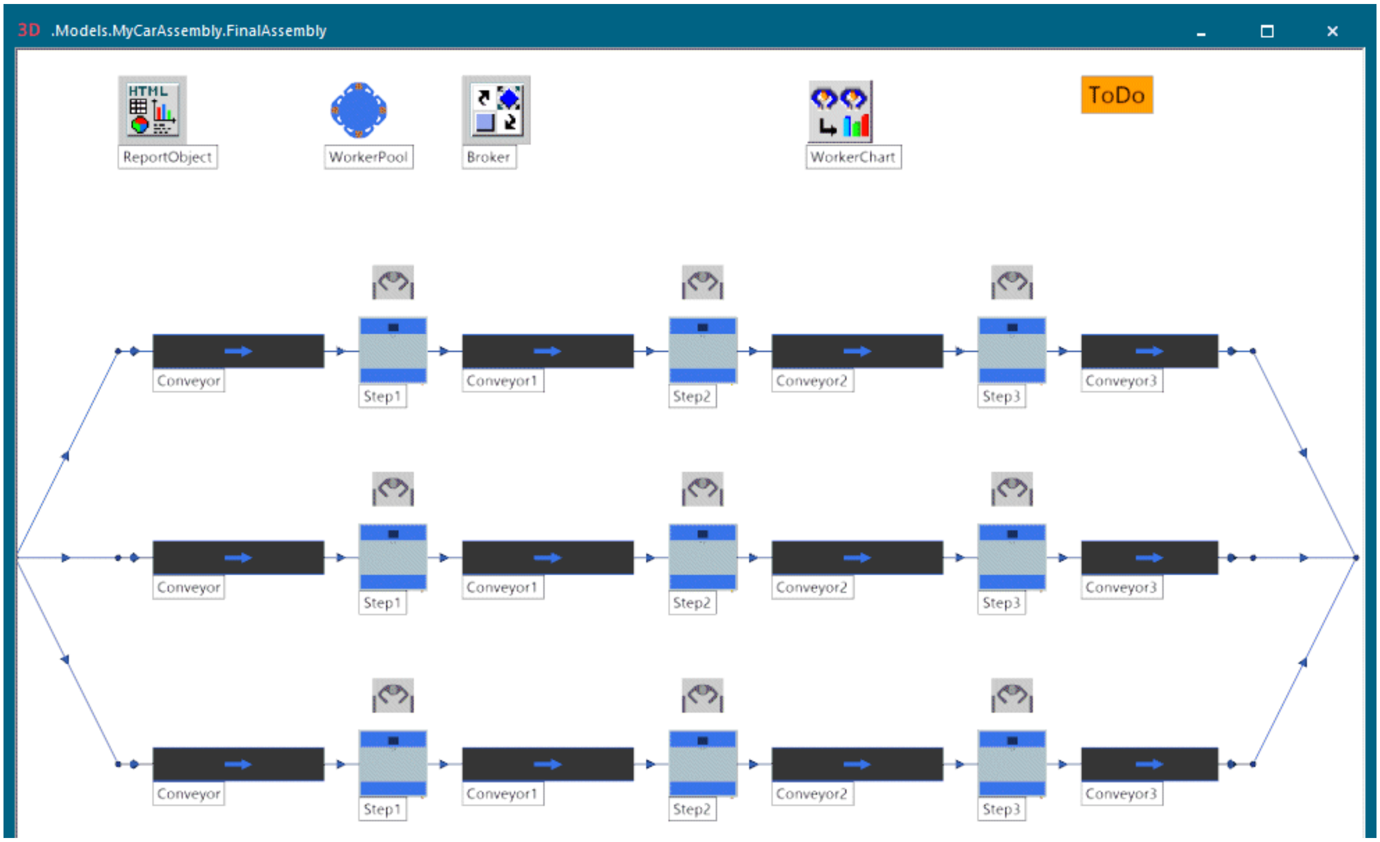
> **Note**
>
> We created the model in a previous version of *Plant Simulation* which provided the now deprecated *WorkerChart*. We can still use it in the model of the current version though as it is part of that model.

- Insert our *Comment* named *ToDo*.

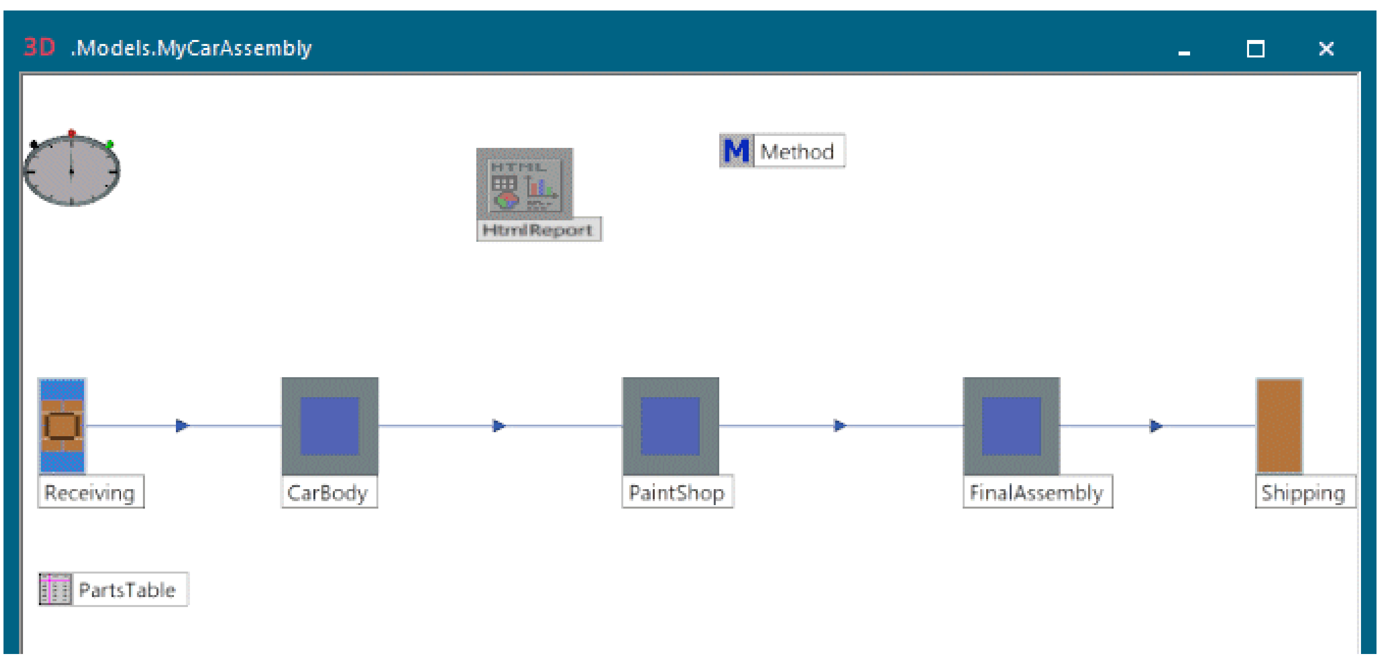
  Type in this comment on the tab **Comment**:

  ```
  Check the availability of AssemblyLine2.
  ```

Our finished *FinalAssembly* with the three assembly lines and the other objects looks like this:




Connect the individual sections in the *Frame MyCarAssembly* with *Connectors*. Our finished car assembly looks like this:




**Back to** **Model the Installation and Configure the Components**

## Configure the HtmlReport

Configure the *HtmlReport* of the car assembly plant as a whole.

We typed in these instructions. We took the **General Information** topic for the most part from the built-in *HtmlReport*.

```
><[!self, Header, *]
# General Information
## Overview
* Model file: [=modelFile]
* Simulation root: [self.location]
* Simulation time: [EventController.simtime]
* End time: [EventController.EndTime]
* Generated on: [=sysdate]
* Contact: <a href="mailto:John.Doe@siemens.com?subject=Simulation
report">John.Doe@siemens.com</a>
## Model
The car assembly model looks like this:
><[current, *]
## Receiving
The receiving section delivers these parts:
><[PartsTable, 1..3]
--
# Statistics
Drain statistics data suffices for our purposes.
## Drain Statistics
Drain statistics shows detailed information about the parts that the
shipping section removed from the plant.
[.MaterialFlow.Drain*]
--
# Production Areas
The car assembly encompasses three production areas:
* CarBody
* PaintShop
* FinalAssembly
[CarBody, #]
[PaintShop, #]
[FinalAssembly, #]
--
# ToDo
These tasks still have to be completed:
[.UserObjects.ToDo*]
Cedar Rapids, [=day(sysdate)].[=month(sysdate)].[=year(sysdate)+1900]
<span style="font-family: Segoe Script; font-size:32pt">>H. Thompson</a>
```

When we run the simulation after we finished modeling, *Plant Simulation* collects the individual *HtmlReports* named *ReportObject* from all *sub-Frames* and shows them in the overall report of the car assembly model.

**Back to Model the Installation and Configure the Components**

# Running the Simulation and Showing the Report

## Run the Simulation and Show the Report

After we finished modeling, we can run the simulation and show its results in the report.

In this step we will:

- Run the Simulation
- Show the HtmlReport

**Compare** Study the HtmlReport

**Compare** Configure the HtmlReport

**Back to** Show Reports in a Hierarchically Structured Model

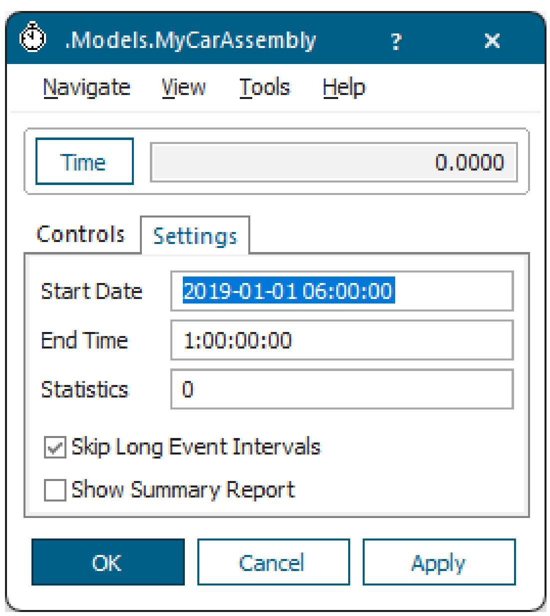
## Run the Simulation

After we finished creating our simulation model, we can run the simulation.

Proceed as follows:

- Double-click the *EventController* in the model *MyCarAssembly*, change to the tab **Settings** and type in the time after which the simulation will be terminated.

  We entered one day, i.e., `1:00:00:00`.




- Change to the tab **Controls** and click  to run the simulation.

**Back to Run the Simulation and Show the Report**

## Show the HtmlReport

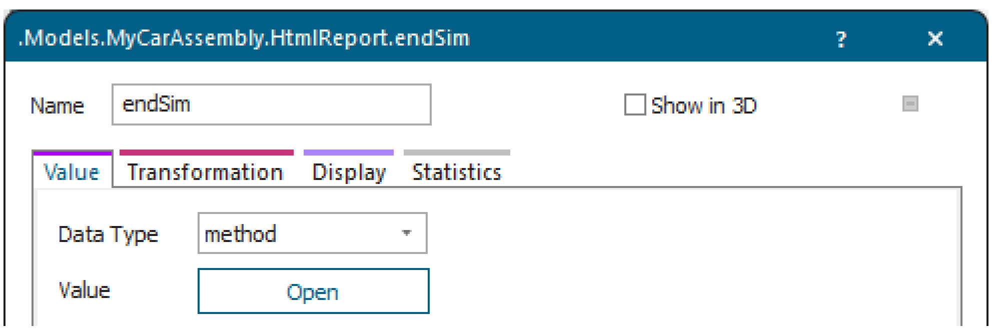
After the simulation has been terminated, we can click the *HtmlReport* with the right mouse button and click **Show** to display it.

During the simulation *Plant Simulation* collects the individual *HtmlReports* named *ReportObject* from all *sub-Frames* and adds these to the overall report of the car assembly model.

The same applies to the individual *Comments* named *ToDo*.

If we want to immediately show the *HtmlReport* after the simulation is over, we can set this in an *endSim method*:

- Open the *HtmlReport* in the model *MyCarAssembly* and change to the tab **User-defined**.
- Click **New** and create a user-defined attribute of **Data Type > Method**.
- Type in `endSim` as name.




- Click **Open** and type in this source code:

```
self.~.show
```

> **Note**
> When the 3D model contains a large number of objects that the *HtmlReport* displays, it can take some time before *Plant Simulation* shows it as it has to render the scene first before taking the screenshots for the *HtmlReport*.

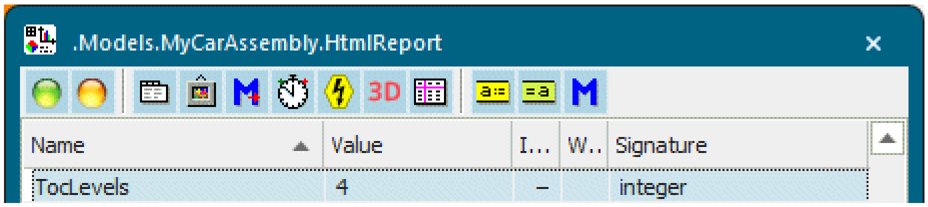
**Back to Run the Simulation and Show the Report**

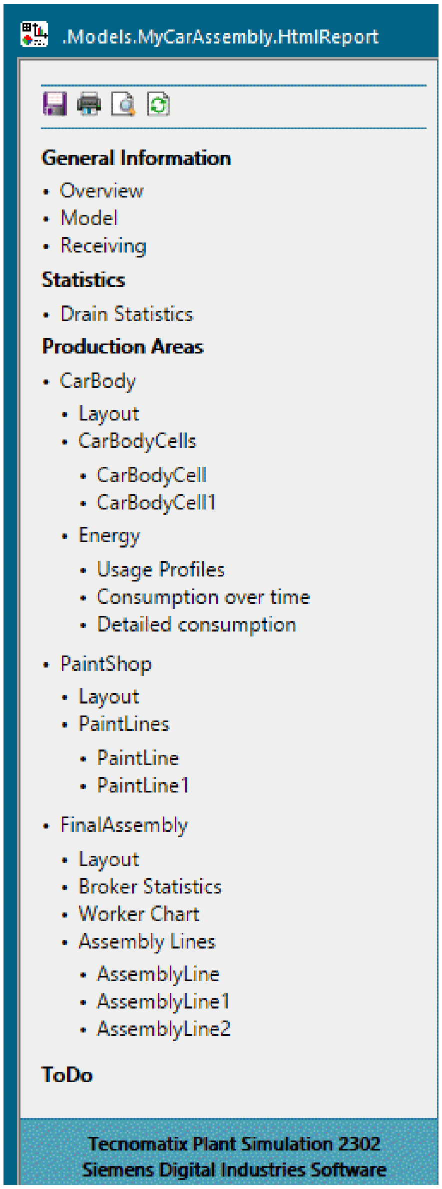
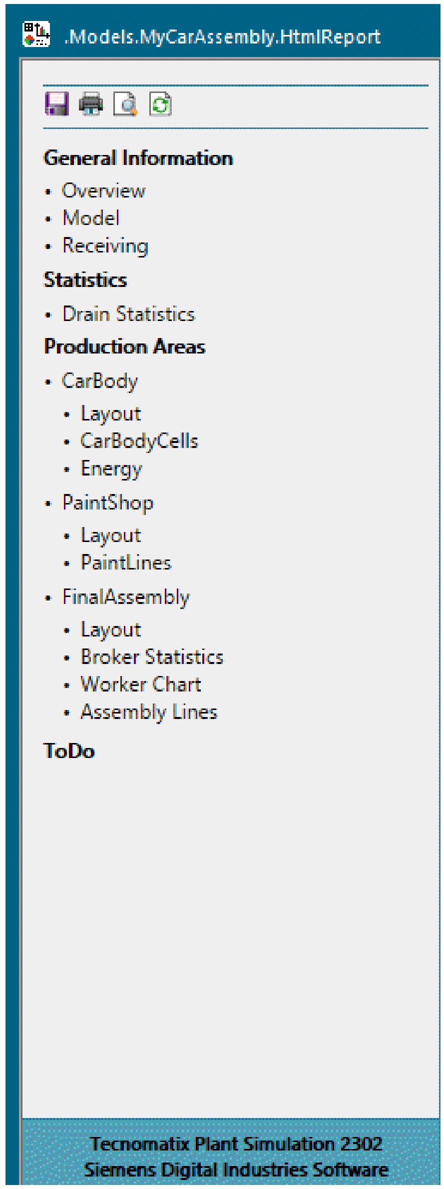
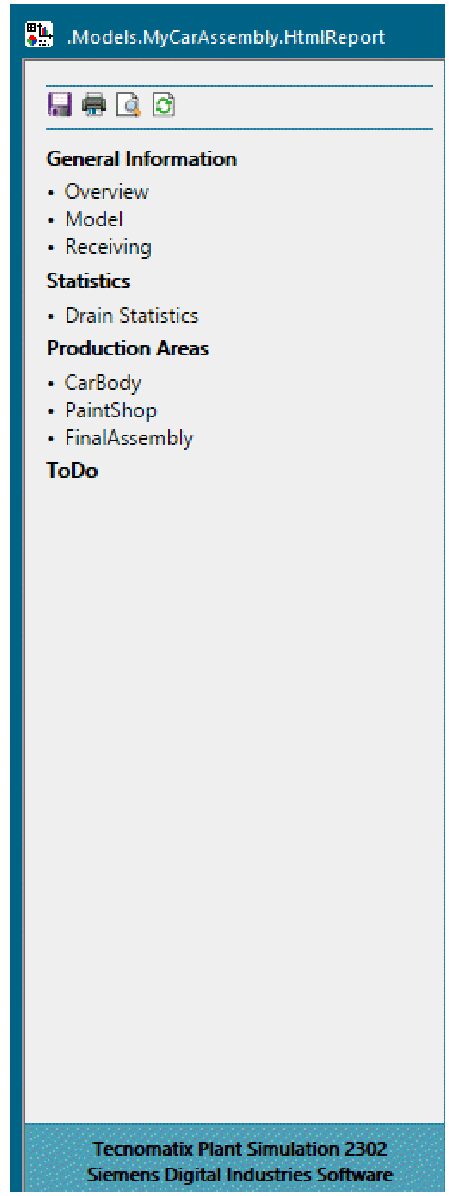
## Studying the HtmlReport

### Study the HtmlReport

When you open the *HtmlReport*, you will notice that the table of contents shows four heading levels by default.

If you want to show fewer heading levels, you can set that with the attribute **TOCLevels**.



| 4 Heading Levels | 3 Heading Levels | 2 Heading Levels |
|---|---|---|
|  |  |  |







We took the topics **General Information** and **Statistics** for the most part from the built-in *HtmlReport*.

In addition we want to:

- **Display the Production Areas**
- **Display the Energy Analysis**
- **Display the ToDos and Change to the Respective Frame**

**Compare Model the Car Body Cell > Configure the ReportObject**

**Compare Configure the Body Assembly > Configure the ReportObject**

**Compare Model the Paint Line > Configure the ReportObject**

**Compare Configure the Paint Shop > Configure the ReportObject**

**Compare Model the Assembly Line > Configure the ReportObject**

**Compare Configure the Final Assembly > Configure the ReportObject**

**Compare Configure the HtmlReport**

**Back to Show Reports in a Hierarchically Structured Model**

### Display the Production Areas

Define how to display the three production areas—*CarBody*, *PaintShop*, and *FinalAssembly*—as second level headings below the first level heading **Production Areas** in our *HtmlReport* in the *Frame MyCarAssembly*.

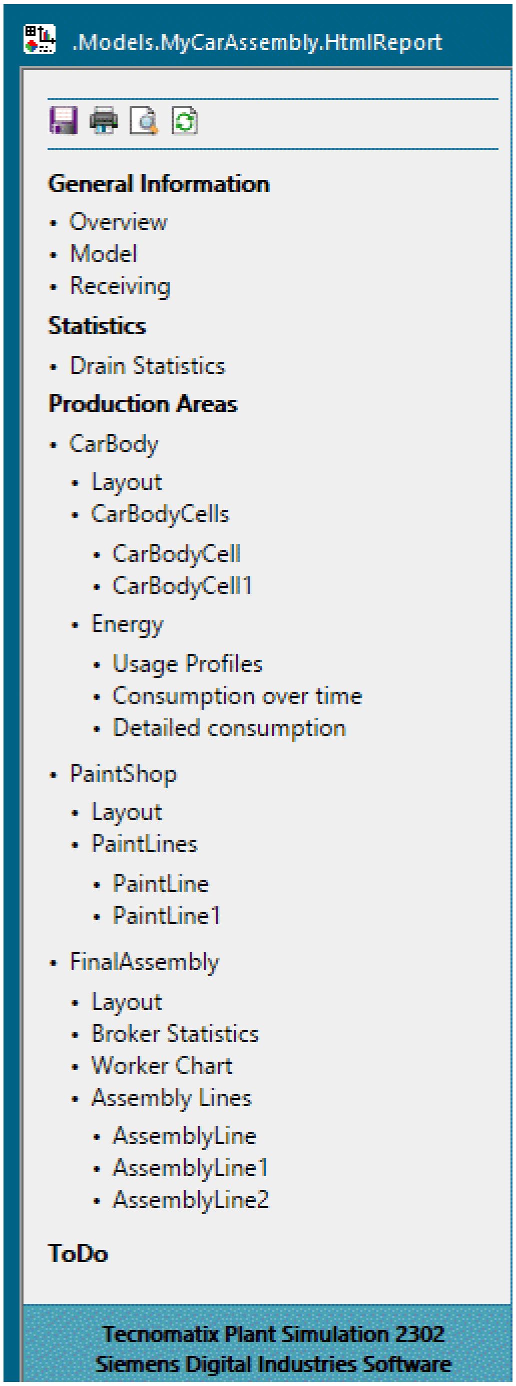
To do so, we typed in the instructions below. The instructions in the *HtmlReport* in the *Frame MyCarAssembly* define how the overall report of the three production areas will be displayed.

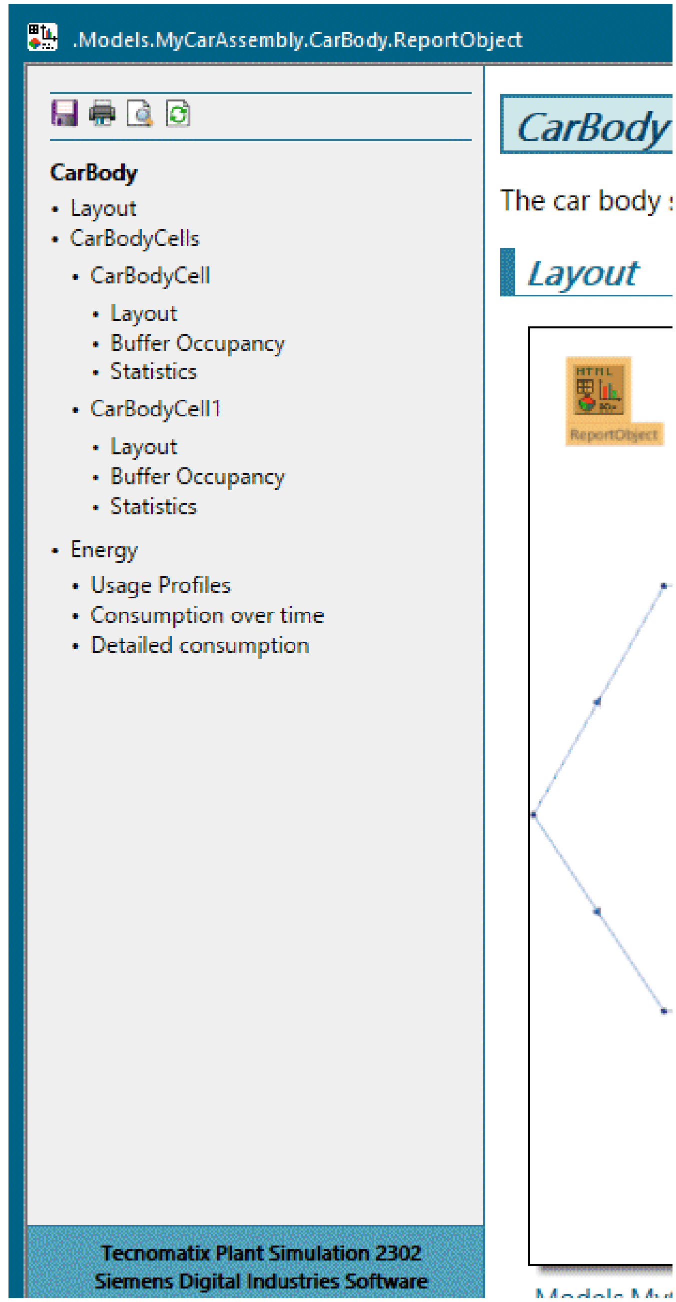
```
# Production Areas
The car assembly encompasses three production areas:
* CarBody
* PaintShop
* FinalAssembly
[CarBody, #]        // Show the HtmlReport of the Frame ‘CarBody’ instead
of the Frame itself
[PaintShop, #]      // Show the HtmlReport of the Frame ‘PaintShop’ instead
of the Frame itself
[FinalAssembly, #]  // Show the HtmlReport of the Frame ‘FinalAssembly’
instead of the Frame itself
```

In our *ReportObject* in the *Frame CarBody*, for example, we show the headings one level higher, compare these instructions:

```
# [current.name]
The car body section of our car assembly looks like this:
## Layout
><[current, *]
## CarBodyCells
The car body section encompasses several CarBodyCells:
[CarBodyCell.Origin*, ##]
[EnergyAnalyzer, #]
```

The instructions in the *ReportObject* in the *Frame CarBody* set which information the overall report displays for the *CarBody* section. The same applies to the *Frames PaintShop* and *FinalAssembly*.

<table>
<tr><th>Heading Levels in the HtmlReport</th><th>Heading Levels in the ReportObject</th></tr>
<tr><td>


</td><td>


</td></tr>
</table>

**Compare Configure the HtmlReport**

**Back to Study the HtmlReport**

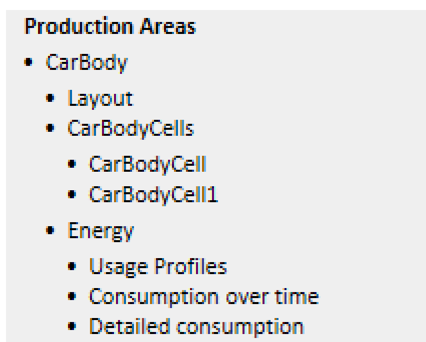
### Display the Energy Analysis

The instruction `[CarBody, #]` in the following code section of the *HtmlReport* shows the content of the *Frame* **.Models.MyCarAssembly.CarBody**.

```
# Production Areas
The car assembly encompasses three production areas:
* CarBody
* PaintShop
* FinalAssembly
[CarBody, #]      // Show the HtmlReport of the Frame ‘CarBody’ instead of
the Frame itself
```

We set in the *ReportObject* in the *Frame* **.Models.MyCarAssembly.CarBody** what exactly the *HtmlReport* for the *CarBody* section shows:

```
# [current.name]
The car body section of our car assembly looks like this:
## Layout
><[current, *]
## CarBodyCells
The car body section encompasses several CarBodyCells:
[CarBodyCell.Origin*, ##]    // Show the heading two levels lower
[EnergyAnalyzer, #]          // Show the heading one level lower
```




Our *HtmlReport* in the *Frame MyCarAssembly* shows the energy consumption of all *AssemblyStations*, which the tool *EnergyAnalyzer* recorded:

- The Chart **Usage profiles**.
- The Chart **Consumption over time**.
- The Table **Detailed consumption**.

**Compare Configure the Body Assembly**

**Compare Configure the HtmlReport**

**Back to Study the HtmlReport**

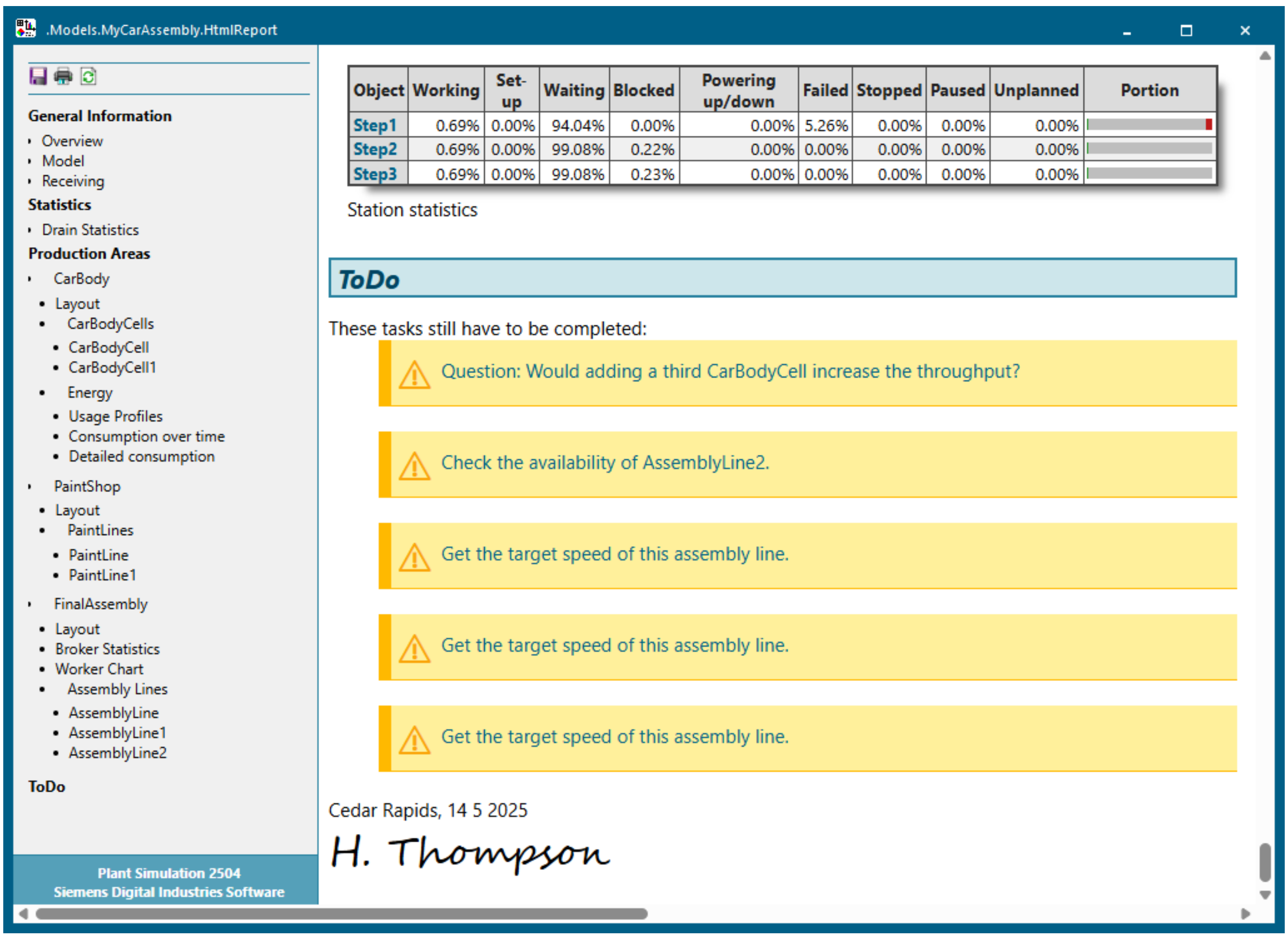
## Display the ToDos and Change to the Respective Frame

We use the object *Comment* to quickly change to the *Frame* in which it is inserted, i.e., its location by selecting **Navigate > Open Location** in its dialog.

The following instructions in the *HtmlReport* collect all objects of type *Comment* named *ToDo* in all *sub-Frames*.

```
# ToDo
These tasks still have to be completed.
[.UserObjects.ToDo*]
```

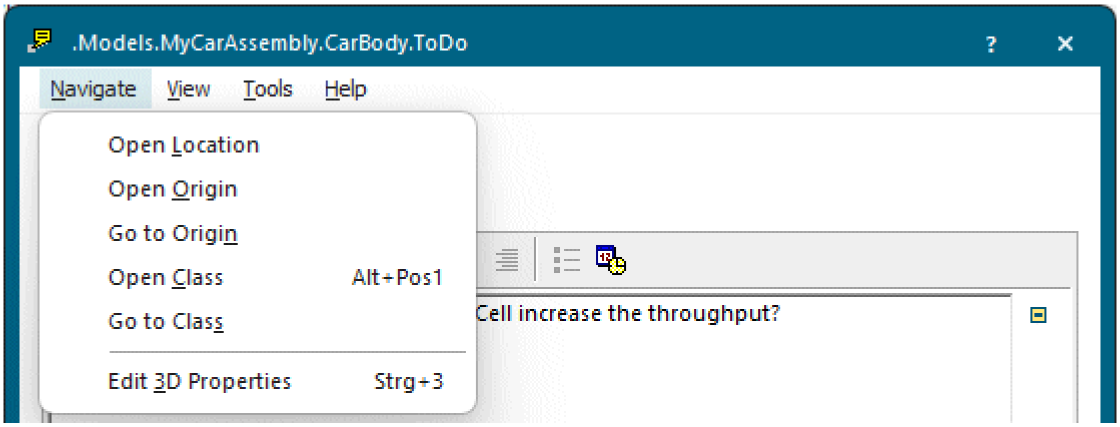
They then show them in the overall report under *ToDo*.



The text that we entered into the individual *Comments* named *ToDo* is a link each to the respective *Comment* in the *Frame* in which it is inserted. The link shows the path in a *Tooltip* when you drag the mouse over it and let it hover.

Click the link to open this *Comment* object.

Then select **Navigate > Open Location** in its dialog to open the **Frame** in which the *Comment* is located.




**Compare** Configure the HtmlReport

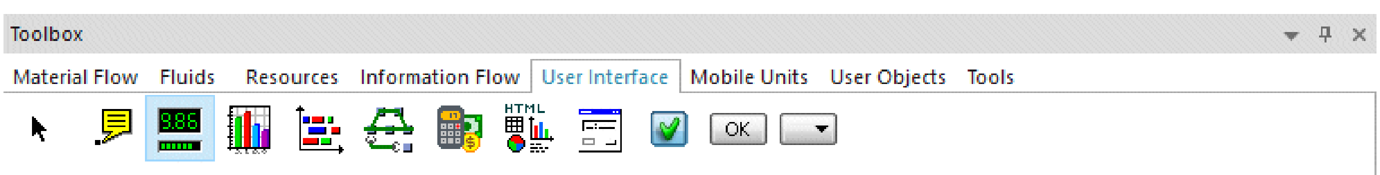
**Back to** Study the HtmlReport

## Showing Values During the Simulation Run with the Display

### Show Values During the Simulation with the Display

If you want to show a value of an *attribute*, of a *method*, or of a *Variable* at all times throughout the entire simulation run, we will use the *Display*. The *Display* is, for example, ideally suited to dynamically show the fill level of a *Buffer*.

You can insert it into your simulation model from the folder **UserInterface** in the *Class Library* or from the toolbar **User Interface** in the *Toolbox*.




- If you deactivate the **Display**, it shows its icon  in the *Frame*, into which you insert it.
- If you activate it by selecting ☑ Active, the *Display* shows a value in the *Frame* window, such as Fill level Buffer 3.

You can:

- **Select Which Data the Display Shows**
- **Select How the Display Shows the Data**

**Compare Viewing the Statistics Report**

**Compare View Statistics in the Dialogs of the Objects**

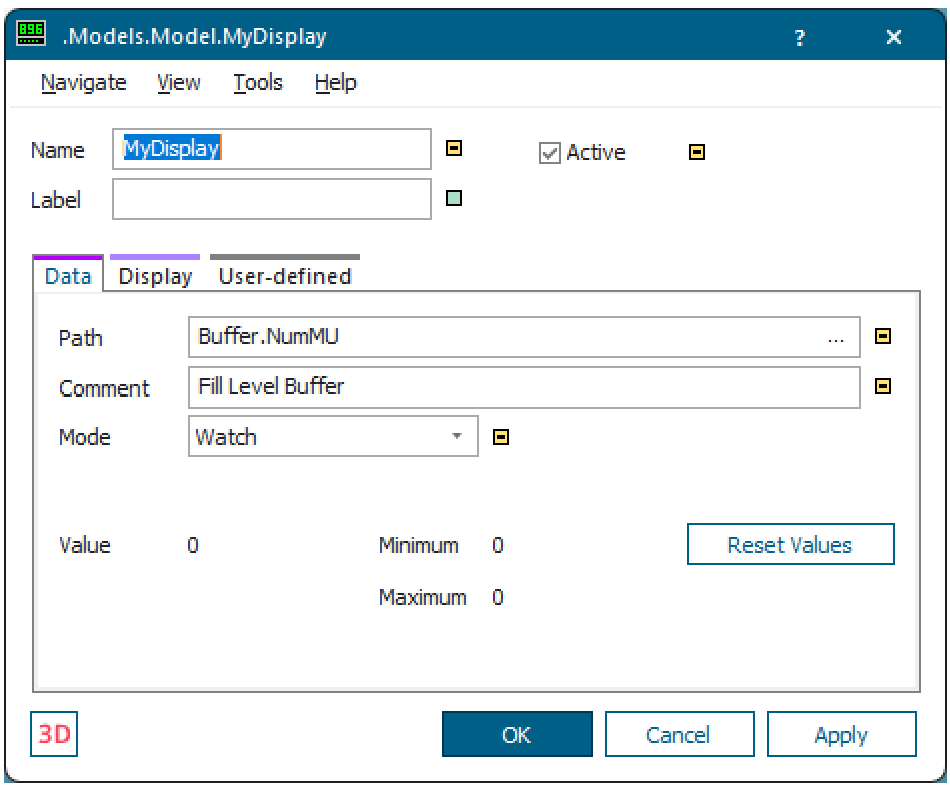
**Compare Show Statistics in a Chart**

**Compare Show Statistics and Other Values in a Report**

**Back to View and Visualize Statistics**

## Select Which Data the Display Shows

Before you start your simulation run, you have to tell the *Display* which value it displays and in which mode it collects the data.




- Click … to type in the path to and the name of the object whose data you want to view. Then, type the value of a global variable (*Variable*), of an *attribute* or of a *method* of an object into the text box after the path. You might, for example, type in `buffer.numMU, .Models.Model.ParallelStation.statNumOut, store.XDim`, etc.

The *Display* supports the data types *boolean, integer, real, string, object, time, money, length, weight, speed, date*, and *datetime*. If the path you type in is invalid, the background of the text box changes to red.

- If you want to, you can also type in a brief description (**Comment**) of the displayed value. You might, for example, type in a description of the attribute that the *Display* displays, such as `Fill Level Buffer`. The *Display* show this text below the value in the *Frame* window.

**Note**

When you select to display the value as **Text**, we recommend to type in one or two spaces after the last word, so that the value does not appear too close to the comment.

- Select which mode the *Display* uses to update the displayed data.
  - **Sample** mode updates the *Display* periodically and takes the next sample after the time span you type into the text box **Interval**.
  - **Watch** mode updates the *Display*, whenever the value of a **watchable** *read-only attribute* changes, for example *NumMU*, *NumMUParts*, *StatMaxNumMU*, *StatNumIn*, *StatNumOut*, etc. or of a watchable *attribute*, for example *Pause*, *Speed*, *Unplanned*, etc.

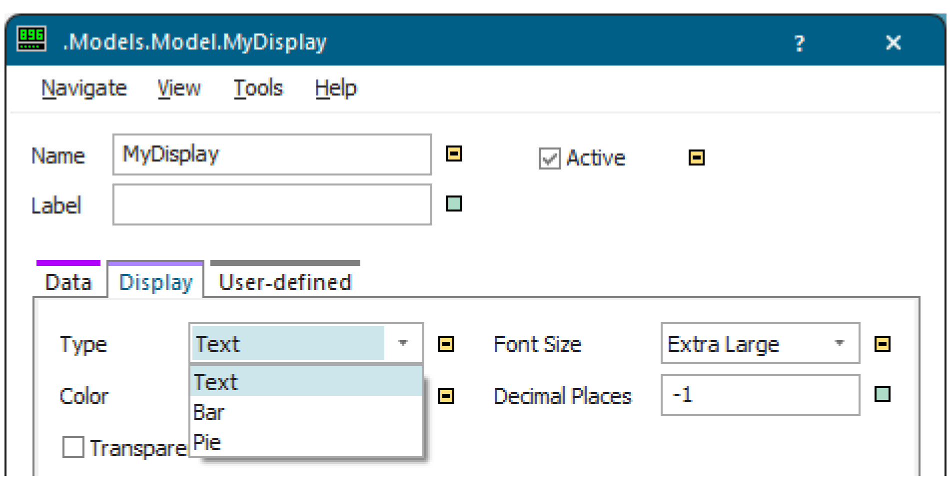
    The column **Watchable** in the window **Show Attributes and Methods** of the individual objects shows all *methods*, *read-only attribute*, and *attributes* which *Plant Simulation* can watch.

**Back to Show Values During the Simulation with the Display**

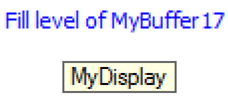
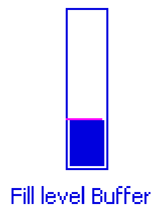
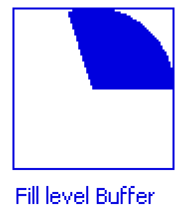
## Select How the Display Shows the Data

Before you start your simulation run, you have to tell the *Display* how to show the value it collects. In **Sample mode** it updates the value periodically, in **Watch mode**, whenever the displayed value changes.

Proceed as follows:



- Select how you want the *Display* to show the value: As **Text**, as a **Bar**, or as a slice of a **Pie**. The *Display* can only show numerical values as a bar and as a slice of a pie. These are easy to comprehend, but not very well suited if accuracy is important.

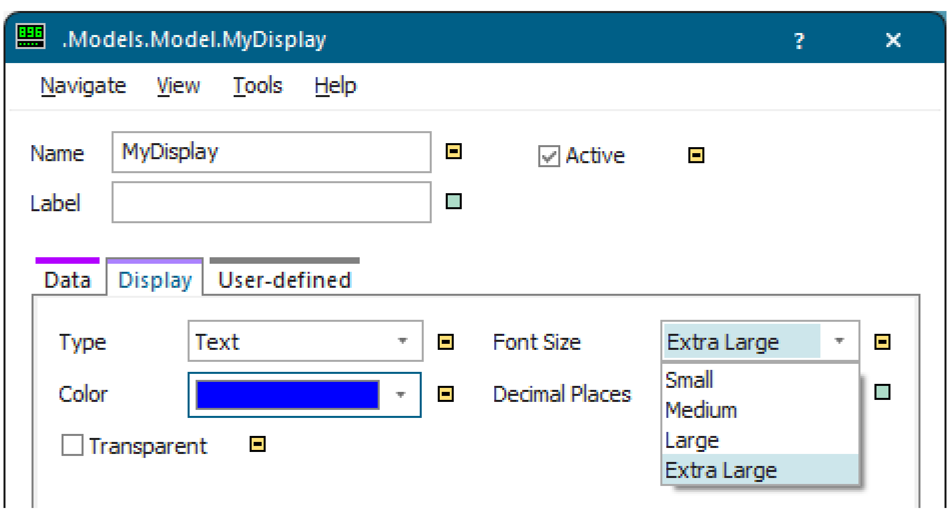
| Text | Bar | Pie |
|---|---|---|
|   |   |   |

  - When you select **Bar** or **Pie**, the *Display* shows the value in relation to the **Minimum** and the **Maximum** value of the range you enter. An empty bar/pie indicates values less than or equal to the lower bound of the range. A full bar/pie represents values greater than or equal to the upper bound of the range. You can also change the height and the width of the bar/pie graphic by dragging its outline with the left mouse button while holding both the **Shift** and the **Ctrl** keys down.

    The *Display* show the actual minimal and maximal values as dashed lines in the *Frame* window, which is not as accurate as you might need it to be. To view the current values, click the tab **Data** and look at the boxes **Minimum** and **Maximum**.

    To reset the values on the tab to 0, click **Reset Values**.

  - When you select **Text**, the *Display* shows the value as a number or as text in the *Frame* window. Then, you can select a **Font size** for displaying the value itself and for displaying the **Comment** you entered. Enter a blank after the text you enter so that the **Comment** value is not shown attached to the text.




- To make the background of the *Display* you inserted into a *Frame* transparent, select **Transparent**.

  Then, the *Display* shows the space around the text in the color of the background image or it shows the grid points of the *Frame*. If you clear **Transparent**, *Plant Simulation* shows the space around the text white.

| Not transparent | Transparent |
|---|---|
| NextNumber= 28 | NextNumber= 28 |

- To select a color for the text the *Display* displays and for the outline of its display graphic, click in the field next to **Color** and select a color in the dialog **Colors**.

**Compare** **Check the Fill Level of the Buffer**

**Back to** **Show Values During the Simulation Run with the Display**

## Accessing Statistics with SimTalk

You can access most of the statistics values with *methods*, *read-only attributes*, and *attributes*.

Compare the following:

- For the *material flow objects* the names of the *read-only attributes* are listed next to the name of the value in the description of the **Statistics Report**.
- For the *Drain* the names of the *read-only attributes* are listed next to the name of the value in the description of the **Statistics Report**.
- For the *Exporter* and the *Worker* the names of the *read-only attributes* are listed next to the name of the value in the description of the **Statistics Report**.
- In addition, you might want to check the topics **Read-Only Attributes for Accessing Statistics** and **Attributes for Statistics**.
- For the *MUs* the names of the *read-only attributes* are listed next to the name of the value in the description of the **Product Statistics of the MUs** in the **Statistics Report**.
- For the object *Transporter* the names of the *read-only attributes* are listed next to the name of the value in the description of the **Driving Statistics of the Transporter**.
- In addition, you might want to check the topics **Methods for Accessing Statistics of the MUs** and **_Attributes of All MUs**.

**Compare** **View Statistics in the Dialogs of the Objects**

**Compare** **Show Statistics in a Chart**

**Compare** **Show Statistics and Other Values in a Report**

**Compare** **Show Values During the Simulation with the Display**

**Back to** **View and Visualize Statistics**

# Showing Part Flows in a SankeyDiagram

## Show Part Flows in a SankeyDiagram

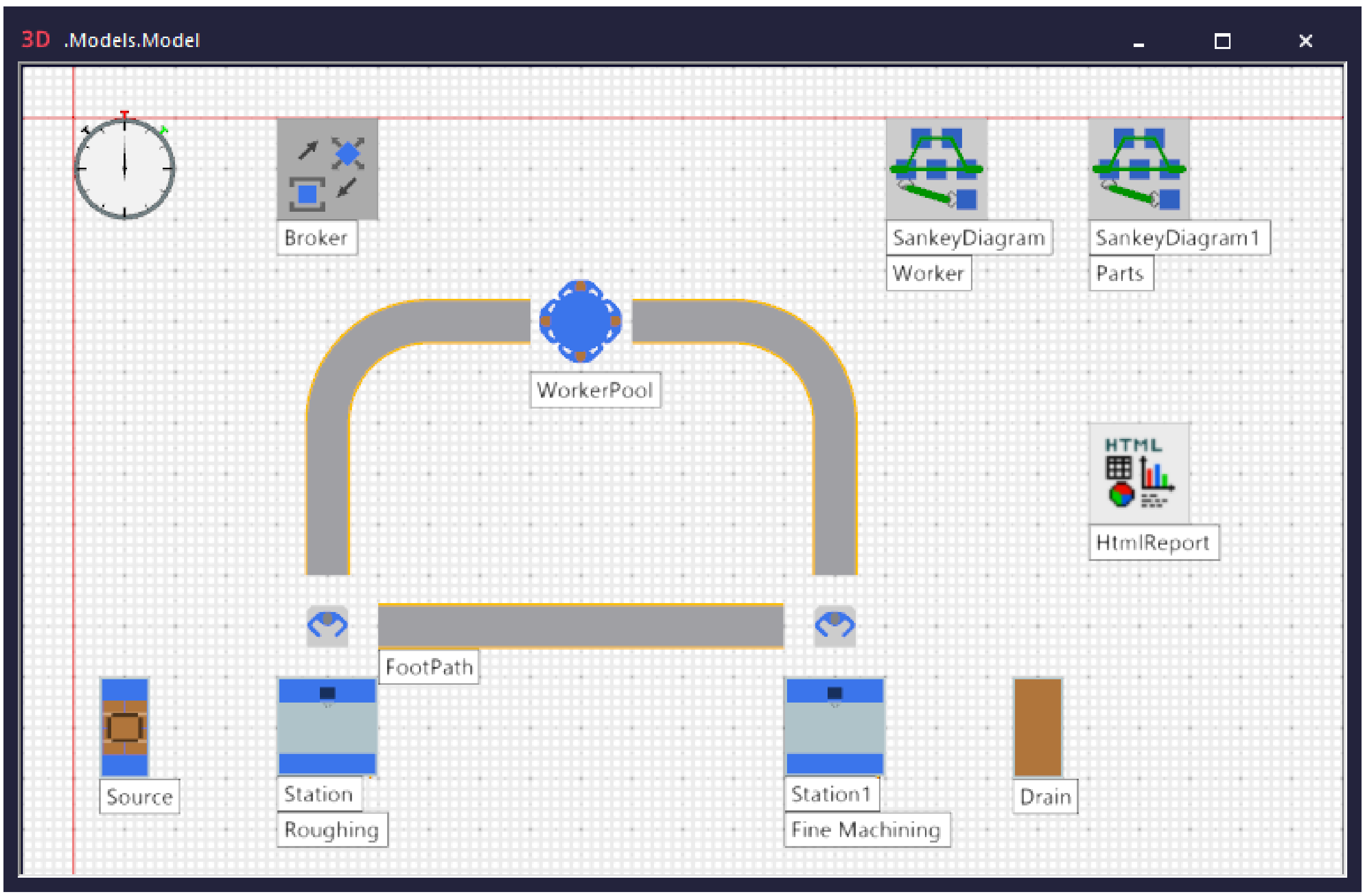
The *SankeyDiagram* visualizes how *parts* or *Workers* move from object to object in your simulation model.

You can insert the **SankeyDiagram** into your simulation model from the folder **User Interface** in the *Class Library* or from the tab **User Interface** in the **Toolbox**.

In our model the *Source* produces parts. The station *Roughing* processes these parts for the defined time. Then the *Worker* carries the processed parts to the station *FineMachining* which adds additional processing steps. The *Drain* removes the parts from the factory.

The *SankeyDiagram* visualizes the flow of the *parts* and the *Workers* after the end of the simulation run.

Our finished simulation model looks like this:



> **Note**
>
> If you also want to show the **names and** the **labels** of the material flow objects in 3D, click the class of the object with the right mouse button and select **Edit 3D Properties**. Then select **Name/Label Activated** on the tab **Captions**.

We will demonstrate how to:

- **Configure the Objects that Produce and Process Parts**
- **Configure the Carrying Operation with Broker, WorkerPool, and Worker**
- **Configure the SankeyDiagrams**

**Back to View and Visualize Statistics**

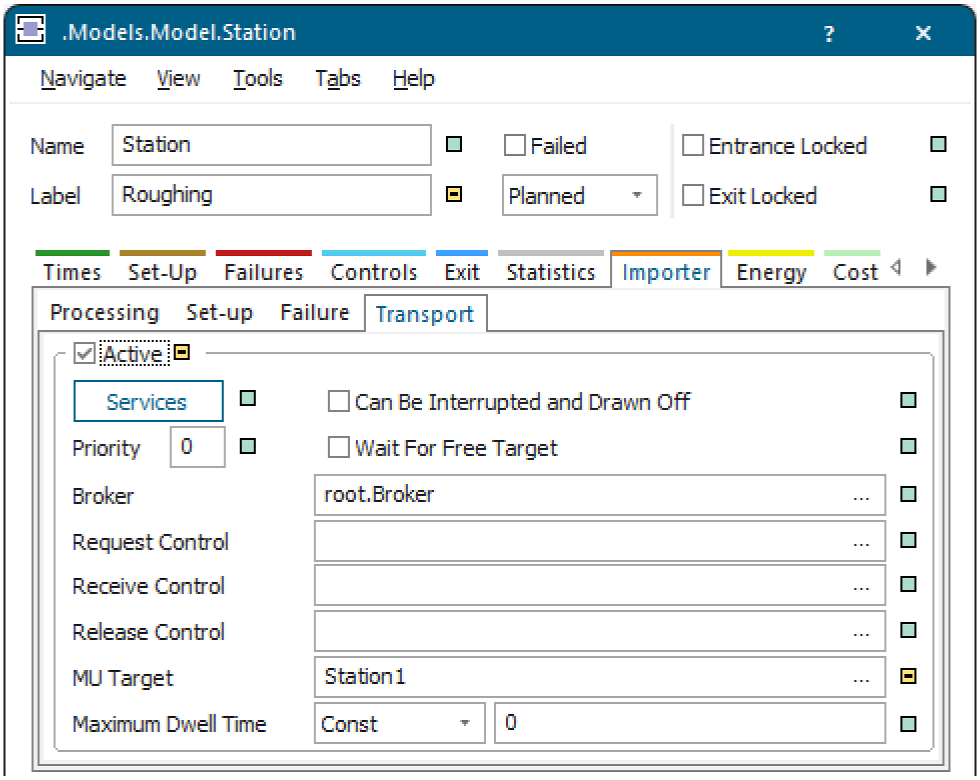
### Configure the Objects that Produce and Process Parts

You can model and configure the material flow in you simulation model.

Proceed as follows:

- A *Source* which produces the parts. We use the *Source* with its default settings.
- A station which roughly machines the parts. For this we used the *Station* with its default settings.

  As we want a *Worker* to carry the processed parts from the *Station* labeled *Roughing* to the *Station* labeled *FineMachining*, we defined these settings for the *transport importer*:




- A second station labeled *FineMachining*, which does the fine machining of the parts. For this we used the *Station* with its default settings as well.
- A *Drain* which removes the parts from the factory.

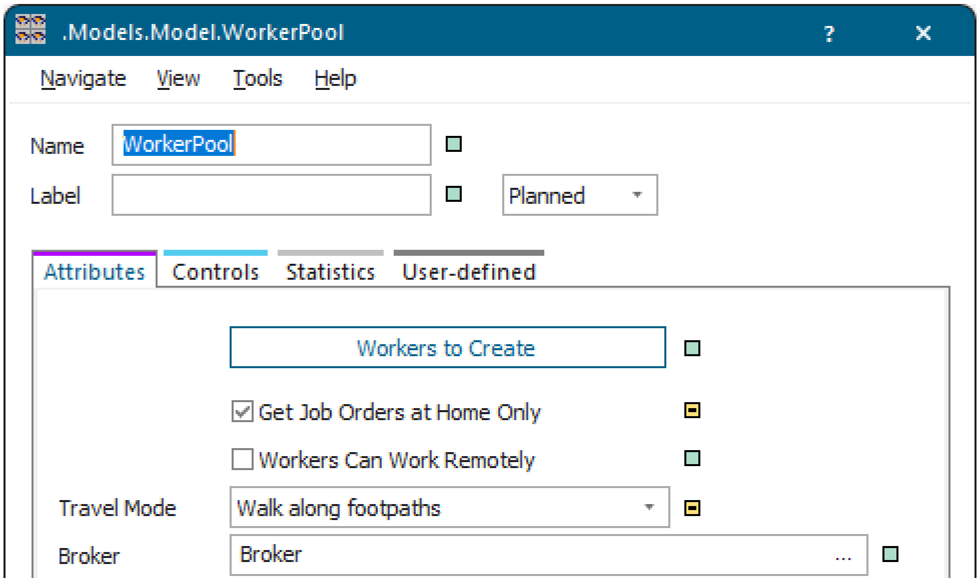
**Back to Show Part Flows in a SankeyDiagram**

## Configure the Carrying Operation with Broker, WorkerPool, and Worker

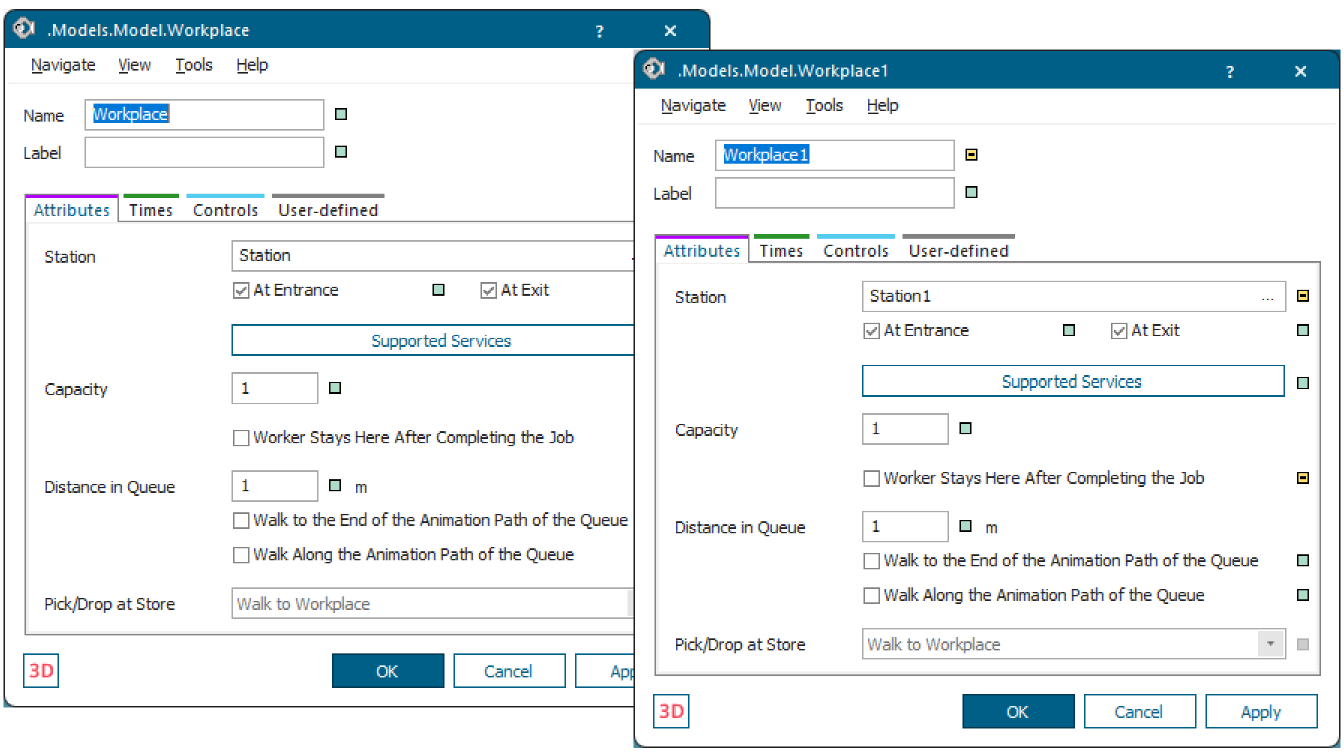
Insert the resource objects into our simulation model to allow the *Worker* to carry parts from the station *Roughing* to the station *Fine Machining*.

Proceed as follows:

- A *Broker* which assigns the required services to the *Worker*. We use the *Broker* with its default settings.
- A *WorkerPool* which sends the *Workers* off to their jobs. Here we selected the **Travel Mode > Walk along footpaths** and we activated **Get Job Orders at Home Only**.




- A *Workplace* each which we attached to the stations *Roughing* and *Fine Machining*. We cleared the check box **Worker Stays Here After Completing the Job**.




- A *FootPath* each between the *Workplaces* and the *WorkerPool* and between the *Workplaces*.

  Make sure that the *Workplaces* are connected with Connectors with the *FootPaths* and the *WorkerPool*.

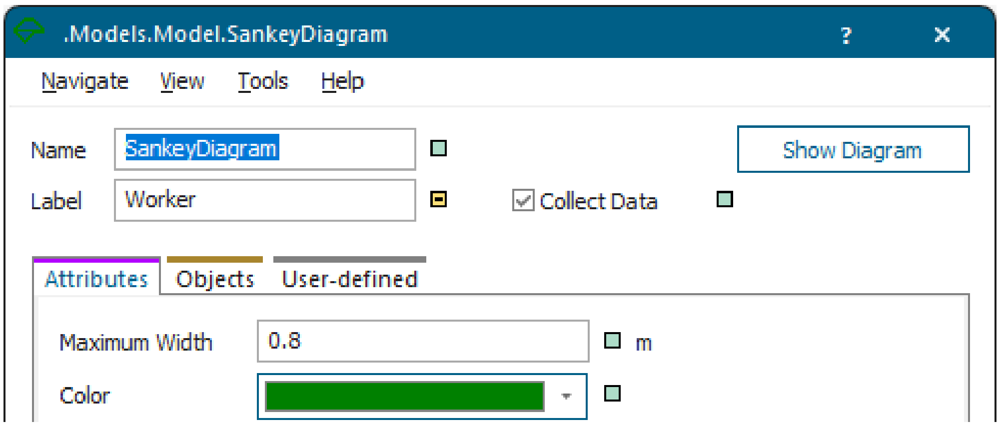
**Back to Show Part Flows in a SankeyDiagram**

## Configure the SankeyDiagrams

We use a *SankeyDiagram* each to visualize the *Sankey flows* of the *Worker* and the *parts*.

We selected these configurations for that:

- For the *Worker SankeyDiagram* we used the default display attributes.



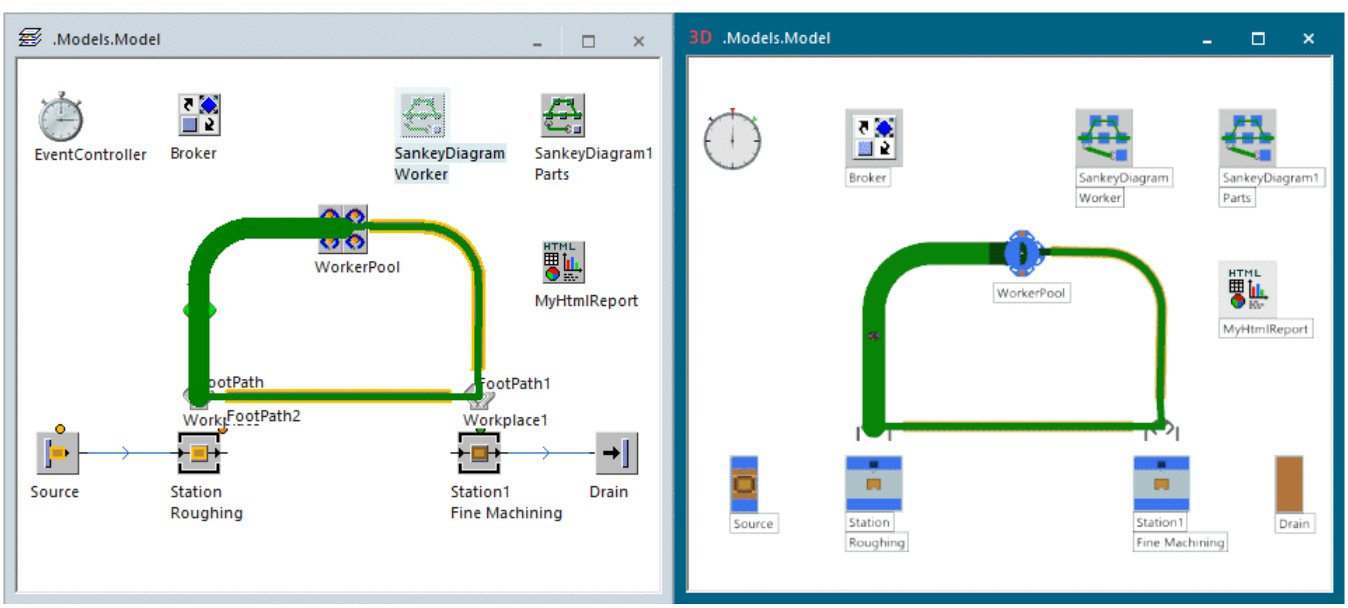

  As we want to show the *Sankey flows* of the *Worker*, we turn the check box for inheritance on the tab **Objects** off and drag the *Worker* from the folder **Resources** in the *Class Library* to this tab.

  We then clicked **OK** in the dialog to apply the new settings.

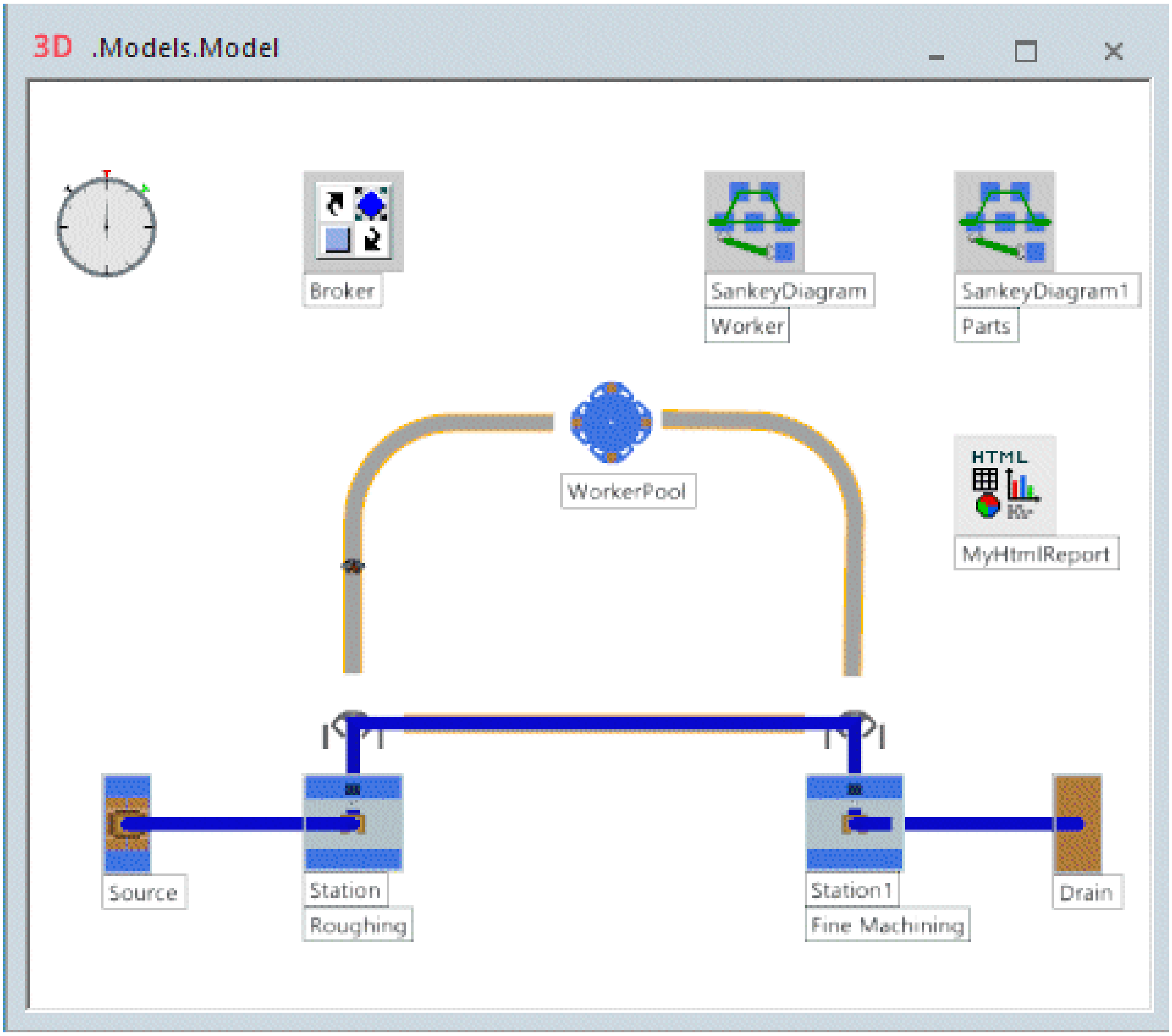
- For the *Parts SankeyDiagram* we only changed the color of the *Sankey flows*.

  We then clicked **OK** in the dialog to apply the new settings.

- Run the simulation over a considerable period of time.
- Click the *SankeyDiagram* labeled *Worker* with the right mouse button and select **Show**. The result looks like this:



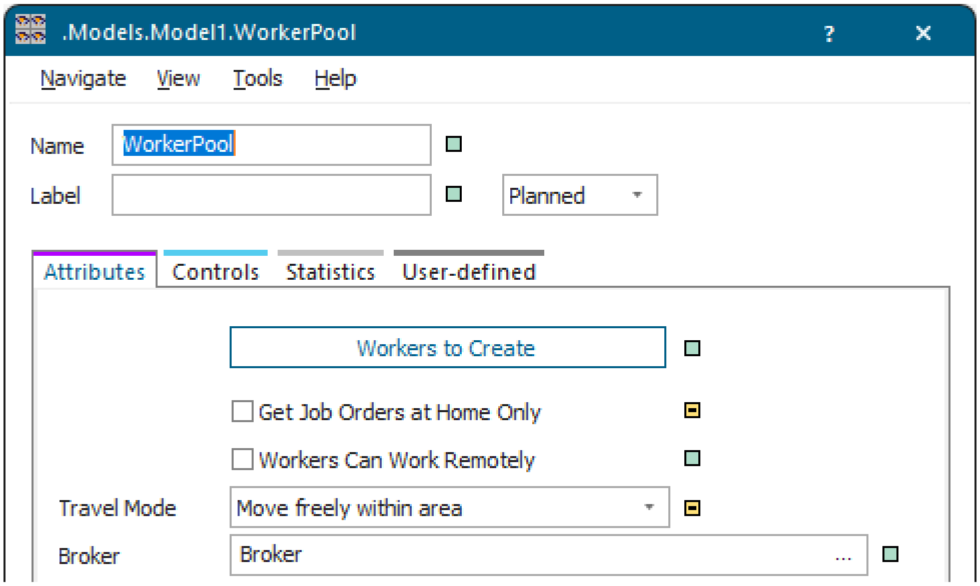
- Click the *SankeyDiagram* labeled *Parts* with the right mouse button and select **Show** to also show the *Sankey flows* of the parts. The result looks like this:



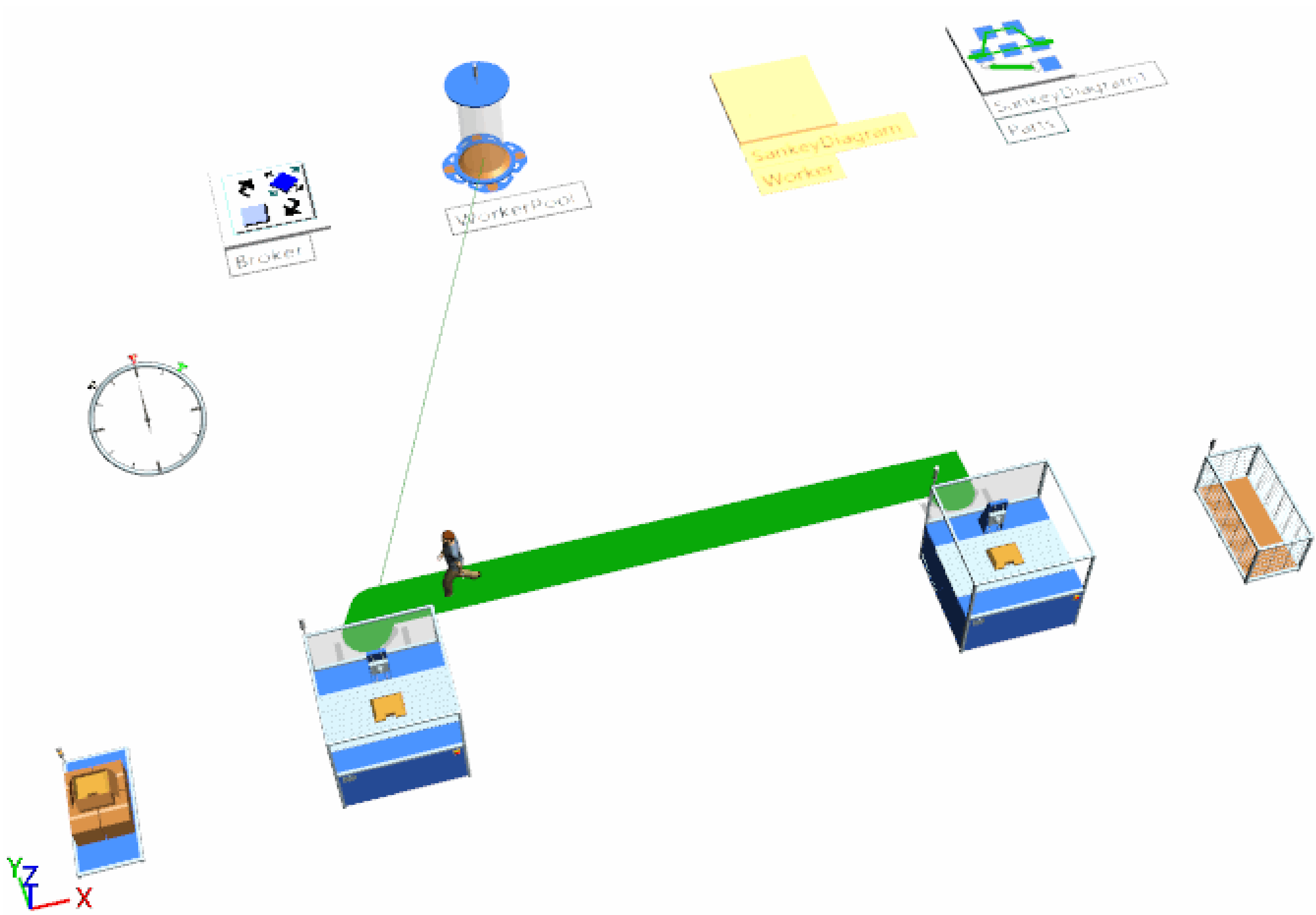

- To hide the *Sankey flows* of a *SankeyDiagram* again, click it with the right mouse button, and select **Hide**. Instead you can also click the button in the dialog of the respective *SankeyDiagram*.

To demonstrate how the *SankeyDiagram* shows the *Sankey flows* of the *Worker*, who walks freely within the area, we duplicated our model. Then we changed these settings in the *WorkerPool*:

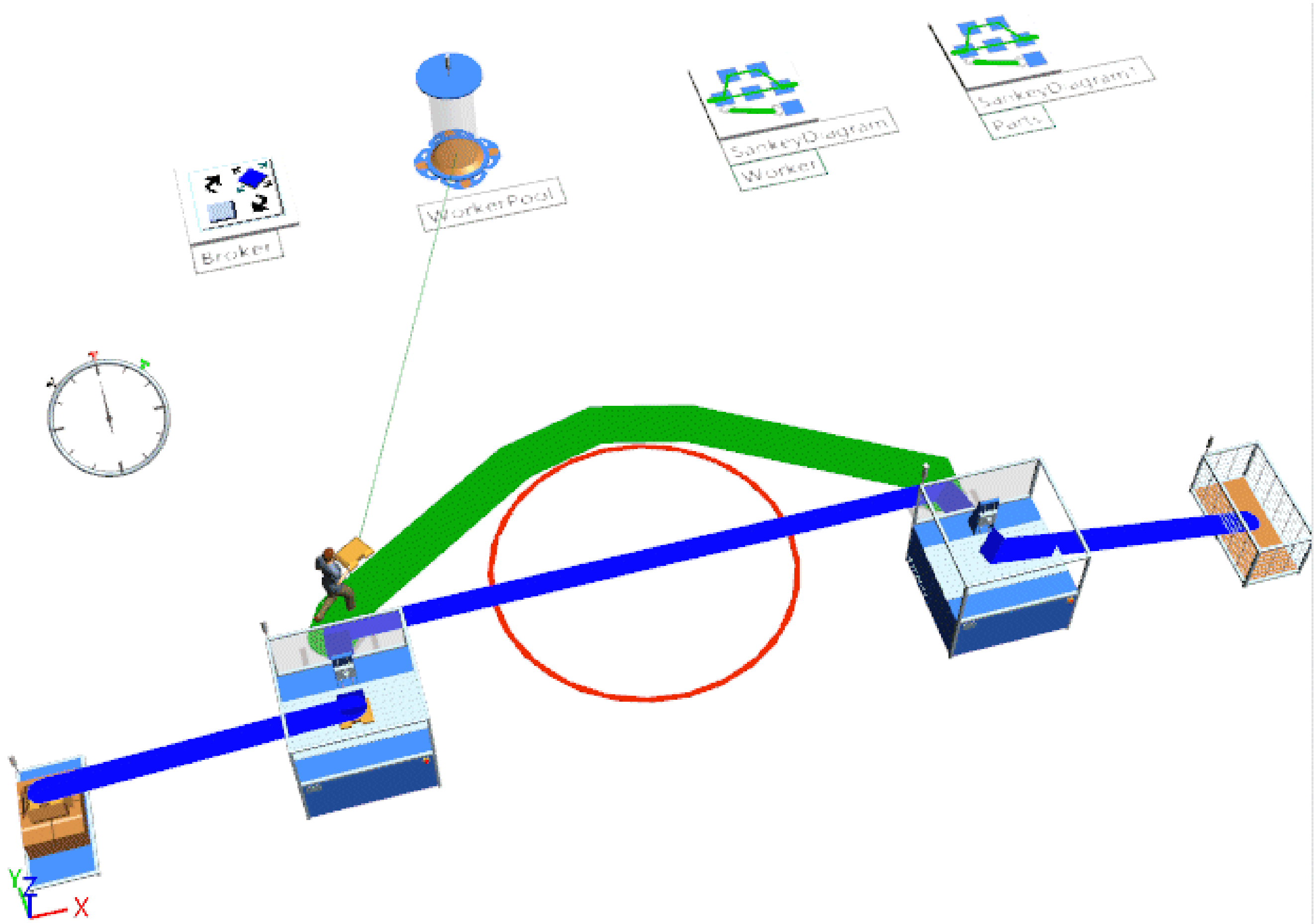
- We cleared **Get Job Orders at Home Only**.
- We selected the **Travel Mode > Move freely within area**.




- The *SankeyDiagram* shows the *Sankey flows* of the *Worker*, who walks freely within the area, like this:



If we insert a barred area, around which the *Worker* has to walk, the *SankeyDiagram* looks like this:




**Back to** [Show Part Flows in a SankeyDiagram]

## Showing AGV Sankey Flows

### Show AGV Flows in a SankeyDiagram

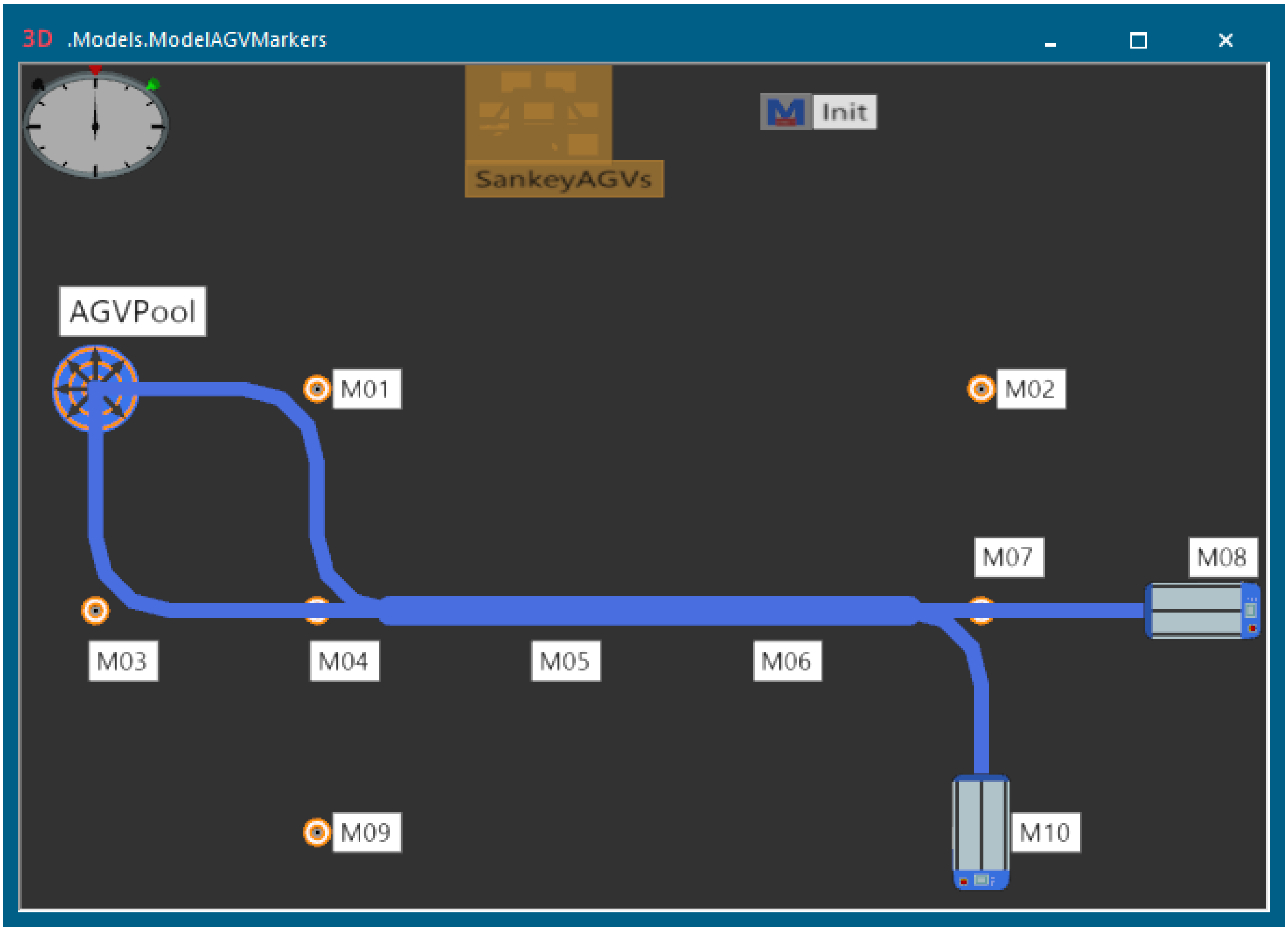
The *SankeyDiagram* visualizes how *AGVs* drive from *Marker* to *Marker* in your simulation model.

You can insert the **SankeyDiagram** into your simulation model from the folder **UserInterface** in the *Class Library* or from the tab **User Interface** in the **Toolbox**.

In our model two *AGV* drive two different, partly overlapping routes along a number of *Markers*.

The *SankeyDiagram* visualizes the flow of the *AGVs* after the end of the simulation run.

Our finished simulation model looks like this:




We will demonstrate how to:

- **Configure AGVPool and Routes of the AGVs**
- **Configure the init Method and the SankeyDiagram**

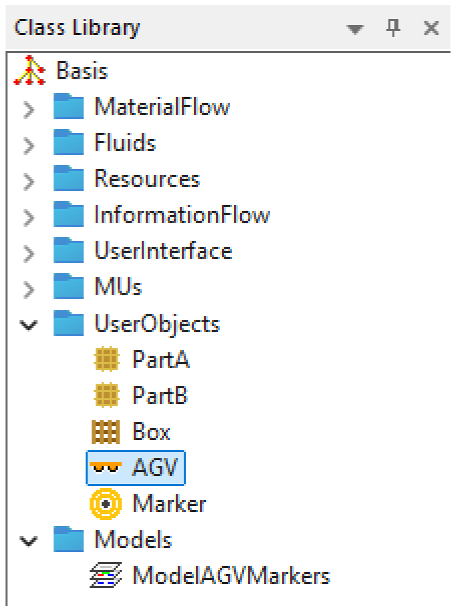
**Back to View and Visualize Statistics**

### Configure AGVPool and Routes of the AGVs

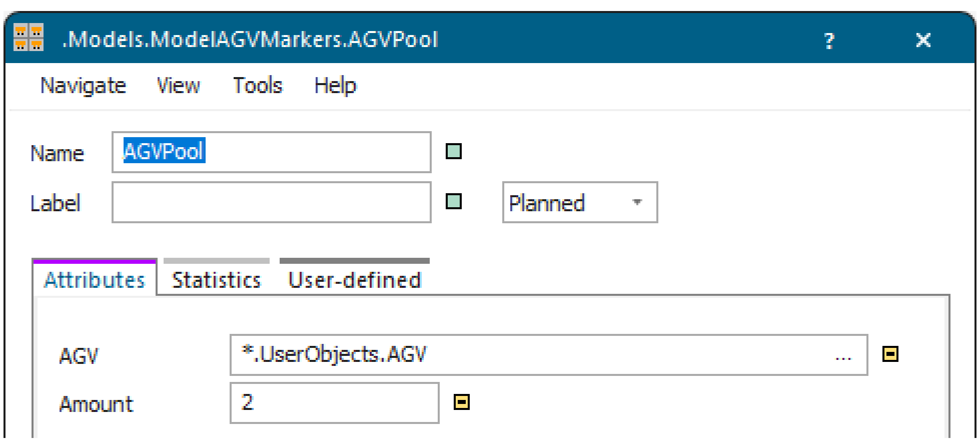
Insert the *AGVPool* from which the two *AGVs* drive along the *Markers* to their respective destination.

Proceed as follows:

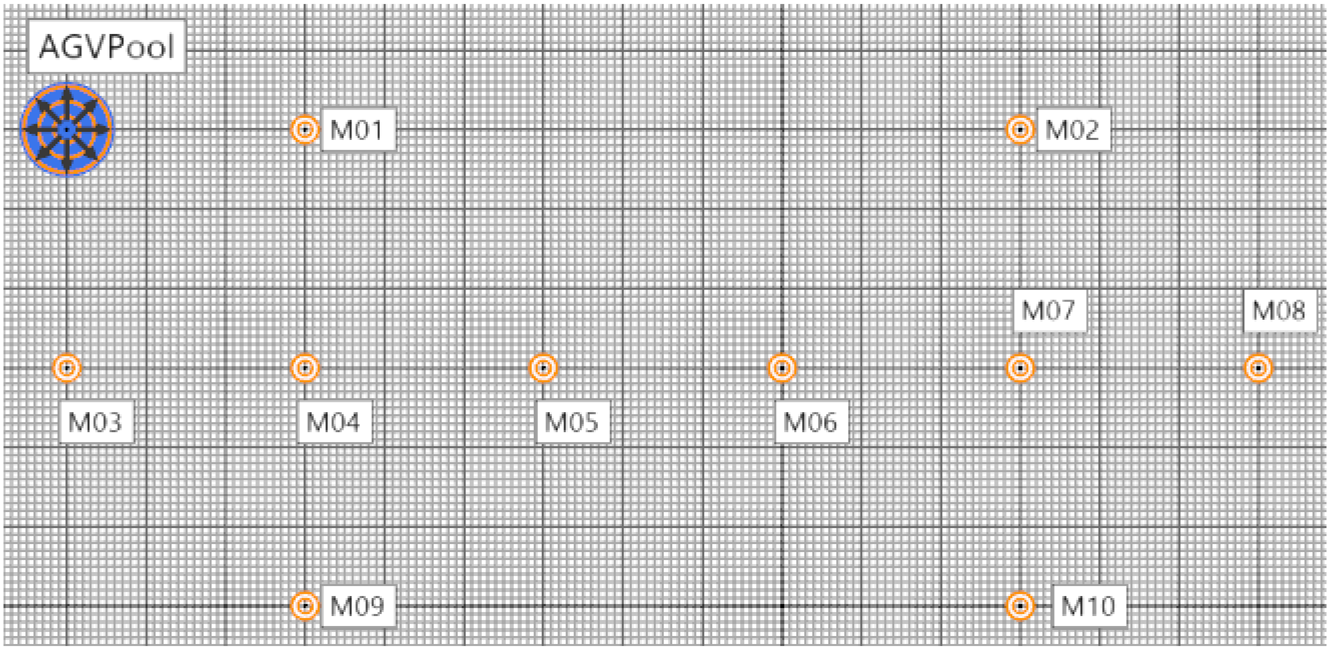
- Duplicate the *AGVPool*, the *Marker*, and the *Transporter*. Rename the *Transporter* to *AGV* in the folder **UserObjects**.




- Insert the *AGVPool*, where the *AGVs* reside, into the model. Configure the *AGVPool*. We selected these settings.




- Insert the *Markers* along which the two *AGVs* drive. We chose these positions.




Back to **Show AGV Flows in a SankeyDiagram**

## Configure the init Method and the SankeyDiagram

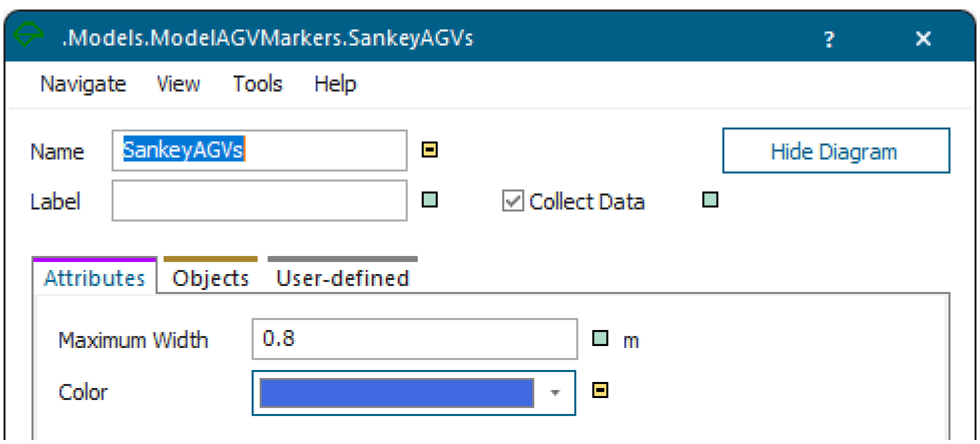
Program the *init* method, which sets the routes on which the *AGVs* drive along the *Markers* to their respective destination.

Insert a *Method* into the model and name it *init*.

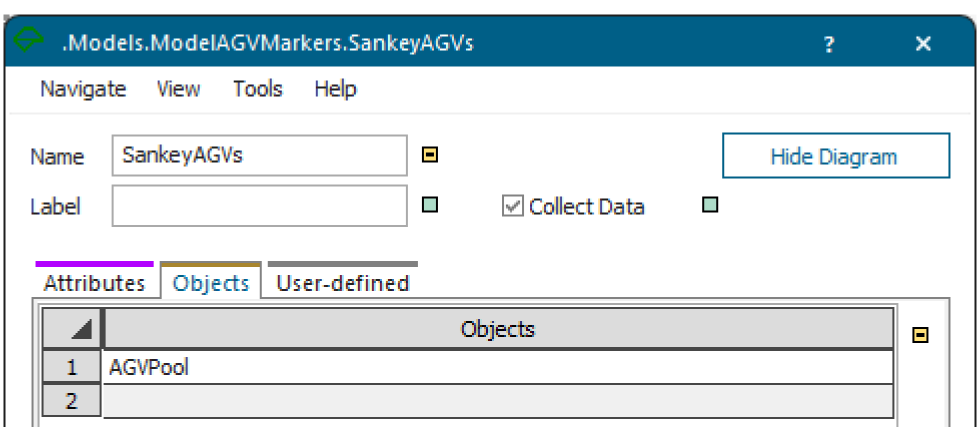
We typed in this source code:

```
var agv1: object := AGVPool.getIdleAGV
agv1.setRoute([M03, M04, M05, M06, M07, M10])

wait 10

var agv2: object := AGVPool.getIdleAGV
agv2.setRoute([M01, M04, M05, M06, M07, M08])
```

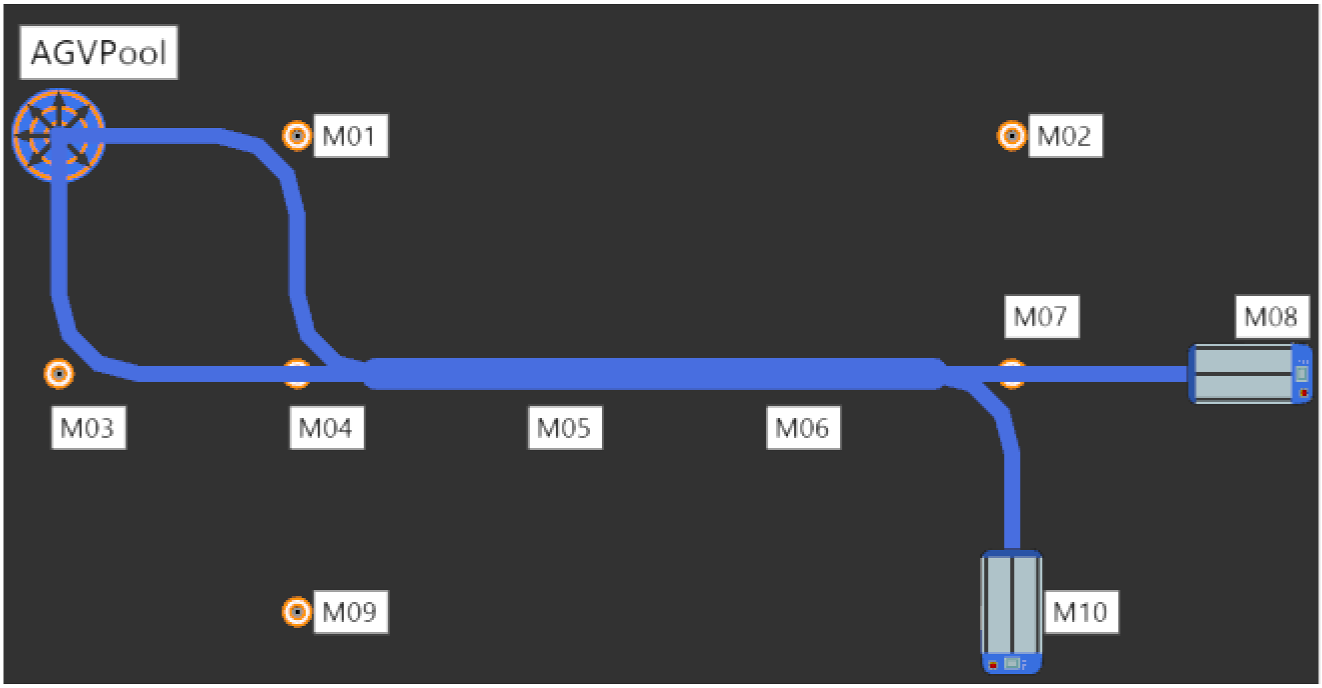
Insert a *SankeyDiagram* into the model. We selected these display settings on the tab **Attributes**.




Drag the *AGVPool* over the *SankeyDiagram* to show the *Sankey flows* of the *AGVs* originating in the *AGVPool*.



Run the simulation and click **Show Diagram** to show the *Sankey flows* of the *AGVs*.




The model then shows the routes, which the *AGVs* took as blue bars. The thicker bar shows the part of the route that both *AGVs* shared.

**Back to Show AGV Flows in a SankeyDiagram**

## Showing Parts on Resources in a GanttChart

### Show Parts on Resources in a GanttChart

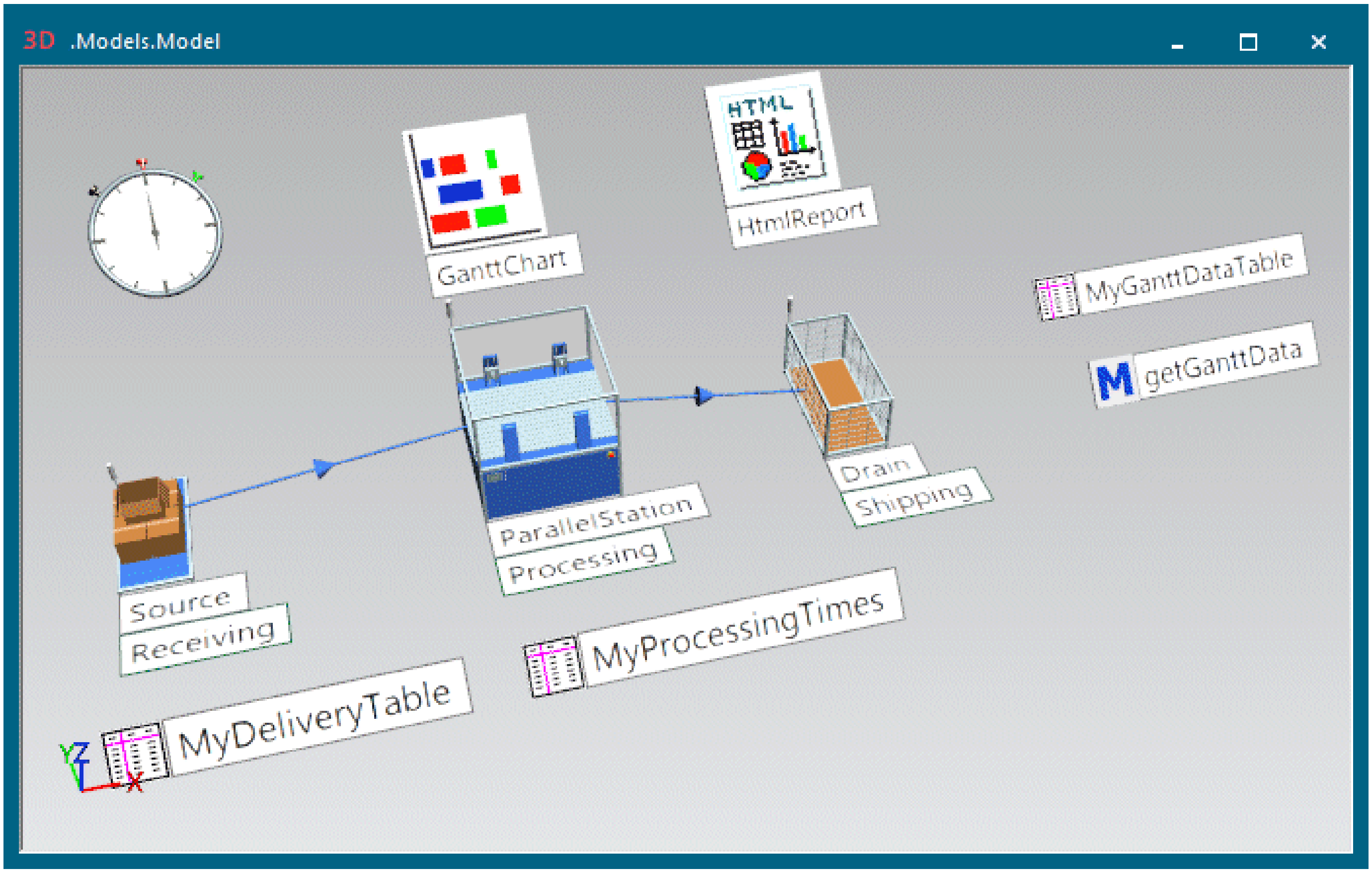
The *GanttChart* shows how parts move from resource to resource during the simulation.

In this context resources are the **material flow objects**, not the **resource objects**.

You can insert the **GanttChart** into your simulation model from the folder **UserInterface** in the *Class Library* or from the toolbar **User Interface** in the *Toolbox*.

In our model the *Source* produces different parts with different processing times according to a delivery table. The *ParallelStation* processes these parts then for the specified times and moves them on to the *Station* and the *Drain*. The *GanttChart* shows the progression of the parts during the simulation.

The finished simulation model looks like this:



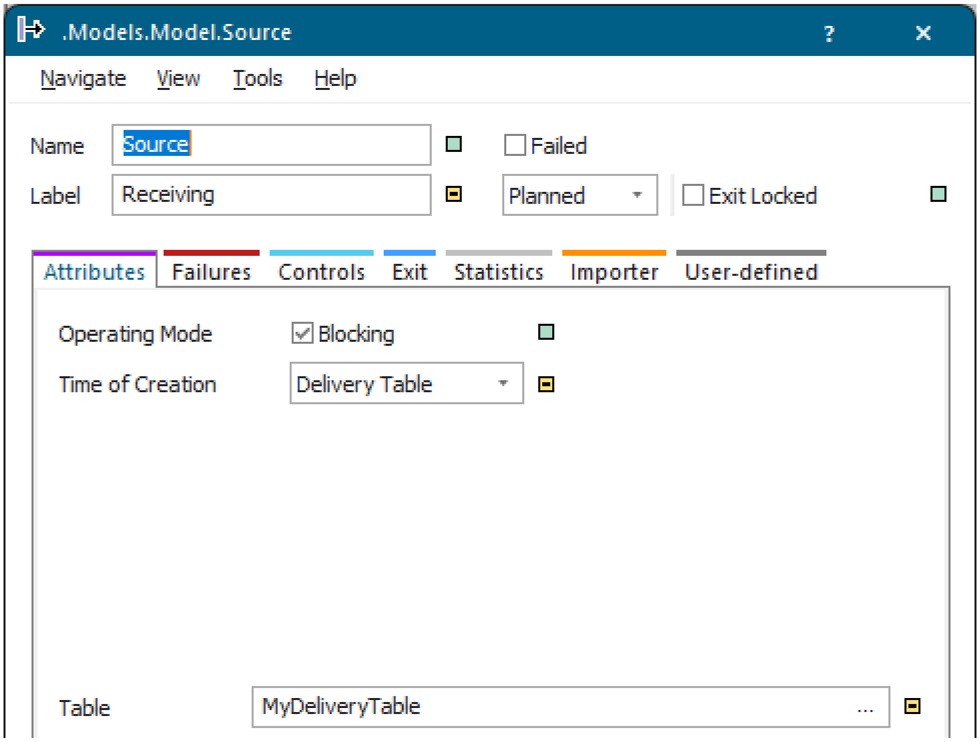

We will demonstrate how to:

- **Configure the Source that Produces the Parts**
- **Configure the ParallelStation that Processes the Parts**
- **Configure the GanttChart**
- **Show the GanttChart**

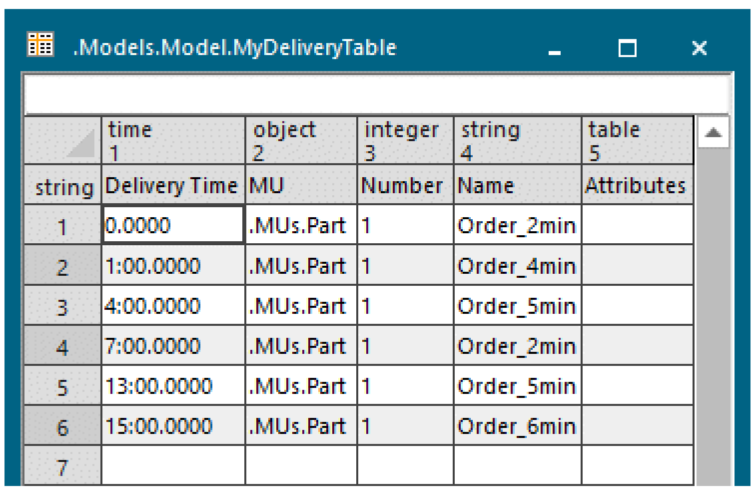
**Back to View and Visualize Statistics**

## Configure the Source that Produces the Parts

Our *Source* produces different parts according to a **Delivery Table**. To accomplish this, we entered the settings below on the tab **Attributes**.




Our **Delivery Table** looks like this:




| | time 1 | object 2 | integer 3 | string 4 | table 5 |
|---|---|---|---|---|---|
| string | Delivery Time | MU | Number | Name | Attributes |
| 1 | 0.0000 | .MUs.Part | 1 | Order_2min | |
| 2 | 1:00.0000 | .MUs.Part | 1 | Order_4min | |
| 3 | 4:00.0000 | .MUs.Part | 1 | Order_5min | |
| 4 | 7:00.0000 | .MUs.Part | 1 | Order_2min | |
| 5 | 13:00.0000 | .MUs.Part | 1 | Order_5min | |
| 6 | 15:00.0000 | .MUs.Part | 1 | Order_6min | |
| 7 | | | | | |

**Back to Show Parts on Resources in a GanttChart**

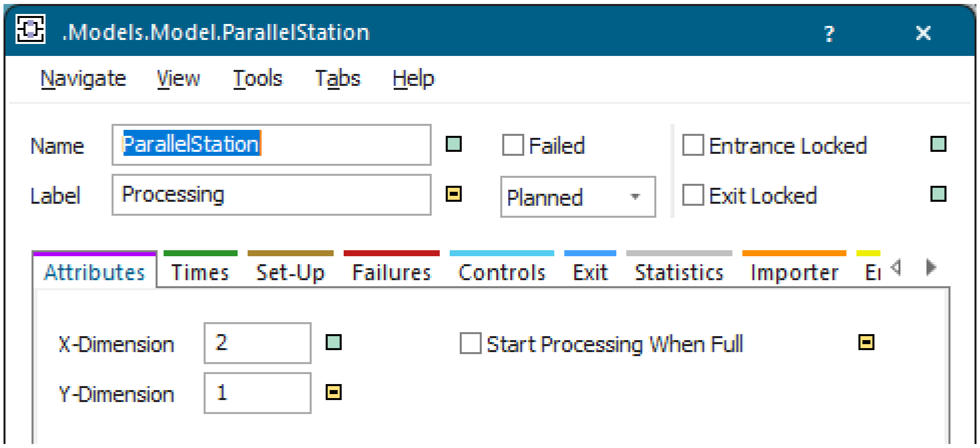
### Configure the ParallelStation that Processes the Parts

Our *ParallelStation* processes the different parts for the times you specify.

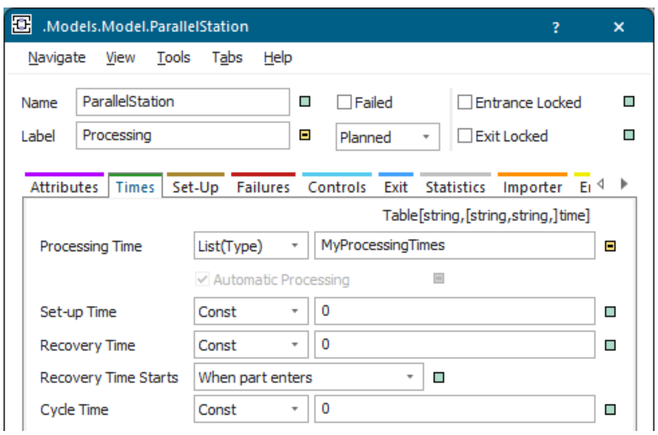
We selected these settings:

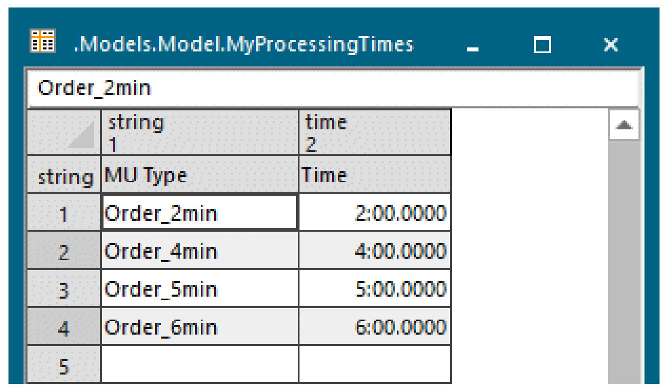
- We entered an **X-Dimension** of 2 and a **Y-Dimension** of 1.

Make sure that the check box **Start Processing When Full** is cleared or your model might not run.

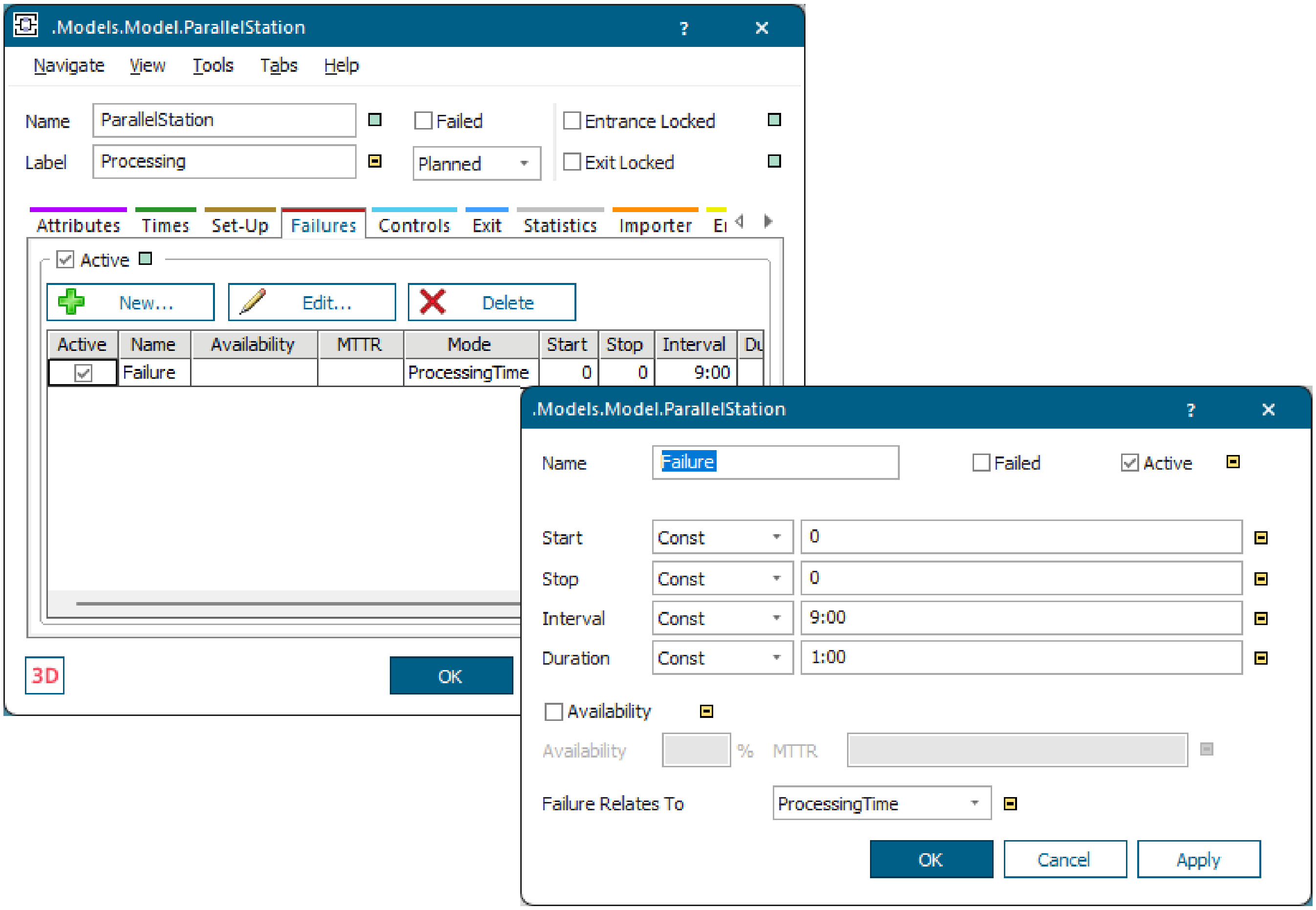



- We set the **Processing Times** of the individual parts in the list *MyProcessingTimes* on the tab **Times**.




- To create a realistic simulation model, we also use **failures**. We selected these settings on the tab **Failures**.




**Back to Show Parts on Resources in a GanttChart**

## Configure the GanttChart

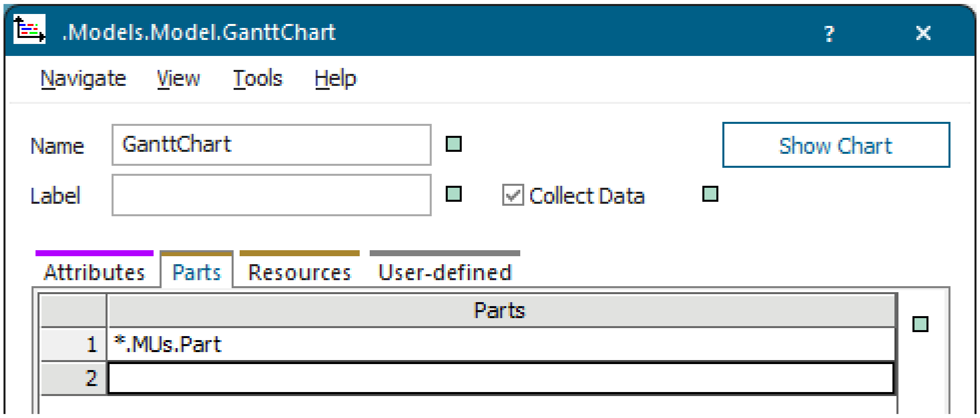
Configure the *GanttChart* to show how the parts move from material flow object to material flow object in the simulation model

Proceed as follows:

- As we want the *GanttChart* to collect data during the simulation, we activated **Collect Data**.
- We would like to watch the part named **Part**. This is already pre-configured, so we do not have to do anything.

  To add another part to the tab **Parts**, select it in the *Class Library*, drag it to the first cell on the tab, and drop it there.

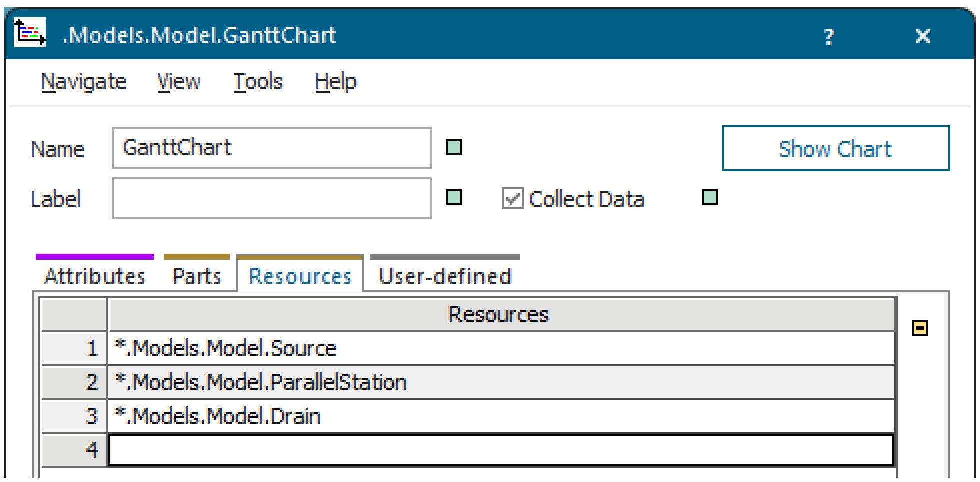
  Instead, you can also select the part in the *Class Library*, drag it on the icon of the *GanttChart*, and drop it there. This automatically deactivates inheritance and adds the part to an empty row on the tab **Parts**.




- We would like to watch all **Resources**. To add them to the tab **Resources**, we selected them in the *Frame*, dragged it to the first cell on the tab, and dropped them there.

  Instead, you can also select one or several resources in the *Frame*, drag it/them on the icon of the *GanttChart*, and drop it/them there. This automatically deactivates inheritance and adds the resources to an empty row each on the tab **Resources**.

  As we want to watch all resources in the simulation model, we could also have left the tab **Resources** empty.



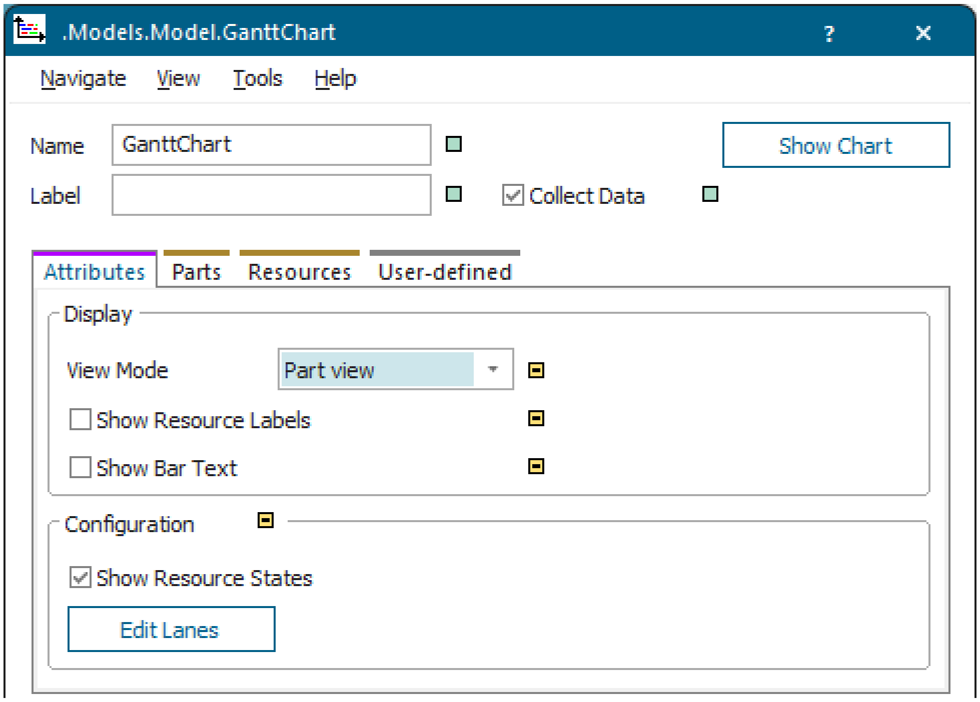

**Back to Show Parts on Resources in a GanttChart**

## Show the GanttChart

Click **Start/Stop Simulation** to run the simulation and let it run for a little while. Then, click **Show Chart** in the open dialog of the *GanttChart*.

Instead, you can also right-click the *GanttChart* in the *Frame* and select **Show** on the context menu.

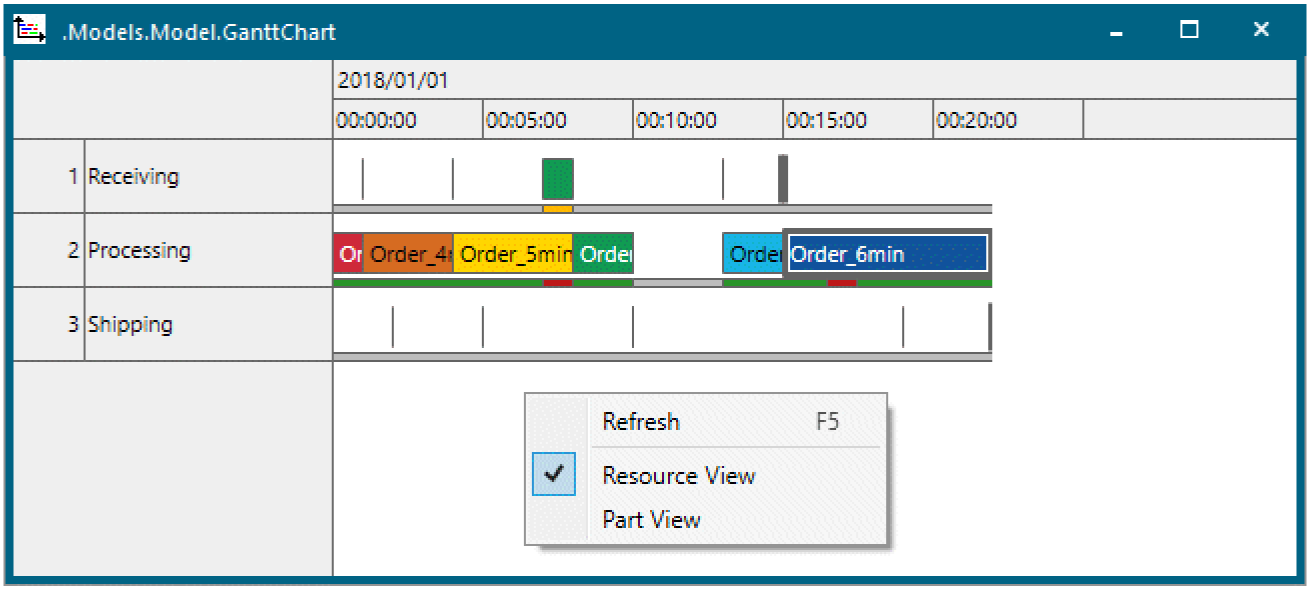
You can select the display settings on the tab **Attributes**.




The *GanttChart* provides two views:

- **Resource view** shows the resources which are occupied by the parts. The *GanttChart* shows the resources along the vertical axis and the elapsing time along the horizontal axis.

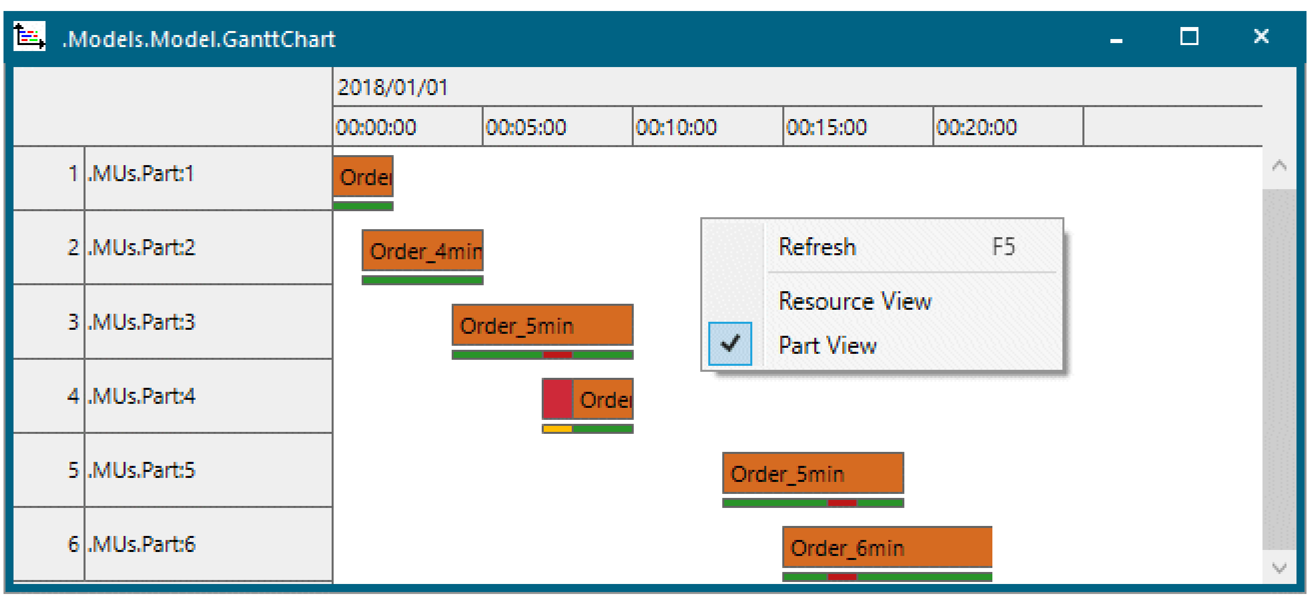
> **Note**
>
> The *GanttChart* shows the resources in the order in which they processed parts for the first time. The resource, which processed a part first, is shown in the first line in the chart. The resource, which processed a part after that, is shown in the line below, etc.
>
> You cannot change the order of the resources.




With the default settings the *GanttChart* shows a part on all resources with the same color.

- **Part view** shows the parts which occupy the resources. The *GanttChart* shows the parts along the vertical axis and the elapsing time along the horizontal axis.

> **Note**
>
> The *GanttChart* shows the parts in the order in which they were processed by a resource for the first time. The part, which was processed by a resource first, is shown in the first line in the chart. The part, which was processed by a resource after that, is shown in the line below, etc.
>
> You cannot change the order of the parts.




With the default settings the *GanttChart* shows a resource for all parts with the same color.

You can select the respective **view mode** on the drop-down list or on the context menu.